ME3201 Machine Design Project

Semester 5

Gearbox for Mechanical Carbon Fiber Impregnation Press

Final Report

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1 Section 1 – Topic: Sliding Mesh Gearbox for an Event Equipment Carrier (230008M)

1.1 Introduction

The event management industry requires frequent transportation of heavy equipment such as loudspeakers, lighting trusses, LED display panels, flight cases, amplifiers and power distribution units during the setup and dismantling of concerts, exhibitions, conventions and sporting events. These items often weight several hundred kilograms and must be transported between trucks, storage areas and stages within limited time. Current transportation methods include manual trolleys, pallet trucks, forklifts and general utility carts. Manual handling is labour intensive and increases the risk of worker injuries while forklifts are bulky require palletised loads and have limited manoeuvrability in narrow backstage areas. Therefore, there is a need for a compact cart type carrier specifically designed for transporting event equipment safely and efficiently.

The proposed Portable Event Equipment Carrier is a compact seated cart type carrier capable of transporting event equipment with a payload of several kilograms. During operation, the machine encounters three different operating conditions. Climbing temporary loading ramps with full load, transporting equipment on level surfaces and returning without a load. A single transmission ratio cannot efficiently satisfy these varying torque and speed requirements. A three-speed sliding mesh gearbox provides a practical solution by allowing high torque for ramp climbing, medium torque and normal speed during loaded flat surface transportation and higher travel speed during unloaded return. The gearbox also incorporates a reverse gear enabling safe manoeuvring during loading and unloading operations.

1.2 Acceptance or rejection statement

During the initial proposal and problem formulation stage, the original concept for a Portable Event Equipment Carrier powered by a small internal combustion engine and a custom gearbox was rejected.

The initial proposal was rejected due to the following reasons:

1. Indoor Emissions Hazard - The integration of an internal combustion (IC) engine generated exhaust emissions and smoke making the carrier completely unsuitable and unsafe for indoor event venues such as exhibition centers, concert halls and enclosed stage areas.

2. Vehicular Scope & Gear Shift Limitation - The initial idea close to conventional vehicle drivetrain rather than a specialized mechanical transmission. Additionally, a sliding mesh gearbox lacks synchronizers, requiring the carrier to come to a complete stop before shifting gears to avoid gear tooth clashing and damage.

3. Reliability of Electric Motors - Electric motors offer significantly higher mechanical reliability than IC engines due to having fewer moving parts and instant torque delivery from a standstill, making them better suited for heavy material handling.

1.3 Design Calculations and Materials Selection

1.3.1 Gear Dimension Analysis

Gear ratio between input shaft and lay shaft was designed using 1st gear ratio :-

$$G = \sqrt{6.25} = 2.25$$

Lay shaft speed = $\frac{10}{2.25}$ = 4 rpm

Table 1: Gear Ratios with Layshaft

Gear Wheel Number	Gear Ratio		
G1	2.5		Lay Shaft
G2			2nd Gear
G3	1.25		
G4			1st Gear
G5	2.5		
G6			3rd Gear
G7	0.63		
G8			Reverse Gear
G9	2.5		
G10		1	
G11			

For the gear ratio selection, standard module values of 1, 1.5, 2, 2.5, and 3 mm and centre distances of 100, 120, 150, and 200 mm were evaluated for the required gear ratios of 2.5, 1.25 and 0.63. The corresponding tooth numbers and gear diameters were calculated, rounded to practical integer tooth numbers and the resulting actual centre distances and dimensional errors were compared. Based on the Excel analysis, a 3 mm module and 120 mm centre distance provided the lowest overall error and was therefore selected for the gearbox design.

Module	3
Centre Distance	120 mm

To avoid interference between gear teeth, [1]

$$T_p \geq \frac{2A_w}{\sqrt{G^2 + sin^2\varphi(1+2G)} - G}$$

A_W = Fraction by which the standard addendum for the wheel should be multiplied,

G = Gear ratio or velocity ratio = T_G / T_P = D_G / D_P,

φ = Pressure angle or angle of obliquity.

According to the reference,

Table 2: Standard proportions of gear systems [1]

S. No.	*Particulars*	*$14\frac{1}{2}^\circ$ composite or full depth involute system*	*20° full depth involute system*	*20° stub involute system*
1.	Addendum	1*m*	1*m*	0.8 *m*
2.	Dedendum	1.25 *m*	1.25 *m*	1 *m*
3.	Working depth	2 *m*	2 *m*	1.60 *m*
4.	Minimum total depth	2.25 *m*	2.25 *m*	1.80 *m*
5.	Tooth thickness	1.5708 *m*	1.5708 *m*	1.5708 *m*
6.	Minimum clearance	0.25 *m*	0.25 *m*	0.2 *m*
7.	Fillet radius at root	0.4 *m*	0.4 *m*	0.4 *m*

$A_w = 1$

$$T_{p,min} = \frac{2\ x\ 1}{\sqrt{2.25^2 + sin^2(20°)x(1+2\ x\ 2.25)} - 2.25} = 14.637 \approx 15$$

Sample calculation for get final gear diameter and teeth count on gear between input shaft and lay shaft,

$$D_g = G \cdot D_p$$

$$D_g = 2.5 . D_p$$

$$C = \frac{Dp + Dg}{2} = 120 \longrightarrow \frac{3.5Dp}{2} = 120$$

$$D_p = 68.5 \text{ mm}$$

$$D_g = 171.43 \text{ mm}$$

$$T = \frac{D}{m} \longrightarrow \begin{matrix} T_p = \frac{68.75}{3} = 22.85 \approx 23 \\ T_g = \frac{171.43}{3} = 57.14 \approx 57 \end{matrix} \quad [7]$$

Final Diamter according to teeth count,

$$D_p = 23 \text{ x } 3 = 69 \text{ mm}$$

$$D_g = 57 \text{ x } 3 = 171 \text{ mm}$$

For the reverse gear choosing same diameter for idler wheel and the pinion wheel,

$$D_p + \frac{D_p + D_g}{2} = 120$$

$$D_g = 2.5.D_p$$

$$3D_p + D_g = 120 \text{ x } 2 \longrightarrow \frac{5.5Dp}{2} = 120$$

$$D_p = 43.63$$

$$D_g = 109.09$$

$$T_p = \frac{43.63}{3} = 14.67 \approx 15$$

$$T_g = \frac{109.09}{3} = 36.33 \approx 36$$

Final Diamter according to teeth count,

$$D_p = 15 \text{ x } 3 = 45 \text{ mm}$$

$$D_g = 36 \text{ x } 3 = 108 \text{ mm}$$

Similarly, all dimensions of the gear wheels can calculate as follows,

Table 3 : Selected values for gears

Gear Wheel	Calculated Diameter	Rounded Diameter	Teeth Count	Final Teeth Count	Final Diameter	Center Distance
G1	68.57	69	23.00	23	69	120
G2	171.43	171	57.00	57	171	
G3	106.67	107	35.67	36	108	120
G4	133.33	133	44.33	44	132	
G5	68.57	69	23.00	23	69	120
G6	171.43	171	57.00	57	171	
G7	147.24	147	49.00	49	147	120
G8	92.76	93	31.00	31	93	
G9	43.64	44	14.67	15	45	120
G10	109.09	109	36.33	36	108	
G11	43.64	44	14.67	15	45	

Considering the layshaft steps, the gear wheels are arranged as illustrated in Figure 1,

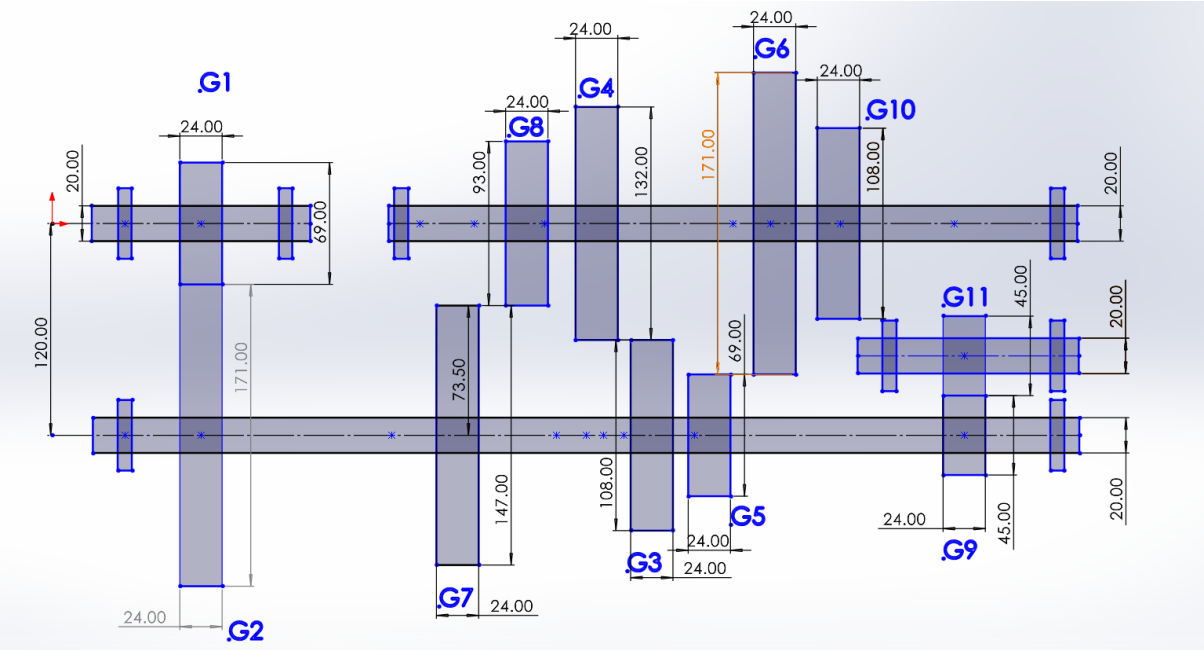


Figure 1: Final Gear Box Layout

1.3.2 Gear Material Selection & Analysis

Since the operating speeds of the gearbox are relatively low, grey cast (medium grade) iron was initially selected as the gear material. It provides adequate strength and wear resistance for the expected operating conditions while offering a low manufacturing cost, making it a suitable choice for the initial gearbox design and calculations.[2] As both meshing gears are made up using same material the pinion wheel is the weaker wheel. [3]

Sample calculation for pinion wheel of 1st Gear (G05 Gear Wheel)

G05 gear wheel is on layshaft.

Angular Speed (ω) = Input shaft angular speed x $\frac{\text{No.of teeth on input shaft gear}}{\text{No.of teeth on lay shaft gear}}$

$$= 10 \text{ x } \frac{23(G01)}{57(G02)} = 4.035 \text{ rpm} \longrightarrow \frac{2\pi}{60} \text{ x } 4.035 = 0.423 \text{ rads}^{-1}$$

Velocity of the G05 gear wheel,

$$V = r\omega$$

$$V = \frac{69}{2} \text{ x } 0.423 \text{ x } 10^{-3} = 0.015 \text{ ms}^{-1}$$

Tangential tooth load is calculated using following equation,

$$W_T = \frac{P}{v} \times C_s$$

P- Power transmitted through gear

For 1st Gear calculated power = 12.5 W

The system is expected to operate under light shock loading conditions for approximately 8 – 10 hours per day. Therefore, a service factor of 1.25 was selected for the gearbox design to account for the expected operating conditions and loading variations.

Table 4: Service Factor Values [1]

Type of load	*Type of service*		
	Intermittent or 3 hours per day	*8-10 hours per day*	*Continuous 24 hours per day*
Steady	0.8	1.00	1.25
Light shock	1.00	1.25	1.54
Medium shock	1.25	1.54	1.80
Heavy shock	1.54	1.80	2.00

$$W_T = \frac{12.5}{0.015} \text{ x } 1.25 = 1071.84 \text{ N}$$

The tangential load was initially determined based on the transmitted power, service factor, and pitch-line velocity. The Lewis equation is then used to determine the allowable tangential tooth load and verify the bending strength of the gear teeth, ensuring that the selected gear dimensions are safe under the expected operating conditions.

$$\sigma_w = \sigma_o \times C_v$$

where, σ_o = Allowable static stress

C_v = Velocity factor. [1]

Velocity factor can be determined by below equation,

$$C_v = \frac{3}{3+v}, \text{ for ordinary cut gears operating at velocities upto 12.5 m / s.}$$

$$= \frac{4.5}{4.5+v}, \text{ for carefully cut gears operating at velocities upto 12.5 m/s.}$$

$$= \frac{6}{6+v}, \text{ for very accurately cut and ground metallic gears operating at velocities upto 20 m / s.}$$

$$= \frac{0.75}{0.75+\sqrt{v}}, \text{ for precision gears cut with high accuracy and operating at velocities upto 20 m / s.}$$

$$= \left(\frac{0.75}{1+v}\right) + 0.25, \text{ for non-metallic gears.}$$

$V = 0.015\ ms^{-1} < 12.5\ ms^{-1} \longrightarrow C_v = \frac{3}{3+V}$

$$C_v = \frac{3}{3+0.015} = 0.095$$

For Cast iron, medium grade allowable static stress = 70 MPa

Table 5: Values of Allowable Static Stress [1]

Material	*Allowable static stress (σ_o) MPa or N/mm²*
Cast iron, ordinary	56
Cast iron, medium grade	70
Cast iron, highest grade	105
Cast steel, untreated	140
Cast steel, heat treated	196
Forged carbon steel-case hardened	126
Forged carbon steel-untreated	140 to 210
Forged carbon steel-heat treated	210 to 245
Alloy steel-case hardened	350
Alloy steel-heat treated	455 to 472
Phosphor bronze	84
Non-metallic materials	
Rawhide, fabroil	42
Bakellite, Micarta, Celoron	56

$\sigma_w = 0.95 \times 70 = 69.661$ MPa

A 20° stub involute tooth profile was selected for the gears due to its higher tooth thickness and improved resistance to bending stresses compared with the full depth involute profile. The stub profile also reduces the risk of interference, making it suitable for the relatively small number of teeth used in the gearbox. [4]

$$y = 0.124 - \frac{0.684}{T}, \text{ for } 14\tfrac{1}{2}^\circ \text{ composite and full depth involute system.}$$

$$= 0.154 - \frac{0.912}{T}, \text{ for } 20^\circ \text{ full depth involute system.}$$

$$= 0.175 - \frac{0.841}{T}, \text{ for } 20^\circ \text{ stub system.}$$

20º sub involute system $\longrightarrow$ $y = 0.175 - \frac{0.841}{T(G5)}$

$$y = 0.175 - \frac{0.841}{23} = 0.138$$

For spur gears, the recommended face width is generally 6 – 12 times the module. Therefore, a face width of 8 times the module was selected, providing a suitable balance between gear strength and material usage while maintaining a safe design for all gears in the gearbox. [3]

$$b = 8 \text{ x } 3 = 24 \text{ mm}$$

Using Lewis equation,

$$W_T = \sigma_W . b . \pi . m . y$$

$$W_T = 69.661 \text{ x } 24 \text{ x } \pi \text{ x } 3 \text{ x } 0.138 = 2181.325 \text{ N}$$

$$1071.84 \text{ N} < 2181.33 \text{ N},$$

Therefore, the gear teeth have sufficient bending strength, and the selected gear design is considered safe under the specified operating conditions. [5]

Similarly, calculation performed for all pinion wheels,

Table 6: Tangential Tooth Load Calculation of Pinion Gears

	G1	G3	G5	G8	G9
Angular Velocity (rpm)	10.000	4.035	4.035	4.035	4.035
Angular Velocity (rad/s)	1.047	0.423	0.423	0.423	0.423
No. of Teeth	23.000	36.000	23.000	49.000	15.000
Diameter (mm)	69.000	108.000	69.000	147.000	45.000
Width (mm)	24.000	24.000	24.000	24.000	24.000
Velocity (m/s)	0.036	0.023	0.015	0.031	0.010
Power(W)	20.000	17.500	12.500	20.000	12.500
Service Factor	1.250	1.250	1.250	1.250	1.250
Tangential Tooth Load (N)	691.978	958.678	1071.814	804.954	1643.448
Velocity Factor	0.988	0.992	0.995	0.990	0.997
Allowable Static Stress	70.000	70.000	70.000	70.000	70.000
Permissible Working Stress	69.167	69.472	69.661	69.283	69.779
y	0.138	0.152	0.138	0.158	0.119
Tangential Tooth Load (N) from Lewis Equation	2165.842	2382.870	2181.325	2473.521	1877.197
State ($W_T < W_{T(Lewis)}$)	Safe	Safe	Safe	Safe	Safe

To calculate the dynamic load following equation is used, [5]

$$W_D = W_T + \frac{21v(bC + W_T)}{21v + \sqrt{bC + W_T}}$$

When calculating W_D service factor is neglected.[3]

$$W_T = \frac{12.5}{0.015} = 833.33 \text{ N}$$

To get the deformation factor (C) following equation used, [1]

$$C = \frac{Ke}{\frac{1}{E_p} + \frac{1}{E_G}}$$

K = A factor depending upon the form of the teeth.
= 0.107, for $14\frac{1}{2}°$ full depth involute system.
= 0.111, for 20° full depth involute system.
= 0.115 for 20° stub system.
E_P = Young's modulus for the material of the pinion in N/mm^2.
E_G = Young's modulus for the material of gear in N/mm^2.
e = Tooth error action in mm.

Table 7: Values of Maximum Allowable Tooth Error in Action [1]

Pitch line velocity (v) m/s	*Tooth error in action (e) mm*	*Pitch line velocity (v) m/s*	*Tooth error in action (e) mm*	*Pitch line velocity (v) m/s*	*Tooth error in action (e) mm*
1.25	0.0925	8.75	0.0425	16.25	0.0200
2.5	0.0800	10	0.0375	17.5	0.0175
3.75	0.0700	11.25	0.0325	20	0.0150
5	0.0600	12.5	0.0300	22.5	0.0150
6.25	0.0525	13.75	0.0250	25 and over	0.0125
7.5	0.0475	15	0.0225		

$V = 0.015$ ms^{-1} $<$ 1.25 ms^{-1} $\longrightarrow$ $e = 0.0925$

Dynamic Factor for G5 Gear wheel,

$$C = \frac{0.115\ x\ 0.0925}{\frac{2}{180\ x\ 1000}} = 957.375$$

Dynamic Load for G5 Gear wheel,

$$W_D = 833.33 + \frac{21\ x\ 0.015\ (24\ x\ 957.375 + 833.33)}{21\ x\ 0.015 + \sqrt{24x\ 957.375 + 833.33}} = 904.621$$

Static tooth load is calculated using following equation, [5]

$$W_S = \sigma_e . b . \pi . m . y$$

$\sigma_e = flexural\ endurance\ limit$

Table 8: Values of Flexural Endurance Limit [1]

Material of pinion and gear	*Brinell hardness number (B.H.N.)*	*Flexural endurance limit (σ_e) in MPa*
Grey cast iron	160	84
Semi-steel	200	126
Phosphor bronze	100	168
Steel	150	252
	200	350
	240	420
	280	490
	300	525
	320	560
	350	595
	360	630
	400 and above	700

The selected medium-grade grey cast iron has a Brinell hardness of 160 HB. According to the material properties provided in the above table, the corresponding flexural endurance limit is 84 MPa. [2]

$$W_S = 84 \text{ x } 24 \text{ x } \pi \text{ x } 3 \text{ x } 0.138 = 2630.31 \text{ N}$$

For steady loads, $W_S \geq 1.25\ W_D$
For pulsating loads, $W_S \geq 1.35\ W_D$
For shock loads, $W_S \geq 1.5\ W_D$ [1]

$$1.5 W_D = 1.5 \text{ x } 904.62 = 1356.93$$

$$2630.31 > 1356.93$$

The dynamic tooth load, multiplied by the specified safety factor of 1.5, was found to be lower than the calculated static tooth load. Therefore, the gear teeth have sufficient strength to withstand the expected dynamic loading, and the gear design is considered safe under the specified operating conditions. [3]

Similarly, above calculation performed for all pinion wheels,

Table 9: Dynamic Tooth Load Calculation of Pinion Gears

	G1	G3	G5	G8	G9
Tangential Tooth Load (N)	553.582	766.942	857.451	643.963	1314.758
Velocity (m/s)	0.036	0.023	0.015	0.031	0.010
Width (mm)	24.000	24.000	24.000	24.000	24.000
Tooth error action	0.093	0.093	0.093	0.093	0.093
Deformation Dynamic Factor	957.375	957.375	957.375	957.375	957.375
Dynamic Tooth Load (W_D)	669.391	840.550	904.621	743.779	1345.837
Static Tooth Load (Ws)	2630.310	2881.192	2630.310	2998.954	2259.775
State ($Ws > 1.5W_D$)	Safe	Safe	Safe	Safe	Safe

Wear tooth load is calculated using following equation, [1]

W_w = Maximum or limiting load for wear in newtons,
D_P = Pitch circle diameter of the pinion in mm,
b = Face width of the pinion in mm,
Q = Ratio factor

$$= \frac{2 \times V.R.}{V.R. + 1} = \frac{2T_G}{T_G + T_P}, \text{ for external gears}$$

$$= \frac{2 \times V.R.}{V.R. - 1} = \frac{2T_G}{T_G - T_P}, \text{ for internal gears.}$$

$V.R.$ = Velocity ratio = T_G / T_P,
K = Load-stress factor (also known as material combination factor) in N/mm^2.

For G5 Gear,

$$Q = \frac{2\ x\ 57}{57+23} = 1.425$$

Value of the K is given by following equation,

$$K = \frac{(\sigma_{es})^2 \sin\phi}{1.4}\left(\frac{1}{E_P} + \frac{1}{E_G}\right)$$

σ_{es} = Surface endurance limit in MPa or N/mm^2,
ϕ = Pressure angle,
E_P = Young's modulus for the material of the pinion in N/mm^2, and
E_G = Young's modulus for the material of the gear in N/mm^2.

Table 10: Values of Surface Endurance Limit [1]

Material of pinion and gear	*Brinell hardness number (B.H.N.)*	*Surface endurance limit (σ_{es}) in N/mm^2*
Grey cast iron	160	630
Semi-steel	200	630
Phosphor bronze	100	630
Steel	150	350
	200	490
	240	616
	280	721
	300	770
	320	826
	350	910
	400	1050

According to the material properties provided in the above table, the corresponding surface endurance limit of grey cast iron is 630 MPa.

$$K = \frac{630^2 \, x \sin(20^o)}{1.4} \, x \, \left(\frac{2}{100 \, x \, 1000}\right) = 1.077$$

$$W_w = 69 \text{ x } 24 \text{ x } 1.425 \text{ x } 1.077 = 2542.36 \text{ N}$$

$W_D = 904.62$ N

$W_D < W_w$

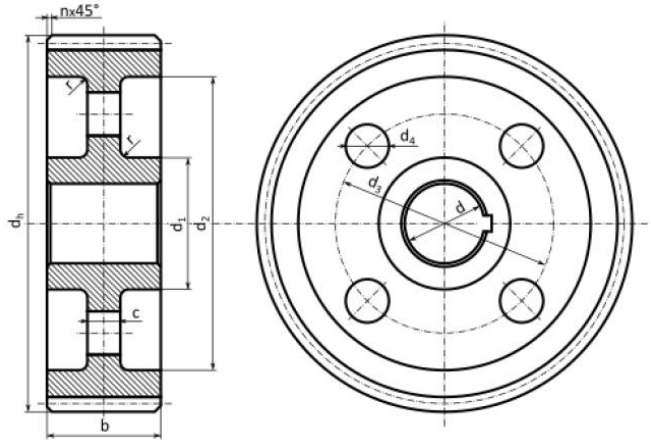
To prevent premature wear of the gear teeth, the allowable wear load must be greater than the dynamic tooth load. Since $W_W > W_D$ for the selected gear, the gear has adequate wear resistance and is considered safe against premature wear. [6]

Similarly, above calculation performed for all pinion wheels,

Table 11: Wear Tooth Load Calculation of Pinion Gears

	G1	G3	G5	G8	G9
Ratio Factor	1.425	1.100	1.425	1.225	1.412
Gear Diameter	69.000	108.000	69.000	93.000	45.000
Load Stress Factor	1.077	1.077	1.077	1.077	1.077
Dynamic Tooth Load (W_D)	669.391	840.550	904.621	743.779	1345.837
Wear Tooth Load (Ww)	2542.362	3071.779	2542.362	2945.727	1642.662
State (W_D < Ww)	Safe	Safe	Safe	Safe	Safe

Weight reduction of the gear wheels are done according to this,



Shaft diameter — d

Hub diameter — $d_1 = 1.6 \cdot d$
Where a is center distance

Inner rim diameter — $d_2 = d_h - 10 \cdot m$
Where m is modulus

Holes pitch diameter — $d_3 = 0.5 \cdot (d_2 + d_1)$

Holes diameter — $d_4 = 0.25 \cdot (d_2 - d_1)$

Plate thickness — $c = 0.32 \cdot b$

Chamfer of the gearing — $n = 0.5 \cdot m$
Where m is modulus

Figure 2: Weight Reduction of Gear Wheels [7]

Sample calculation for G2 Gear wheel,

Shaft diameter (d) = 40 mm

d_h = 178.5 mm

d_1 = 1.6 x 40 = 64mm

d_2 = 178.5 – 10 x 3 = 145.8 mm

d_3 = 0.5 x (64 +145.8) = 107.7 mm

d_4 = 0.25 x (145.8 – 64) = 20.45 mm

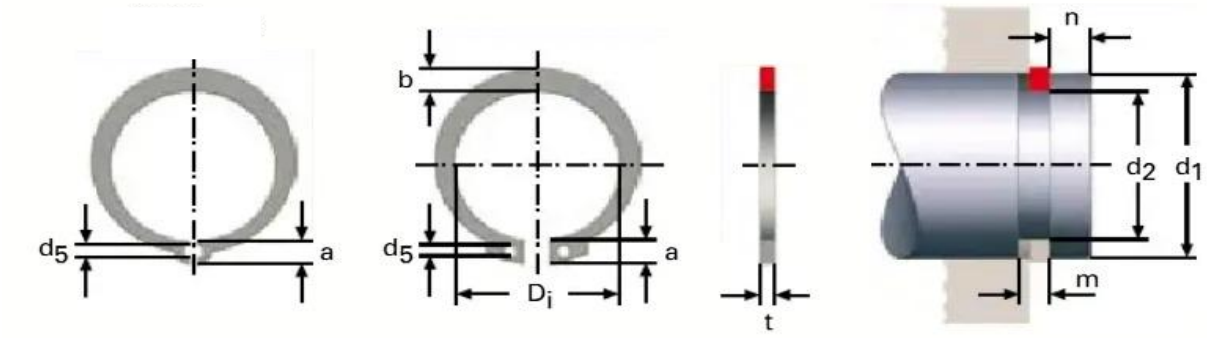
c = 0.32 x 24 = 8 mm

Assumptions:

- 20° sub involute spur gears are used for all gear pairs.
- The gears are assumed to be ordinary cut gears manufactured with standard tooth proportions.
- The face width of each gear is taken as 8 times the module.
- The gears are assumed to operate under steady state loading conditions.
- The load is assumed to be uniformly distributed across the gear face width.

1.3.3 Circlip Design Analysis

Circlips are designed according to fallowing standard, [8]



Shaft d_1	Ring dimension D_i	t h11	b	a Max.	d_5 Min.	Groove dimension d_2	d_2 Tol.	m Min.
3	2,7	0,4	0,8	1,9	1	2,8	+0 / -0,04	0,5
4	3,7	0,4	0,9	2,2	1	3,8	+0 / -0,04	0,5
5	4,7	0,6	1,1	2,5	1	4,8	+0 / -0,04	0,7
6	5,6	0,7	1,3	2,7	1,2	5,7	+0 / -0,06	0,8
7	6,5	0,8	1,4	3,1	1,2	6,7	+0 / -0,06	0,9
8	7,4	0,8	1,5	3,2	1,2	7,6	+0 / -0,06	0,9
9	8,4	1	1,7	3,3	1,2	8,6	+0 / -0,06	1,1
10	9,3	1	1,8	3,3	1,5	9,6	+0 / -0,11	1,1
12	11	1	1,8	3,3	1,7	11,5	+0 / -0,11	1,1
14	12,9	1	2,1	3,5	1,7	13,4	+0 / -0,11	1,1
15	13,8	1	2,2	3,6	1,7	14,3	+0 / -0,11	1,1
16	14,7	1	2,2	3,7	1,7	15,2	+0 / -0,11	1,1
18	16,5	1,2	2,4	3,9	2	17	+0 / -0,11	1,3
20	18,5	1,2	2,6	4	2	19	+0 / -0,15	1,3
22	20,5	1,2	2,8	4,2	2	21	+0 / -0,15	1,3
24	22,2	1,2	3	4,4	2	22,9	+0 / -0,15	1,3
25	23,2	1,2	3	4,4	2	23,9	+0 / -0,21	1,3
26	24,2	1,2	3,1	3,5	2	24,9	+0 / -0,21	1,3
28	25,9	1,5	3,2	4,7	2	26,6	+0 / -0,21	1,6
30	27,9	1,5	3,4	5	2	28,6	+0 / -0,21	1,6
32	29,6	1,5	3,6	5,2	2,5	30,3	+0 / -0,25	1,6
35	32,2	1,5	3,9	5,6	2,5	33	+0 / -0,25	1,6
38	35,2	1,75	4,4	5,8	2,5	36	+0 / -0,25	1,85
40	36,5	1,75	4,4	6	2,5	37,5	+0 / -0,25	1,85
42	38,5	1,75	4,5	6,5	2,5	39,5	+0 / -0,25	1,85
45	41,5	1,75	4,7	6,7	2,5	42,5	+0 / -0,25	1,85
48	44,5	1,75	5	6,9	2,5	45,5	+0 / -0,25	1,85

Figure 3: Circlip Design Catalogue

1.3.4 Circlip Material Selection

Carbon spring steel was selected as the material for the circlips in the gearbox due to its high yield strength, elasticity and excellent spring properties. Circlips must undergo elastic deformation during installation and removal while returning to their original shape without permanent deformation. Carbon spring steel provides sufficient flexibility and resistance to permanent deformation, enabling the circlips to securely retain the gears on the shafts and prevent unwanted axial movement during operation. Furthermore, it offers good fatigue resistance and wear resistance, making it suitable for repeated loading and vibration encountered in gearbox applications. Its relatively low cost, wide availability and ease of manufacturing also make it an economical choice for the gearbox. [8]

1.3.5 Endcap Design Analysis

The extended end cap was designed to accommodate two bearings at separate axial positions to provide improved support for the output shaft, which is subjected to radial loads and bending moments generated by the chain drive. Unlike a conventional end cap with a single bearing this design distributes the shaft reactions across two bearing locations, reducing shaft deflection and improving alignment during operation. The extended cone type body transfers the bearing loads to the mounting flange, while the longitudinal ribs increase structural stiffness and help resist bending and deformation. The mounting flange and circumferentially arranged bolt holes provide secure attachment to the gearbox housing and facilitate assembly and maintenance. Overall, the design improves shaft support, bearing alignment and load distribution while reducing the reliance on the gearbox housing alone to resist the effects of the chain drive loading.

The end caps for the layshaft and idler shaft were designed primarily to close the openings in the gearbox housing, retain the lubricant, and protect the internal components from dust and external contaminants. They provide a secure closure while allowing the shafts to remain properly supported by the bearing arrangement. The end caps are attached to the housing using circumferentially arranged bolts, facilitating easy removal during inspection, maintenance, and gear replacement.

1.3.6 Endcap Material Selection

Aluminium alloy was selected as the preferred material for the end cap because the gearbox is designed to be lightweight. Its low density reduces the overall mass of the gearbox, which suits a weight sensitive application. The alloy also has good machinability so the bearing bores, mounting flange and stiffening ribs can be produced accurately with standard machining operations and the bearing seats can hold tight tolerances. Its good corrosion resistance means the end cap can be used without heavy protective treatment, which reduces maintenance and extends service life.

1.4 Final Design

1.4.1 Gear Wheels

The involute profile was used for the gear teeth because it maintains a constant velocity ratio, giving smooth power transmission with low vibration and noise.[4] The pressure angle and line of action stay constant during the mesh, so tooth loading is uniform. The velocity ratio is also unaffected by small centre distance errors, which allows for manufacturing and assembly tolerances. In addition, involute gears are easy to manufacture with standard cutters and hobs, so they are economical and interchangeable. Removing material from the hub is useful as it can reduce the weight of the gear wheel so less force applied on bearings.

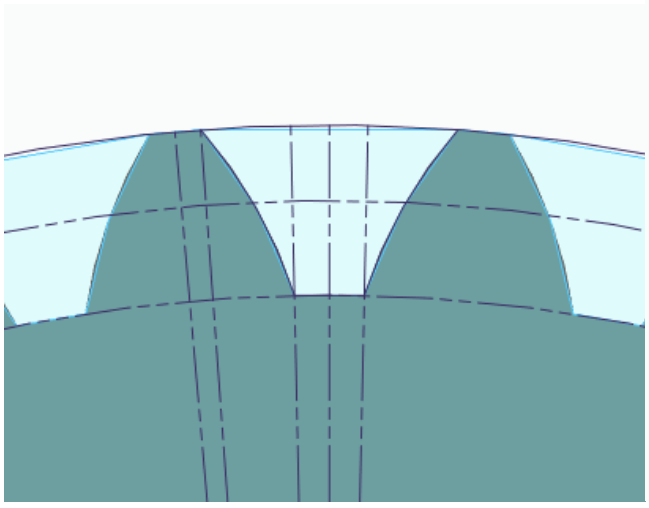


Figure 4: Gear Teeth Profile



Figure 5: Gear Wheel Front View

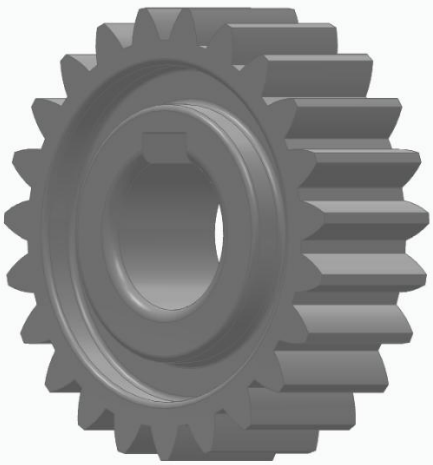


Figure 6: Gear Wheel Isometric View

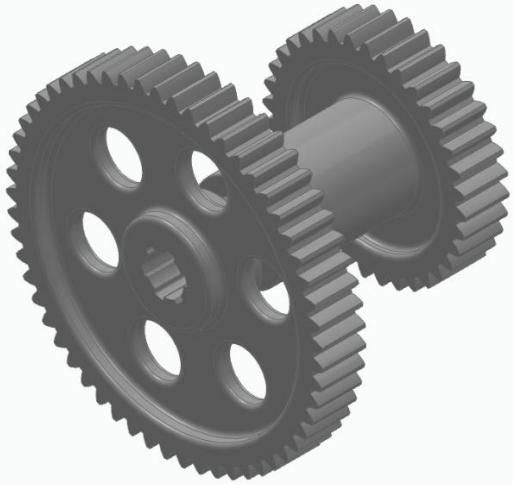


Figure 7: Gear Wheel Isometric View (in Spline Shaft)

A chamfer was added to the edge of the bore to make it easier to insert the shaft. It guides the shaft into the bore during assembly.

1.4.2 Circlips

A circlip is a semi flexible metal fastener installed into a machined groove on a shaft or inside a housing to prevent the axial (sideways) movement of components like gears, bearings, and pulleys. These Circlips are designed according to standard catalogue. [8]



Figure 8: 3D Design of a Circlip

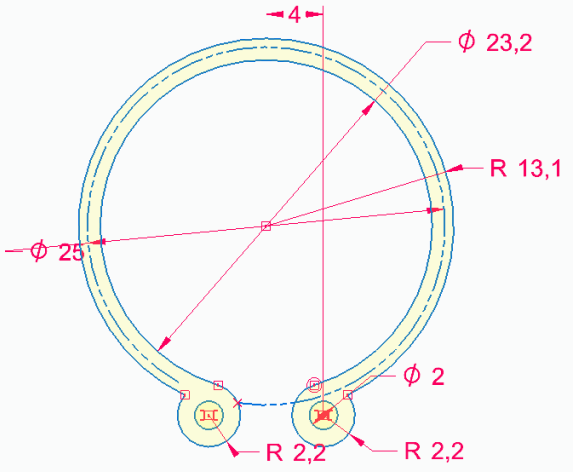


Figure 9: Design Sketch of Circlip

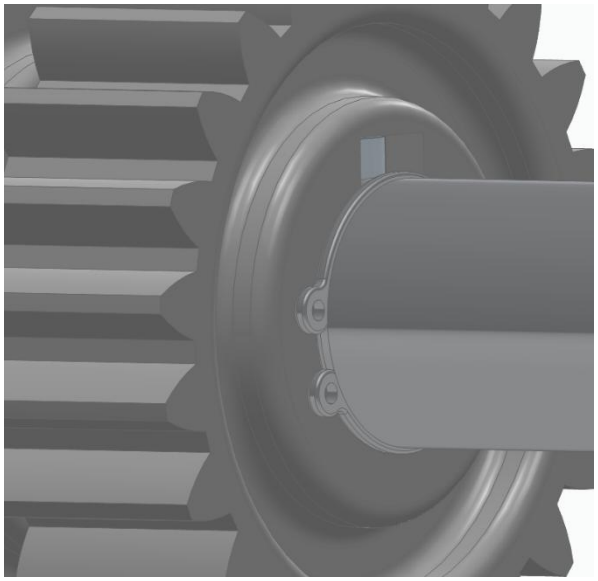


Figure 10: Circlip Assembly with Gear & Shaft

1.4.3 Endcaps

The extended end cap holds two bearings at separate positions to support the output shaft against the radial loads and bending moments from the chain drive. This reduces shaft deflection and improves alignment. Its cone shaped body and longitudinal ribs carry the loads to the bolted mounting flange, so the housing does not have to resist them alone.

The layshaft and idler shaft end caps mainly close the housing openings, retain the lubricant and keep out contaminants. They are bolted on, so they are easy to remove for inspection and maintenance.

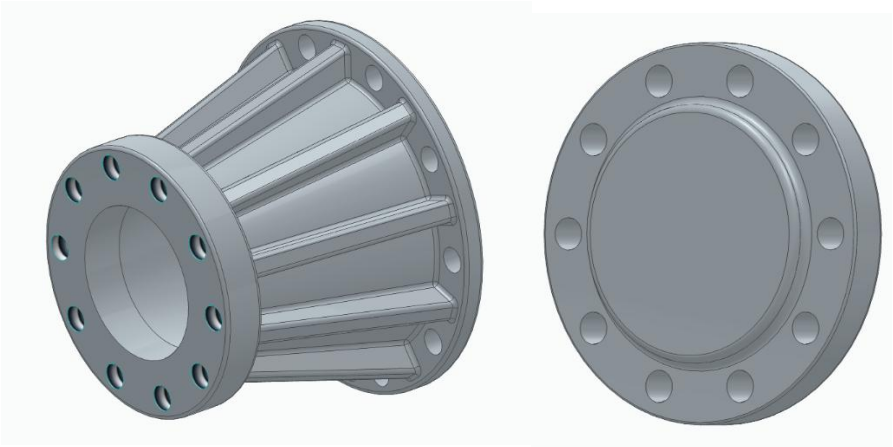


Figure 11: Extended Endcap **Figure 12: Flat Endcap**

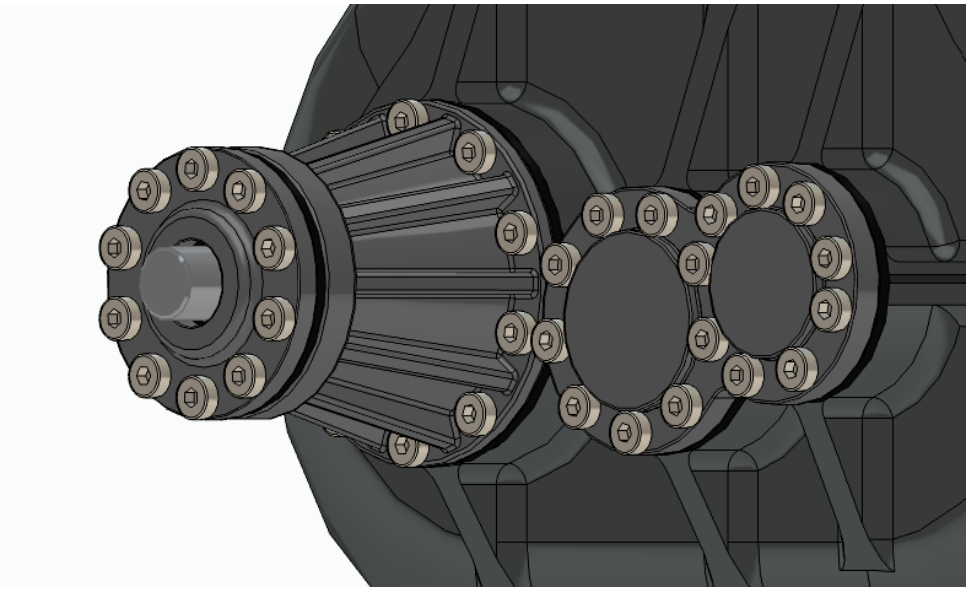


Figure 13: Endcap Assembly

1.5 Future work

Gear wheels

- Run a finite element analysis of tooth bending and contact stress to verify the weight reduced gear wheels.
- Optimize the lightening pattern (hole size, web thickness) with topology optimization to cut mass further without losing strength.
- Consider surface treatments such as carburizing, nitriding or shot peening to improve fatigue and wear life.

Circlips

- Check the groove strength and ring thrust capacity experimentally under the real axial loads.

End caps

- Run finite element analysis on the extended end cap to check the deflection and stress under the chain drive loads and use the results to refine rib layout and wall thickness.
- Study thermal expansion of the aluminum against the steel shafts and bearings, and how it affects bearing fits and preload.

1.6 References

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2 Section 2 – Gear Box for Mechanical Carbon Fiber Impregnation Press (230690T)

2.1 Introduction

Carbon fiber composites are widely used in lightweight, high strength applications such as aerodynamic body panels, nose cones, and marine structures. Achieving consistent resin impregnation and a controlled resin to fiber ratio is essential for structural integrity. However, commercially manufactured prepreg materials are expensive and rely heavily on imports, making them less accessible for small scale workshops. Furthermore, conventional manual manufacturing methods like wet layup and vacuum bagging are labor intensive and often result in uneven resin distribution, air voids, and excessive resin content.

To solve these limitations, this project proposes the development of a custom-built carbon fiber impregnation roller press incorporating a multi speed gearbox. The system uses a single-speed motor coupled with a sliding mesh gearbox to provide different output speeds and torque levels. This allows the pulling forces, processing speeds, and compaction pressures to be precisely adjusted for different carbon fiber grades (such as 3K, 6K, and 12K fabrics). Ultimately, the integration of this gearbox aims to provide a cost effective, mechanized method for producing high-quality prepreg material locally while reducing processing time and material waste.

2.2 Acceptance or rejection statement

The proposed carbon fiber impregnation press gearbox was accepted by the advisors for further development during the problem formulation stage. The project was commended for addressing a highly relevant industrial requirement, specifically the need for localized, cost-effective prepreg manufacturing.

The advisors noted that the problem was highly feasible for the scope of the machine design project. A key factor in the project's acceptance was that our group members already possessed significant practical experience and technical understanding of composite manufacturing processes. This prior hands on exposure, combined with the wide availability of technical details and industrial data regarding the impregnation process, provided a strong, realistic foundation for the design and calculation phases. Therefore, the topic was approved to proceed into detailed design.

2.3 Design Calculations and Materials Selection for the Shafts.

2.3.1 Shaft Material Selection

Using the results obtained in the detailed design and progress review calculations, to achieve the ability to withstand stress concentrations generated due to various steps, splines and groove machined to the shaft and to achieve the expected compactness of the gear box, a strength enough material was selected.

Material Selection

Selected Material – AISI 4340 Carbon Steel [1]

Material Properties

Yield Stress – 1586 MPa

Ultimate Tensile Stress – 1931 MPa

- Base allowable shear stress = $\min(0.3S_y, 0.18S_{ut}) = \min(475.8,347.58) = 347.58$
- Base allowable tensile stress = $\min(0.6S_y, 0.36S_{ut}) = min(951.6,895.16 = 895.16$

(**Base permissible stresses were calculated according to the ASME standards accounting for safety.)

2.3.2 Lay Shaft

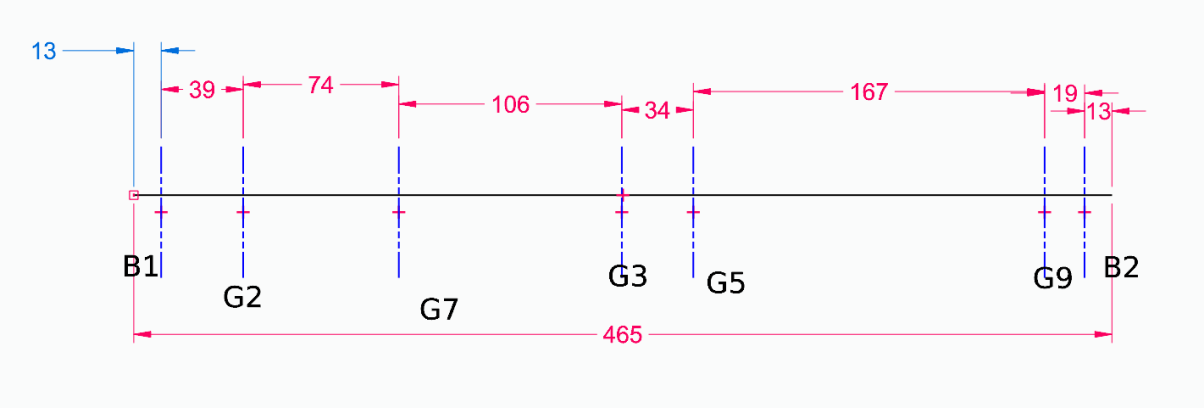


Figure 14: Gear layout - Layshaft

Initial lay shaft configuration was determined considering the gear meshing positions between the input shaft, lay shaft, splined output shaft and the shifting mechanism configuration. The gear box is planned to place horizontally and, the longitudinal axes of the lay shaft, input shaft and splined output shaft will be on a common plane.

2.3.2.1 Weights of gears connected to lay shaft

- Gear material – Medium grey cast iron
- Material density – 7250 kg/m^3
- Gear face width - 0.024 m
- Gear diameter – D

$$Weight\ of\ a\ gear = \frac{\pi \times D^2}{4} \times 0.024 \times 7250 \times 9.81$$

Note:

- These calculations for the gears were done, without considering the weight reduction of the gears.
- All gears have same face width and same material.

Table 12: Layshaft Gear Weights

Gear	Diameter(m)	Weight(N)
G2	0.171	39.20
G5	0.069	6.38
G3	0.108	15.64
G7	0.147	28.97
G9	0.045	2.71

2.3.2.2 Tangential and radial tooth loads

According to the gear calculations, the maximum power experienced by the shaft during the operation is 20Ws. For the further calculations regarding the lay shaft, the safety factor of 1.3 is being used and 26W (20 × 1.3) input power is considered.

- Lay shaft rpm = 4.035
- Lay shaft angular speed (ω) = 0.423 rad/s
- Power input for the lay shaft = 26 W

$$Power = \tau \times \omega$$

- Lay shaft torque (τ) = 61.532 Nm

For any gear on lay shaft,

$$Tangential\ tooth\ load = \frac{\tau}{Diameter/2}$$

$$Radial\ tooth\ load = Tangential\ tooth\ load \times \tan(20°)$$

Note:

- Because the shaft arrangement is horizontal, tangential tooth load has become a vertical load and parallel to the gear weight, while the radial tooth load becomes a horizontal load.
- **Assumption**: Own weight of the shafts is neglected at this stage because the material is not selected and required shaft diameter is not yet identified.

Adhering to above general equations and calculations following values were obtained:

Table 13: Forces on layshaft

Gear Type		Diameter(m)	Weight(N)	Tangential Force(N)	Radial Force(N)
Neutral	**Gear 2**	0.171	39.2	719.67	261.94
1	**Gear 5**	0.069	6.383	1783.54	649.15
2	**Gear 3**	0.108	15.637	1139.48	414.74
3	**Gear 7**	0.147	28.97	837.17	304.71
Reverse	**Gear 9**	0.045	2.71	2734.76	995.37

Note:

- Assume when looking at the gear box from input shaft side, the input wheel rotates counterclockwise.
- Tangential tooth load and weight of the gearwheel G2, acting on the same direction.

2.3.2.3 Bending moment calculation on neutral position

Forces acting on lay shaft during neutral position:

Table 14: Weight & Forces of Gears on Layshaft

Gear	Tangential Force(N)	Radial Force(N)	Weight(N)
Gear 2	719.67	261.94	39.20
Gear 7	-	-	6.38
Gear 3	-	-	15.64
Gear 5	-	-	28.97
Gear 9	-	-	2.71

Vertical bending moment:

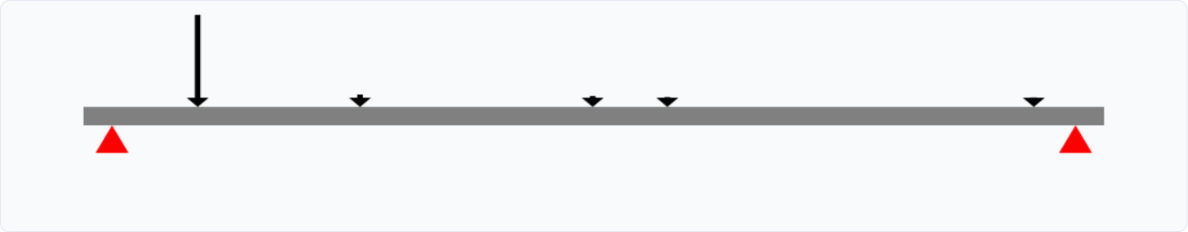


Figure 15: Free body diagram (Vertical Forces) of the Layshaft [2]

Table 15: Beam Input Parameters of the Layshaft

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 465	
Support-pinned	13	
Support-pinned	452	
Point Load	52	758.87
Point Load	126	28.97
Point Load	232	15.64
Point Load	266	6.38
Point Load	433	2.71

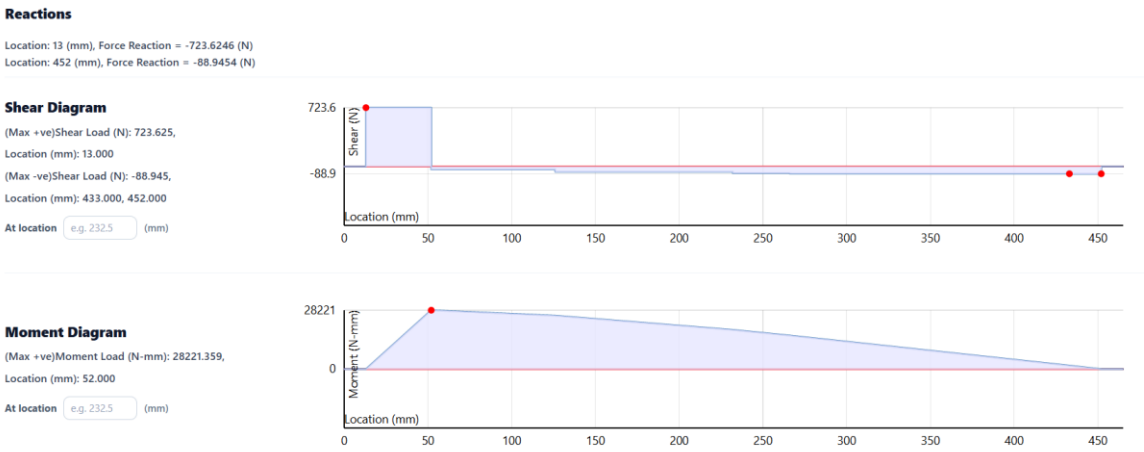


Figure 16: Shear force and Bending moment diagrams

Maximum vertical bending moment: **28.2 Nm**

Horizontal bending moment:

Only the radial tooth load of gear wheel 2 is acting horizontally at this position;



Figure 17: Free body diagram (Horizontal Forces) of the Layshaft

Table 16: Beam Input Parameters of the Layshaft

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 465	
Support-pinned	13	
Support-pinned	452	
Point Load	52	261.94

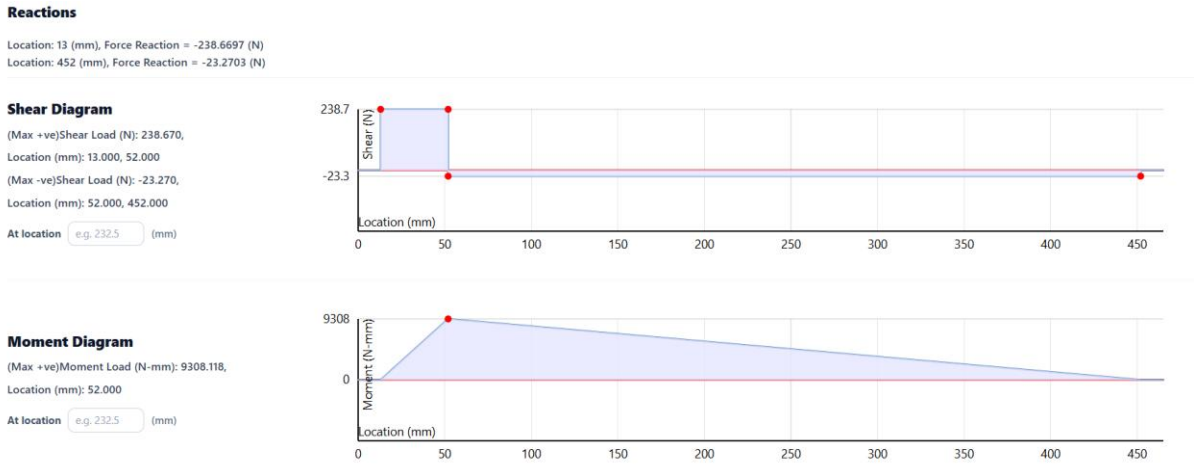


Figure 18: Shear force and bending moment diagram

Maximum horizontal bending moment: **9.3 Nm**

$$Resultant\ bending\ moment = \sqrt{Horizontal\ Moment^2 + Vertical\ Moment^2}$$

Resultant Bending Moment = **29.7 Nm**

2.3.2.4 Bending moment calculation on 1st gear

Table 17: Forces acting on lay shaft during first gear

Gear	Tangential Force(N)	Radial Force(N)	Weight(N)
Gear 2	719.67	261.94	39.20
Gear 7	-	-	6.38
Gear 3	-	-	15.64
Gear 5	1783.54	649.15	28.97
Gear 9	-	-	2.71

Vertical bending moment:

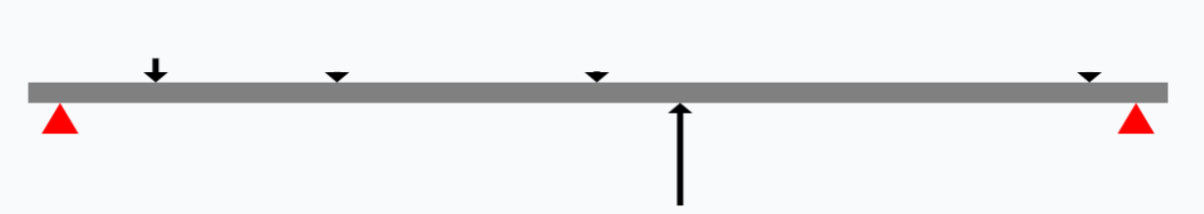


Figure 19: Free body diagram (Vertical Forces) of the Layshaft (1st Gear Engaged)

Table 18: Beam Input Parameters of the Layshaft (1st Gear)

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 465	
Support-pinned	13	
Support-pinned	452	
Point Load	52	261.94
Point Load	232	15.84
Point Load	126	6.38
Point Load	266	-1754.57
Point Load	433	2.71

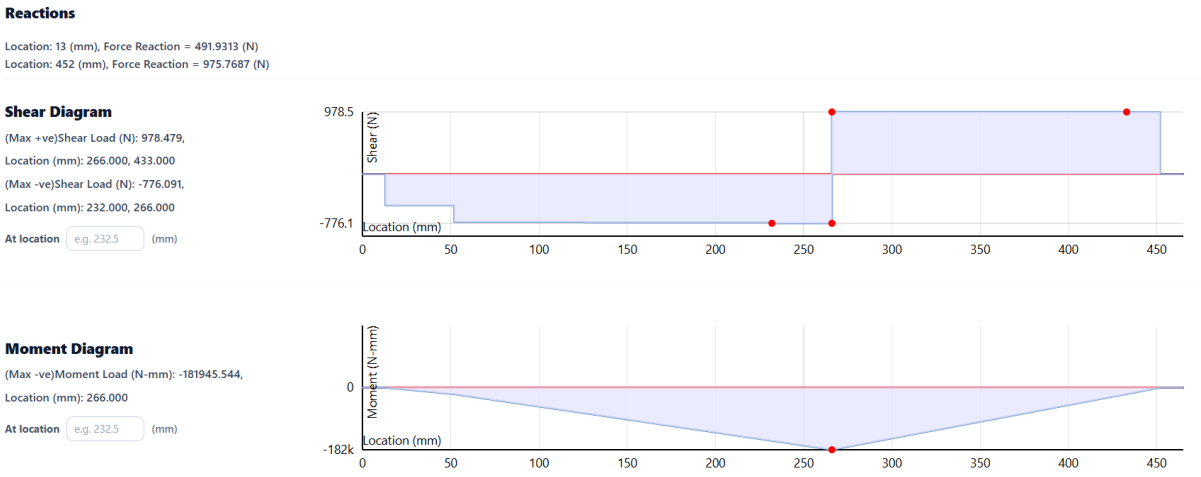


Figure 20: Shear force and bending moment diagrams: (1st Gear)

Maximum vertical bending moment = **181.9 Nm**

Horizontal bending moment:

Free body diagram:

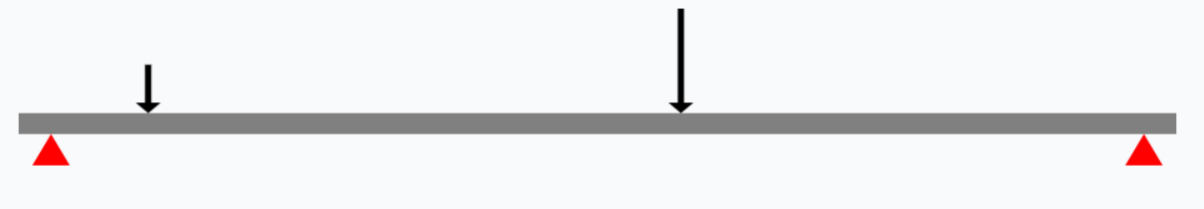


Figure 21: Free body diagram (Horizontal Forces) of the Layshaft (1st Gear Engaged)

Table 19: Beam Input Parameters of the Layshaft (1st Gear)

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 465	
Support-pinned	13	
Support-pinned	452	
Point Load	52	261.94
Point Load	266	649.15

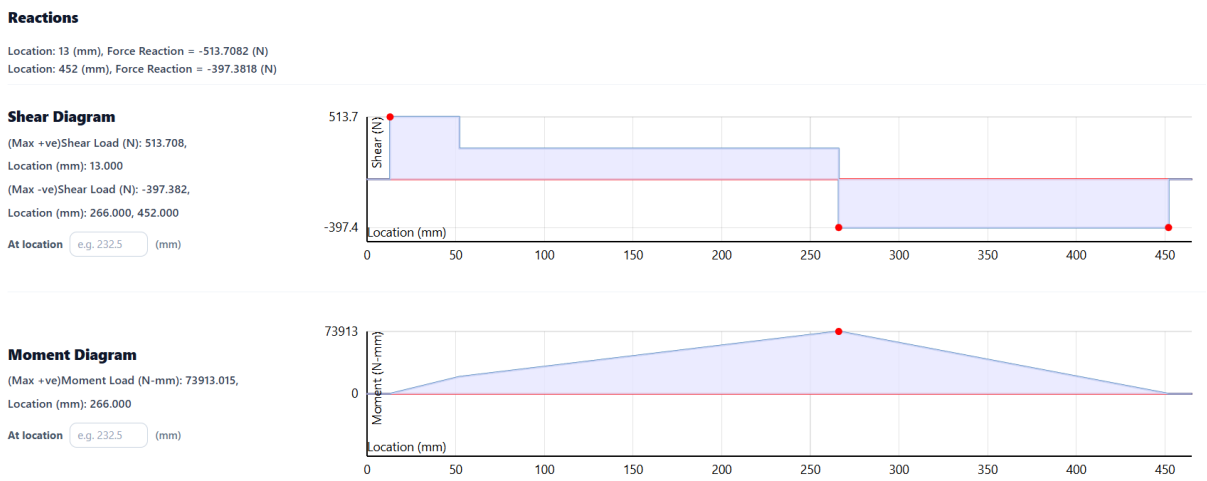


Figure 22: Shear force and bending moment diagrams: (1st Gear)

Maximum horizontal bending moment = **73.911 Nm**

Resultant bending moment: **196.3 Nm**

2.3.2.5 Bending moment calculation on 2nd gear

Table 20: Forces acting on lay shaft during second gear

Gear	Tangential Force(N)	Radial Force(N)	Weight(N)
Gear 2	719.67	261.94	39.20
Gear 7	-	-	6.38
Gear 3	1139.48	414.74	15.64
Gear 5	-	-	28.97
Gear 9	-	-	2.71

Vertical Bending Moments:

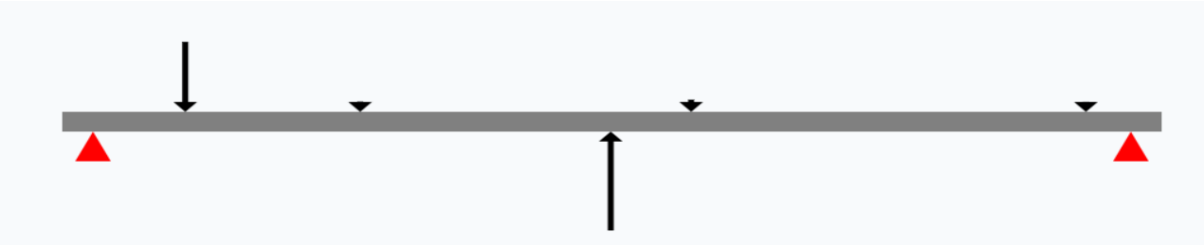


Figure 23: Free body diagram (Vertical Forces) of the Layshaft (2nd Gear Engaged)

Table 21: Beam Input Parameters of the Layshaft (2nd Gear)

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 465	
Support-pinned	13	
Support-pinned	452	
Point Load	52	758.87
Point Load	126	6.38
Point Load	232	-1123.84
Point Load	266	28.97
Point Load	433	2.71

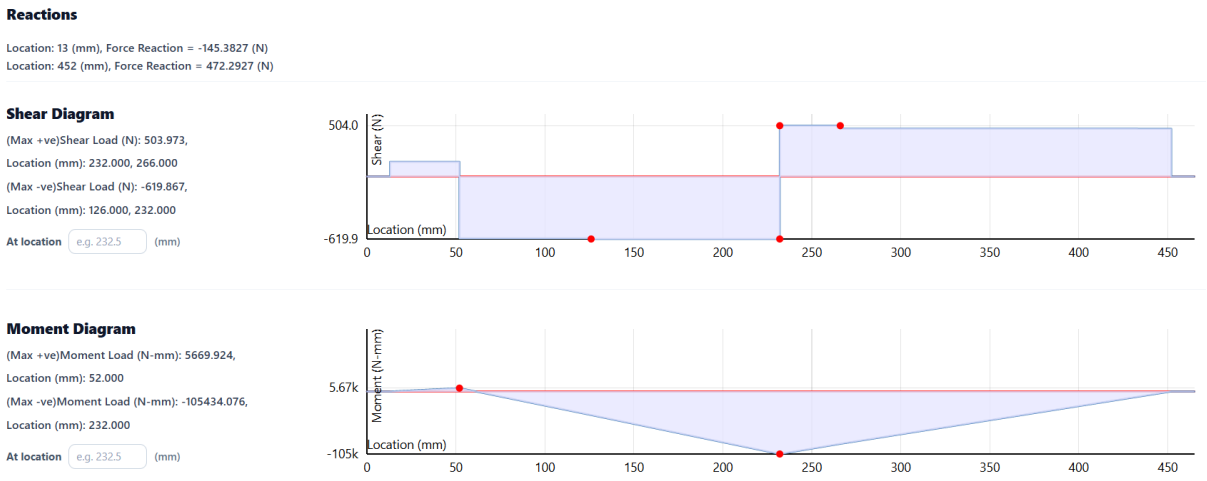


Figure 24: Shear force and bending moment diagrams: (2nd Gear)

Maximum vertical bending moment: - 105.4 Nm

Horizontal bending moment:

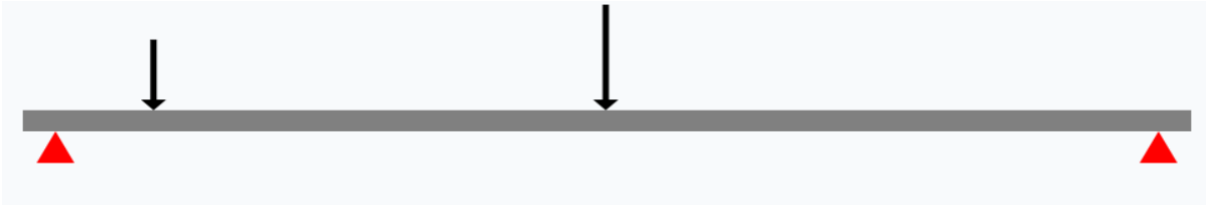


Figure 25: Free body diagram (Horizontal Forces) of the Layshaft (2nd Gear Engaged)

Table 22: Beam Input Parameters of the Layshaft (2nd Gear)

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 465	
Support-pinned	13	
Support-pinned	452	
Point Load	52	261.94
Point Load	232	414.74

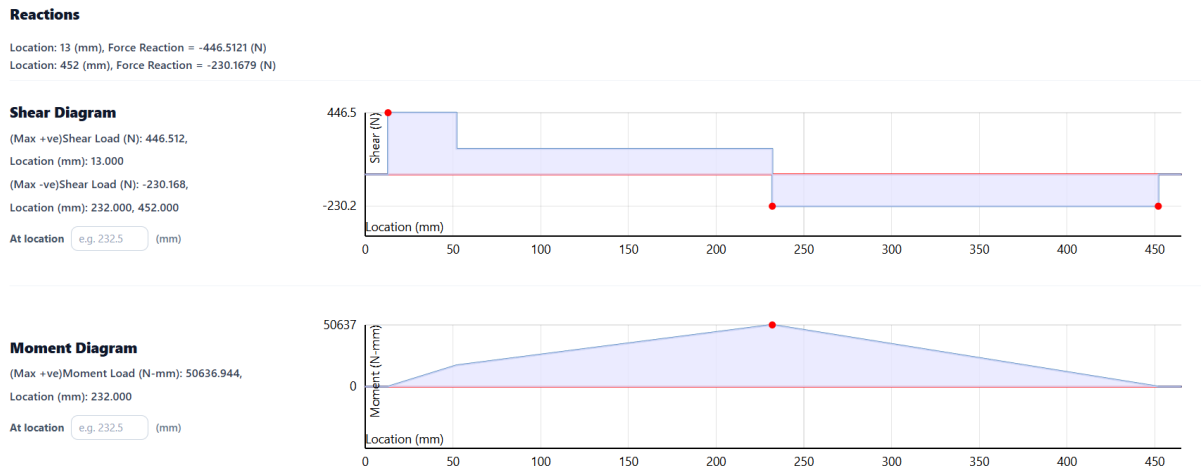


Figure 26: Shear force and bending moment diagrams (2nd Gear)

Maximum horizontal bending moment = 50.6 Nm

Resultant bending moment = **116.9 Nm**

2.3.2.6 Bending moment calculation on 3rd gear

Table 23: Forces acting on lay shaft during 3rd gear

Gear	Tangential Force(N)	Radial Force(N)	Weight(N)
Gear 2	719.67	261.94	39.20
Gear 7	837.17	304.71	6.38
Gear 3	-	-	15.64
Gear 5	-	-	28.97
Gear 9	-	-	2.71

Vertical bending moment:

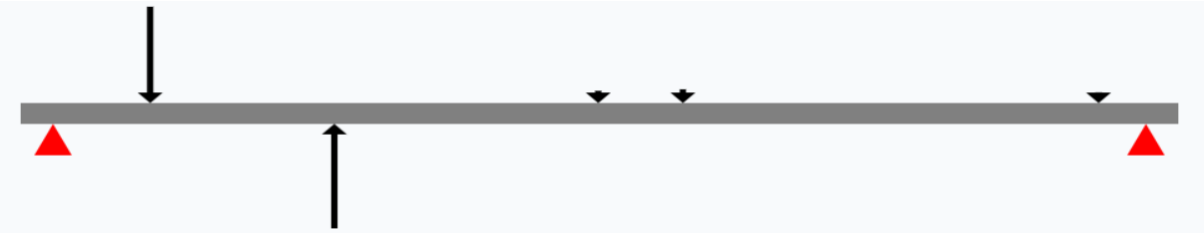


Figure 27: Free body diagram (Vertical Forces) of the Layshaft (3rd Gear Engaged)

Table 24: Beam Input Parameters of the Layshaft (3rd Gear)

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 465	
Support-pinned	13	
Support-pinned	452	
Point Load	52	758.87
Point Load	232	15.64
Point Load	126	-830.79
Point Load	266	28.97
Point Load	433	2.71

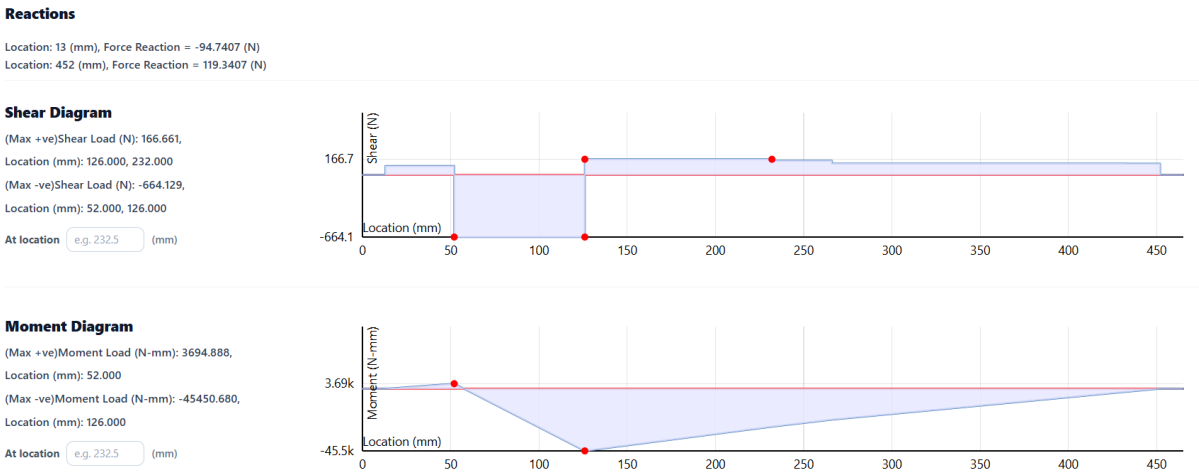


Figure 28: Shear force and bending moment diagrams (3rd Gear)

Maximum vertical bending moment: **-45.4 Nm**

Horizontal bending moment:

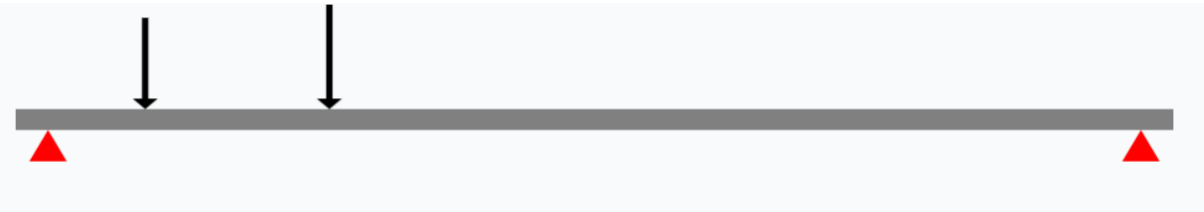


Figure 29: Free body diagram (Horizontal Forces) of the Layshaft (3rd Gear Engaged)

Table 25: Beam Input Parameters of the Layshaft (3rd Gear)

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 465	
Support-pinned	13	
Support-pinned	452	
Point Load	52	261.94
Point Load	126	304.71

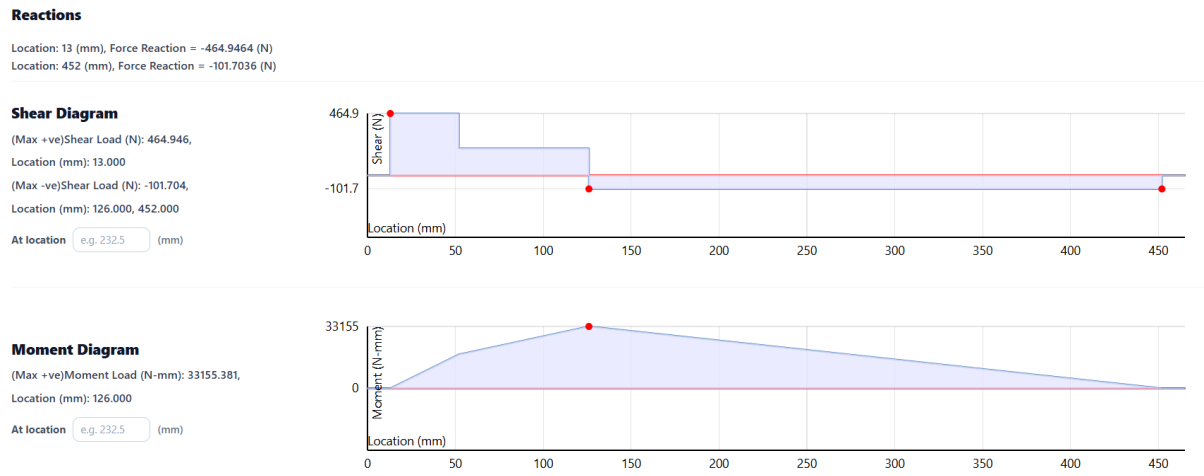


Figure 30: Shear force and bending moment diagrams (3rd Gear)

Maximum horizontal bending moment = **33.15 Nm**

Resultant bending moment = **56.2 Nm**

2.3.2.7 Bending moment calculation on reverse gear

Smaller angle of the idle shaft was neglected due to small change in forces.

Forces acting on lay shaft at reverse gear:

Gear	Tangential Force(N)	Radial Force(N)	Weight(N)
Gear 2	719.67	261.94	39.20
Gear 7	-	-	6.38
Gear 3	-	-	15.64
Gear 5	-	-	28.97
Gear 9	2734.76	995.37	2.71

Table 19

Vertical bending moments:

Free body diagram:

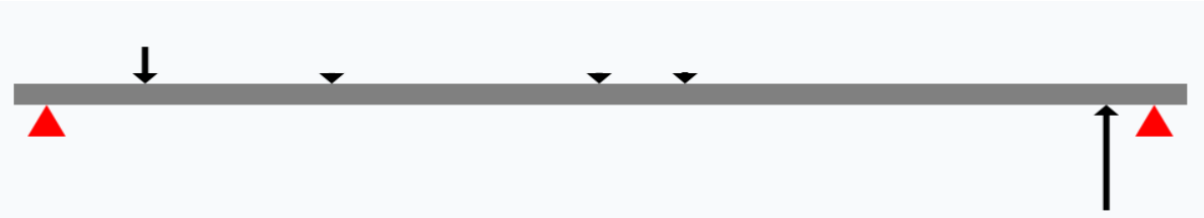


Figure 31: Free body diagram (Horizontal Forces) of the Layshaft (Reverse Gear Engaged)

Table 26: Beam Input Parameters of the Layshaft (Reverse Gear)

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 465	
Support-pinned	13	
Support-pinned	452	
Point Load	52	758.87
Point Load	126	6.38
Point Load	232	12.64
Point Load	266	28.97
Point Load	433	-2732.05

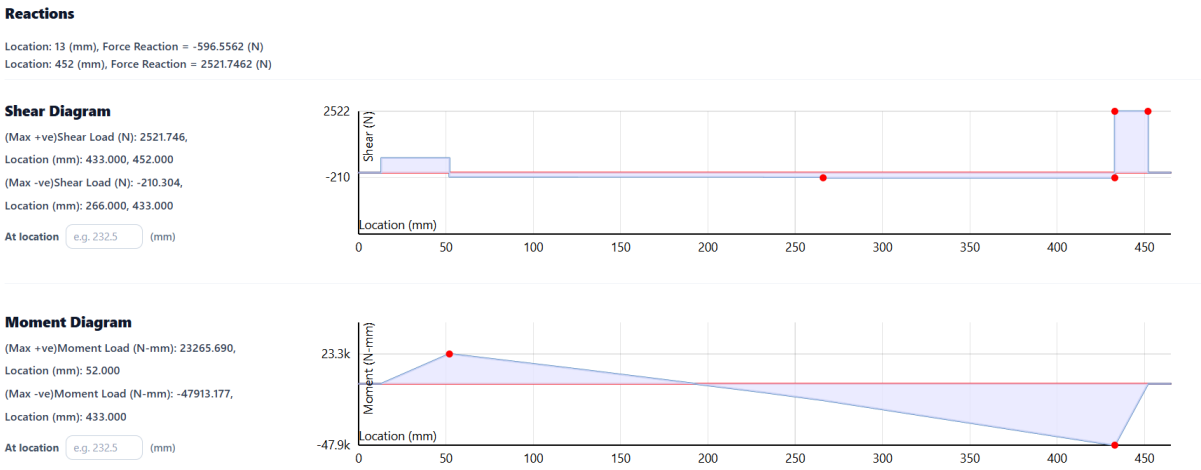


Figure 32: Shear force and bending moment diagrams (Reverse Gear)

Maximum vertical bending moment= **-47.91 Nm**

Horizontal bending moment:

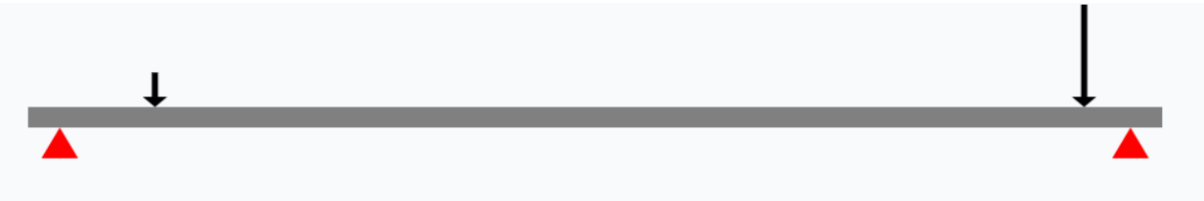


Figure 33: Free body diagram (Horizontal Forces) of the Layshaft (Reverse Gear Engaged)

Table 27: Beam Input Parameters of the Layshaft (Reverse Gear)

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 465	
Support-pinned	13	
Support-pinned	452	
Point Load	52	261.94
Point Load	433	995.37

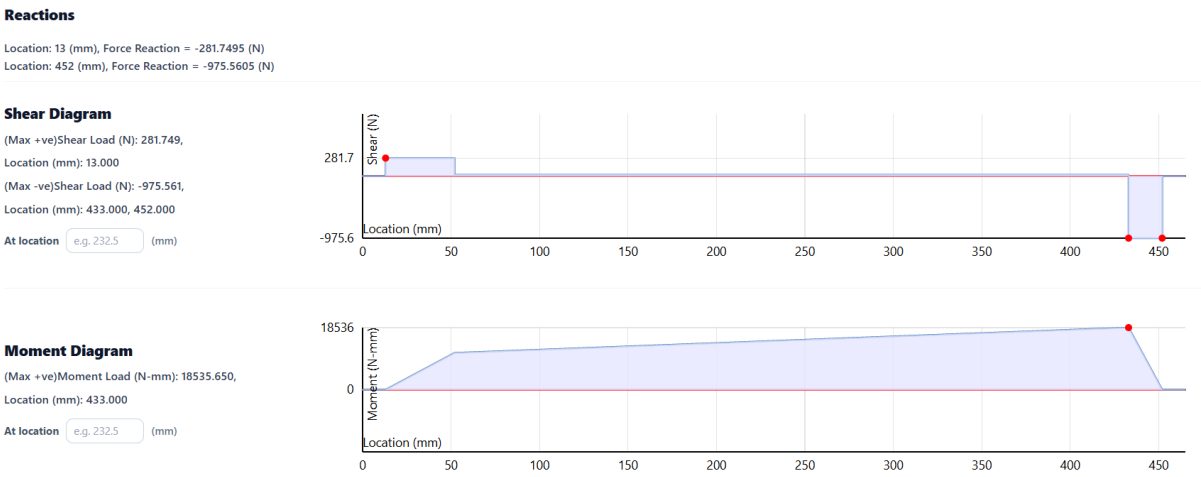


Figure 34: Shear force and bending moment diagrams (Reverse Gear)

Maximum horizontal bending moment: **18.5 Nm**

Resultant bending moment = **51.4 Nm**

2.3.2.8 Equivalent Bending moment and twisting calculation

Table 28: Summary of the resulting bending moments

Setting	Gear	Maximum Resultant BM(Nm)
Neutral	Gear 2	29.7
Third	Gear 7	56.2
Second	Gear 3	116.9
First	Gear 5	196.3
Reverse	Gear 9	51.4

Maximum bending moment shaft experience at any time is; **196.3 Nm.**

Equivalent bending moment of the shaft (M_e): (Rankine theory) [3]

$$M_e = \frac{1}{2} \times [(K_m \times M) + T_e]$$

Equivalent twisting (T_e): (Guest Theory)

$$T_e = \sqrt{(K_m \times M)^2 + (K_t \times T)^2}$$

- **Km** = Shock factor for bending
- **Kt** = Shock factor for twisting
- **M** = Maximum Bending moment of the shaft

Table 29: Recommended values for Km and Kr

Table 14.2. Recommended values for K_m and K_t

Nature of load	K_m	K_t
1. Stationary shafts		
(*a*) Gradually applied load	1.0	1.0
(*b*) Suddenly applied load	1.5 to 2.0	1.5 to 2.0
2. Rotating shafts		
(*a*) Gradually applied or steady load	1.5	1.0
(*b*) Suddenly applied load with minor shocks only	1.5 to 2.0	1.5 to 2.0
(*c*) Suddenly applied load with heavy shocks	2.0 to 3.0	1.5 to 3.0

From above referenced table, suddenly applied load with minor shocks only is used, because the lay shaft rotates and experiences sudden load changes when the sliding gears engage.

Because of that;

- K_m = 1.5
- K_t = 1.5

Following values were obtained;

	Nm
Equivalent Twisting	308.6
Equivalent Bending	301.5

2.3.2.9 Minimum diameter required calculation

Checking diameter for the twist: (d_1)

$$d_1 = \left(\frac{16 \times T_e}{\pi \times \tau_{allow}}\right)^{1/3}$$

Checking diameter for the bending:(d_2)

$$d_2 = \left(\frac{32 \times M_e}{\pi \times \sigma_{allow}}\right)^{1/3}$$

Using above equations, following values were obtained;

	mm
d1	16.53
d2	15.08

According to the theoretical analysis the diameter of the lay shaft must be greater than **16.53 mm.**

For the lay shaft, 17mm minimal diameter was chosen to compact the design and to be able to fit with 17mm bearing.

2.3.2.10 Calculation for the shaft steps

To prevent the axial movement of gearwheels during the meshing process, it is decided to use a stepped lay shaft under ASME B106.1M (Design of Transmission Shafting) standard. [4]

Since the standard shaft diameters increased by 5mm steps, it is decided to test two approaches.

- Shaft diameter increased by 10 mm steps.
- Shaft diameter increased by 5mm steps.

In the portions where the gears fit with the shaft, the minimum diameter to be set to 20mm and then decided to increase, as such the largest gear wheels to the side and diameter of gear wheels are decreasing gradually along the shaft axis.

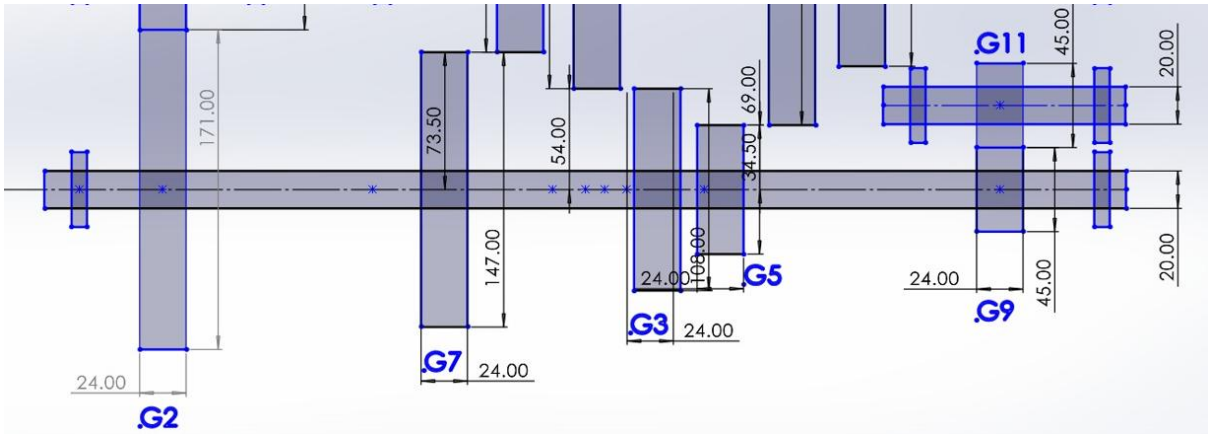


Figure 35: Layshaft Layout

It is understood that machining steps on the shaft creates stress concentration points along the shaft which makes the shaft weak. This calculation is done to ensure the shaft will not fracture during operation, because of machined steps.

First for each set of diameters, bending stress of the shaft was calculated.

$$Bending\ stress(\sigma_b) = \frac{32\ \times M}{\pi \times d^3}$$

Lay shaft	Nm
Max moment	196.3

Table 30: Bending Stress Summery

d(mm)	20	25	30	35	40	45	50	60
m	0.02	0.025	0.03	0.035	0.04	0.045	0.05	0.06
Sigma b(MPa)	249.94	127.97	74.06	46.64	31.24	21.94	16.00	9.26

Under these steps, and applied fillet radii to reduce the stress concentration, respective stress multiplication factors (Kt) were obtained using the Peterson charts in the Shigley's Mechanical Engineering Design reference. [5]

Figure A–15–9

Round shaft with shoulder fillet in bending. $\sigma_0 = Mc/I$, where $c = d/2$ and $I = \pi d^4/64$.

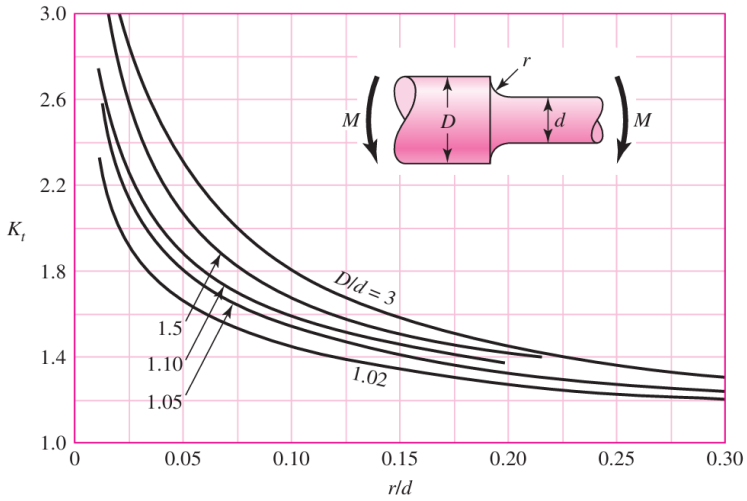


Figure 36: Shoulder Fillet Radii

For both step setups, the fillet radius(r) of the step was chosen as 1mm.

Table 31: Fillet Radii (10 mm increasing of diameter)

Opt 1	20	30	40	50	60
D/d	1.50	1.33	1.25	1.20	1.17
r/d	0.05	0.03	0.03	0.02	0.02
Kt	1.7	1.9	2	2.05	2.03
Stress(MPa)	424.89	140.71	62.48	32.79	18.79

Table 32: Fillet Radii (5 mm increasing of diameter)

Opt 2	20	25	30	35	40
D/d	1.25	1.20	1.17	1.14	1.13
r/d	0.05	0.04	0.03	0.03	0.03
Kt	1.7	1.85	1.9	1.9	1.9
Stress(MPa)	424.89	236.74	140.71	88.61	59.36

All above multiplied stress values are lower than,

Base allowable tensile stress = 895.16 MPa

Because of that both options are safe, but considering the more compact, robust and effective gear shifting mechanism it is decided to use 5mm increase of steps for the lay shaft.

Note:

For other cross sections of the layshaft except the gearwheel steps (Ex: for bearings), the step value is always lesser than 5mms and if larger the fillet radii used for such cross sections were sufficiently increased.

During the design of the gears, the fillet of the gear hub wheel edge which contact the shaft is set to 2mm.Since the shaft step fillet radii is 1mm, it is observed that, the gear hub perfectly fits with the step wall. Hence decided that additional groove for the step is not necessary. Additionally, it tested that, this step of 2.5mm, which stops the axial movement of the gear wheel is whether able to stop the axial movement of a wheel without fracturing. To do that, it is imagined that sudden shock load of 5000N (500kg) was applied on the shaft axially and the small strip of cross section can withstand that shock stress.

Assumed 5000N s of Sudden Axial Force

d inner	24	29	34	39	44
Area contact	38.48	46.34	54.19	62.05	69.90
Stress	129.92	107.90	92.26	80.58	71.53

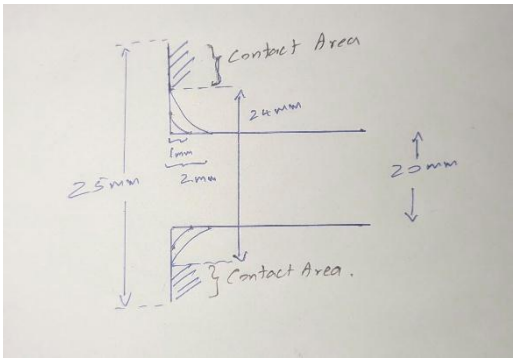


Figure 37: Fillet Design

Obtained stress values are clearly less than the tensile stress of the shaft = 895.16 MPa. The chosen steps are sound and safe for the application of the layshaft.

2.3.2.11 Standards used for the bearing undercut

On the layshaft, where NSK 6003 ,17mm inner ring bearing fits with the shaft, the step down of the shaft diameter is from 20mm to 17mm. The bearing comes with a standard fillet radius of 0.3mm and in order to fit the bearing inner ring wall firmly with the shaft step, the fillet radius on the step must be numerically lesser than 0.3mm, which is not practical and does not reduce the stress concentration firmly.

To address this issue under the internationally recognized standard DIN 509, an undercut was designed, where both tasks of, [6]

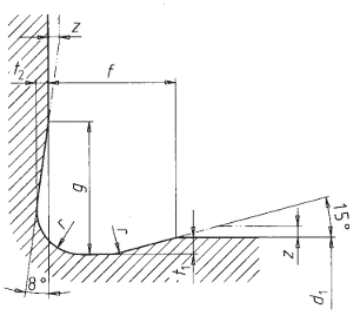
1. The shaft step wall firmly contacts the bearing inner wall.
2. Stress concentration of the step edge getting reduced.

addressed correctly.

For the lay shaft step, the **Type F** undercut was chosen as the surfaces are perpendicular.

4.2 Type F

Type F undercuts are to be used on workpieces with surfaces which are perpendicular to each other, intended for subsequent machining.



z Machining allowance
d_1 Workpiece diameter

Figure 2: Undercut for workpieces with shoulder and a cylindrical surface intended for subsequent machining

Figure 38: Type F Undercut

Dimensions were obtained using the following table along the 0.4mm row, because should be slightly larger than the bearing fillet radius.

Table 33: Undercut Dimensions

Page 4
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Table 1: Undercut dimensions

Type	$r^{1)}$ ±0,1 Series 1	$r^{1)}$ ±0,1 Series 2	t_1 +0,1 0	f +0,2 0	g	t_2 +0,05 0	Corresponding diameter $d_1^{2)}$ for workpieces: subjected to a normal stress concentration	Corresponding diameter $d_1^{2)}$ for workpieces: subjected to higher fatigue loads
E and F	—	0,2	0,1	1	(0,9)	0,1	Over 1,6 up to 3	—
	0,4	—	0,2	2	(1,1)	0,1	over 3 up to 18	
G				1	(1,2)	0,2		
E and F	—	0,6		2	(1,4)	0,1	over 10 up to 18	
	0,8	—	0,3	2,5	(2,1)	0,2	over 18 up to 80	
					(2,4)			
H				2	(1,1)	0,05		
E and F	1,2	1	0,2	2,5	(1,8)	0,1	—	Over 18 up to 50
			0,4	4	(3,2)	0,3	over 80	—
		—	0,2	2,5	(2)	0,1	—	over 18 up to 50
			0,4	4	(3,4)	0,3	over 80	—
H			0,3	2,5	(1,5)	0,05	—	over 18 up to 50
E and F	1,6	—	0,3	4	(3,1)	0,2		over 50 up to 80
	2,5		0,4	5	(4,8)	0,3		over 80 up to 125
	4		0,5	7	(6,4)			over 125

1) Undercuts with series 1 radii as specified in DIN 250 are to be given preference. For types G and H, the radii conform to those specified for indexable hardmetal inserts as in DIN 4967, DIN 4768 and DIN 4769-1.
2) Components with a short shoulder and thin-walled components are excepted. When a workpiece has different diameters, it may be convenient for manufacturing reasons to use the same form and size of undercut at several points.

Following above international standards, following profile was drawn to the shaft sketch to generate the undercut.

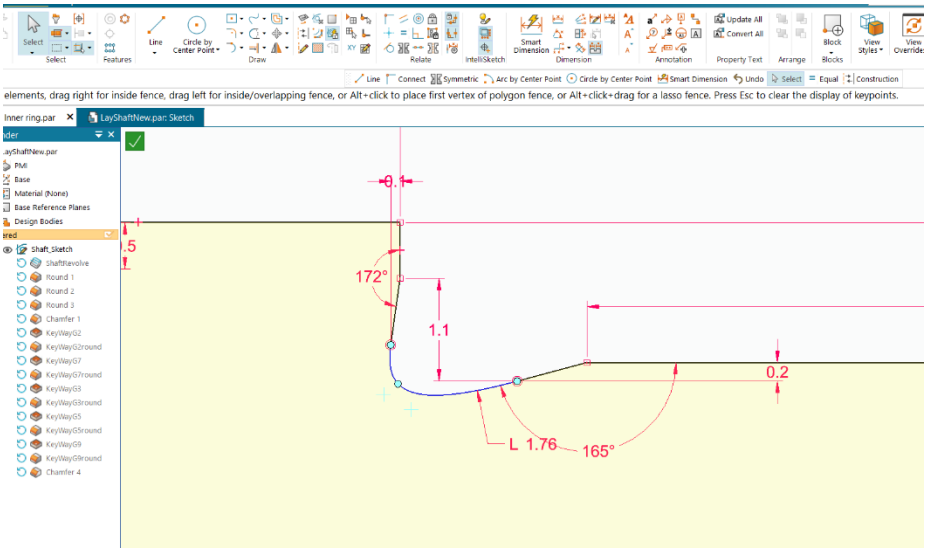


Figure 39: Undercut Sketch Profile

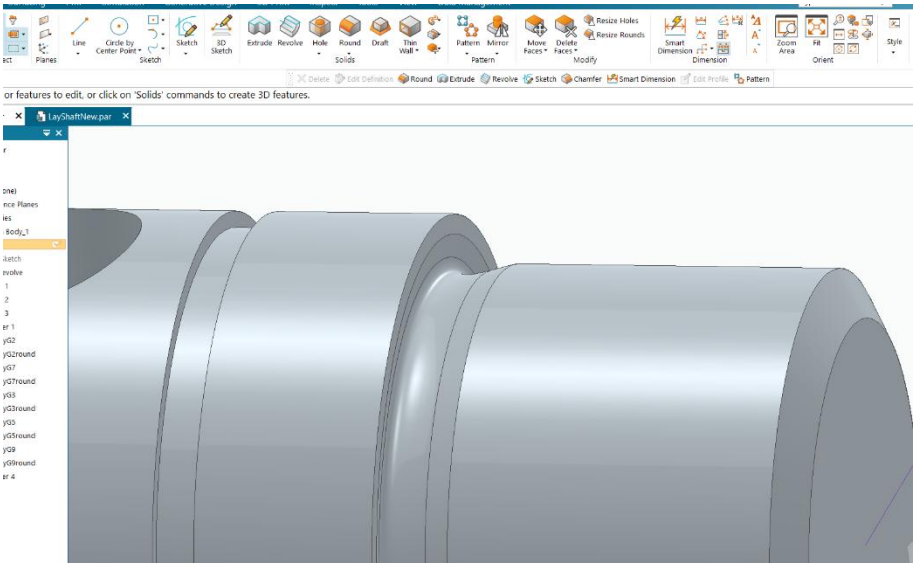


Figure 40: 3-dimensional view of the undercut

Apart from above mentioned major features, the ends of the shaft were chamfered, in order to make the easy access for fitting the shaft to the housing, bearings and gears to the shaft in a maintenance without damaging each component. Both ends were chamfered by 30 degrees, 1.5mm step.

2.3.2.12 Finalized Layshaft CAD model

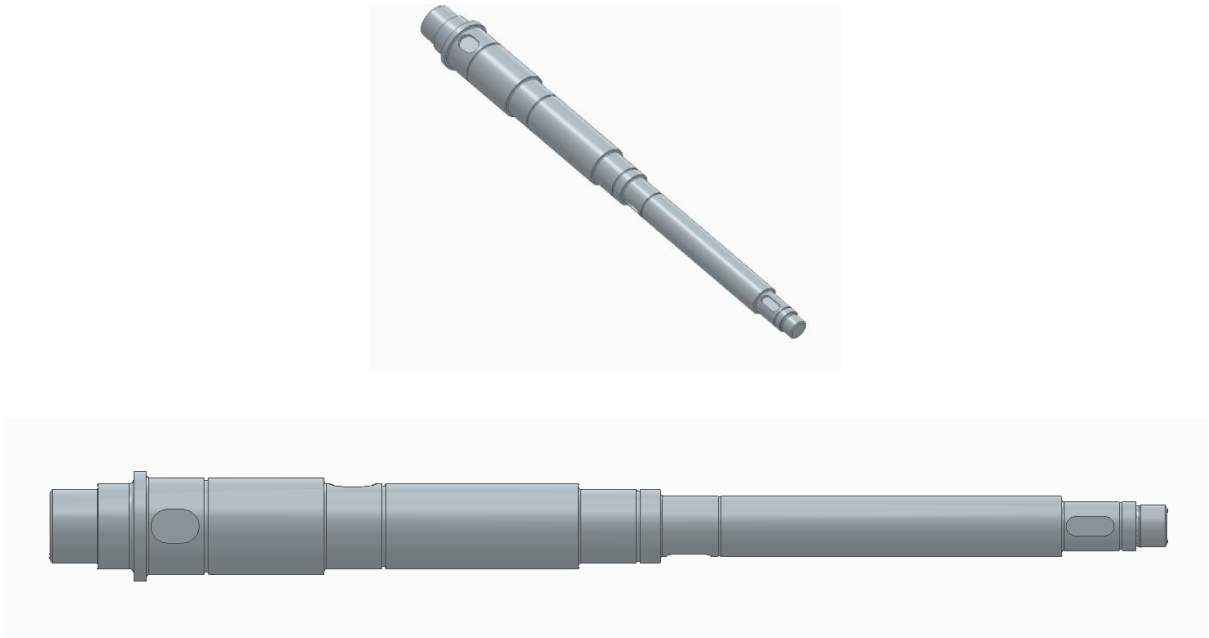


Figure 41: Finalized Layshaft CAD model with keyways and c-clip grooves

2.3.3 Splined Output Shaft

For the splined output shaft, its' gear wheel position orientation, distance between gear wheels and the overall shaft length were decided after repeatedly analysing the meshing process and maintaining sufficient gap to have smooth shifting mechanism.

The length of the splined output shaft is designed as 450mm.

2.3.3.1 Weights of the gears connected to the output shaft

Table 34: Gear weights of the Output Shaft

	Diameter(m)	Weight(N)
Gear 6	0.171	39.20
Gear 4	0.132	23.36
Gear 8	0.093	11.60
Gear 10	0.108	15.64

**These values were obtained using the equations defined in section 1.3.2.1

2.3.3.2 Torque values, tangential tooth loads and radial loads of the gears

According to the gear calculations, the maximum power experienced by the shaft during the operation is 20Ws. For the further calculations regarding the output shaft, the safety factor of 1.3 is being used and 26W (20 × 1.3) input power is considered.

The rpm of the output shaft rotates differ on the gear setting it is engaged at.

Table 35: Forces on Output Shaft

Gear Setting	Gear Number	Output rpm	Torque(Nm)	Tangential Force(N)	Radial Force(N)
1	**Gear 6**	1.63	152.49	1783.50	649.14
2	**Gear 4**	3.30	75.20	1139.46	414.73
3	**Gear 8**	6.38	38.93	837.15	304.70
Reverse	**Gear 10**	1.68	147.79	2736.79	996.11

**These numerical values were obtained using the equations defined in section 1.3.2.2

Note:

- Here also the shaft own weight was neglected.
- Each gear wheel travels by 28mm distance from neutral position depending on the gear setting.

2.3.3.3 Bending moment analysis under the neutral position

At this position, only the weight of the gears acting on the output shaft and the shaft remains stationary without rotating. Hence the bending moment analysis of this state was neglected, since it is obviously lower than the rest.

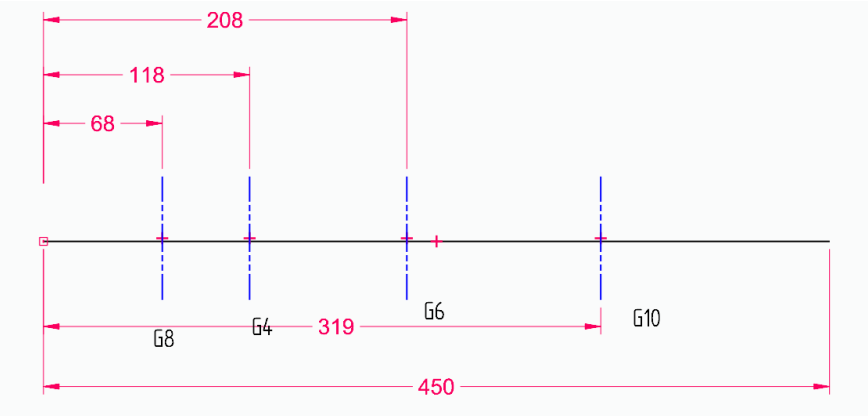


Figure 42: Shaft layout under neutral position

2.3.3.4 Bending moment analysis under the 1st gear

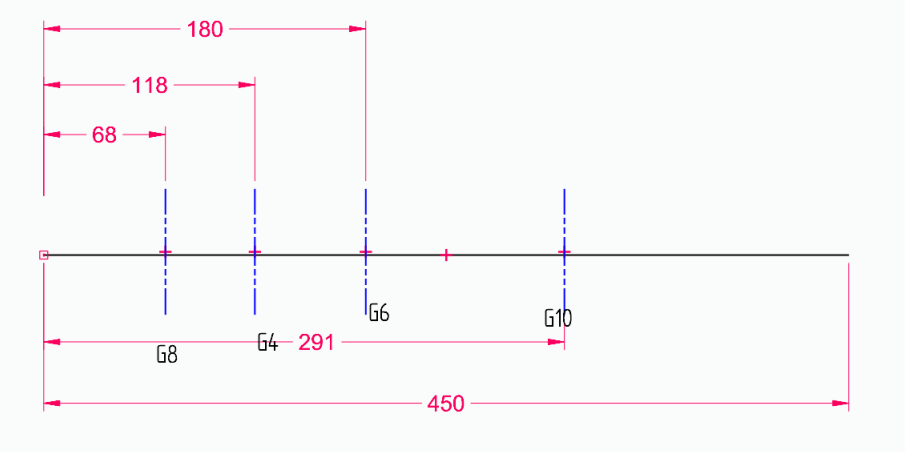


Figure 43: Shaft layout under first gear

Table 36: Forces acting on output shaft during the first gear

Gear	Tangential Force(N)	Radial Force(N)	Weight(N)
Gear 6	1783.50	649.14	39.20
Gear 4	-	-	23.36
Gear 8	-	-	11.60
Gear 10	-	-	2.71

Assume when looking at the gear box from input shaft side, the input wheel rotates counterclockwise; this will apply to further all calculations as previous.

Vertical bending moment;



Figure 44: Free body diagram (Vertical Forces) of the Output shaft (1st Gear Engaged)

Table 37: Beam Input Parameters of the Output Shaft

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 450	
Support-pinned	8	
Support-pinned	375	
Point Load	68	11.6
Point Load	118	23.36
Point Load	180	1822.7
Point Load	291	2.71

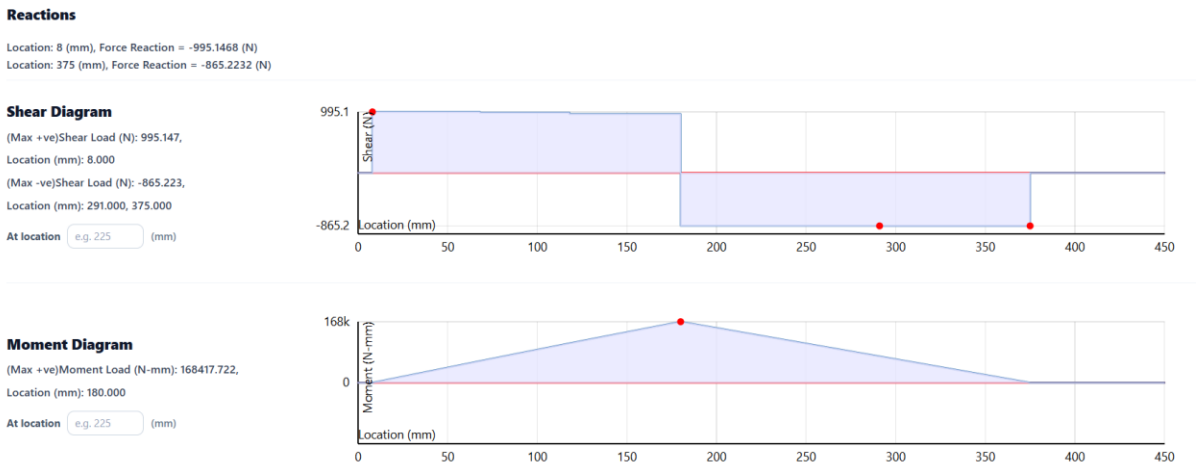


Figure 45: Shear force and bending moment diagrams (1st Gear)

Maximum vertical bending moment = 168.4 Nm

Horizontal bending moment; (Only radial forces acting)



Figure 46: Free body diagram (Horizontal Forces) of the Output shaft (1st Gear Engaged)

Table 38: Beam Input Parameters of the Output Shaft

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 450	
Support-pinned	8	
Support-pinned	375	
Point Load	180	649.14

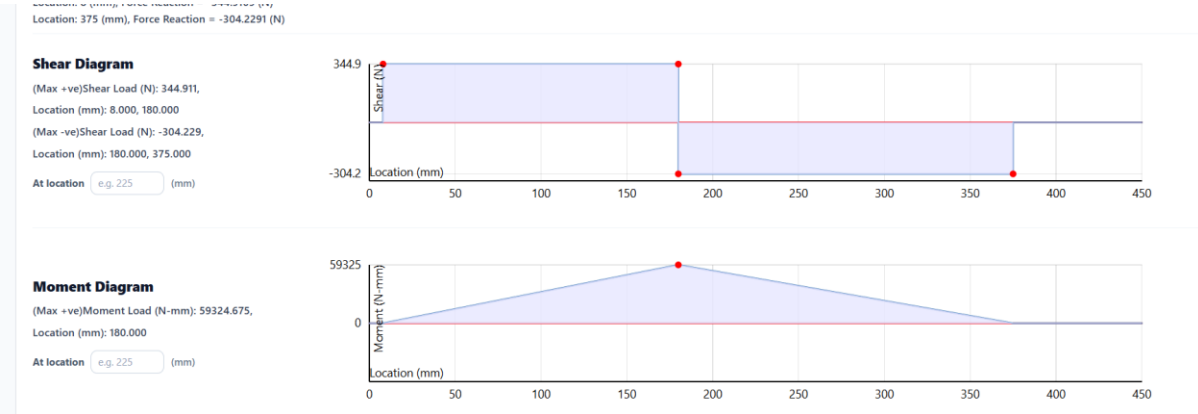


Figure 47: Shear force and bending moment diagrams (1st Gear)

Maximum horizontal bending moment = 59.3 Nm

Maximum resultant bending moment = **178.5 Nm**

2.3.3.5 Bending moment analysis under the 2nd gear

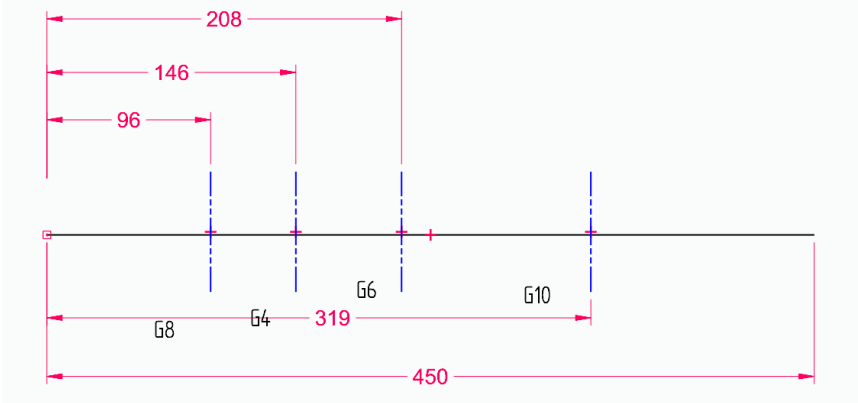


Figure 48: Gear layout for second gear

Table 39: Forces acting on the shaft under second gear

Gear	Tangential Force(N)	Radial Force(N)	Weight(N)
Gear 6	-	-	39.20
Gear 4	1139.46	414.73	23.36
Gear 8	-	-	11.60
Gear 10	-	-	2.71

Vertical bending moment;

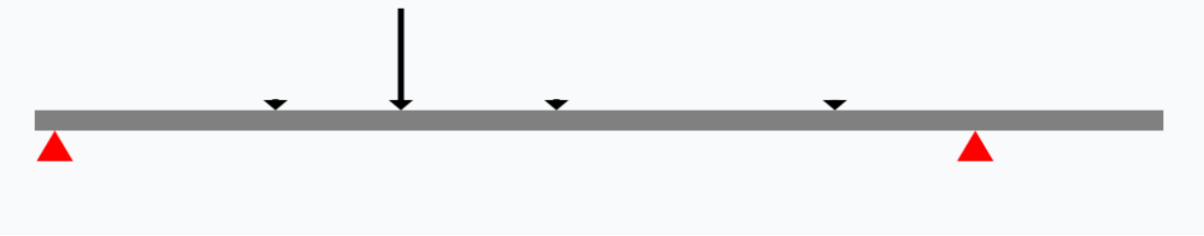


Figure 49: Free body diagram (Vertical Forces) of the Output shaft (2nd Gear Engaged)

Table 40: Beam Input Parameters of the Output Shaft

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 450	
Support-pinned	8	
Support-pinned	375	
Point Load	96	11.6
Point Load	146	1162.82
Point Load	208	11.6
Point Load	319	2.71

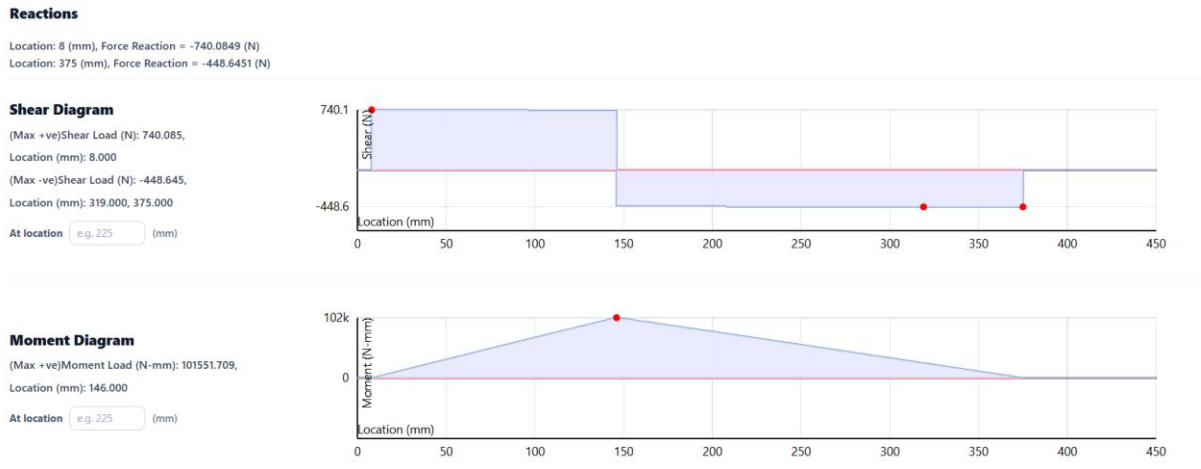


Figure 50: Shear force and bending moment diagrams (2nd Gear)

Maximum vertical bending moment = 101.55 Nm

Horizontal bending moment;

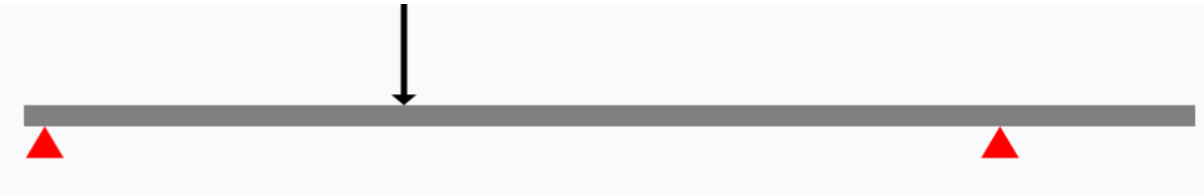


Figure 51: Free body diagram (Horizontal Forces) of the Output shaft (2nd Gear Engaged)

Table 41: Beam Input Parameters of the Output Shaft

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 450	
Support-pinned	8	
Support-pinned	375	
Point Load	146	414.73

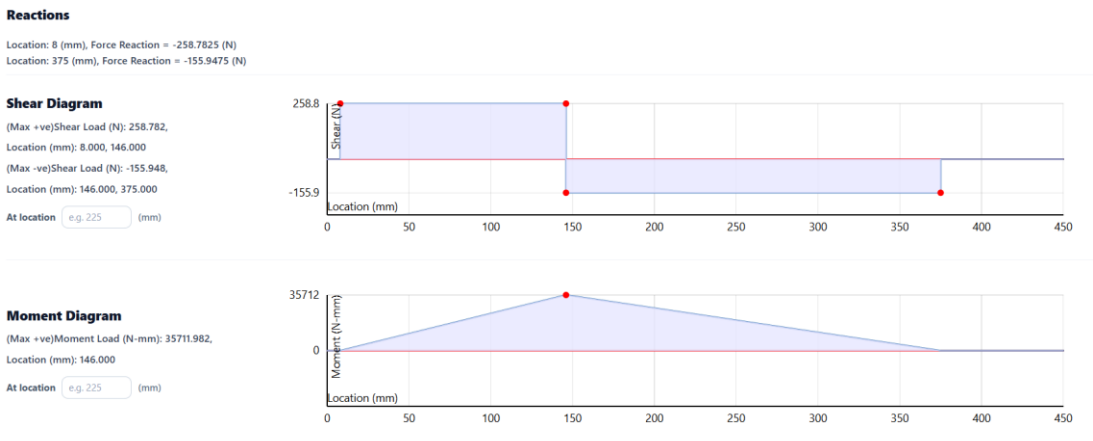


Figure 52: Shear force and bending moment diagrams (2nd Gear)

Maximum horizontal bending moment = 35.7 Nm

Resultant maximum bending moment = **<u>107.6 Nm</u>**

2.3.3.6 Bending moment analysis under the 3rd gear

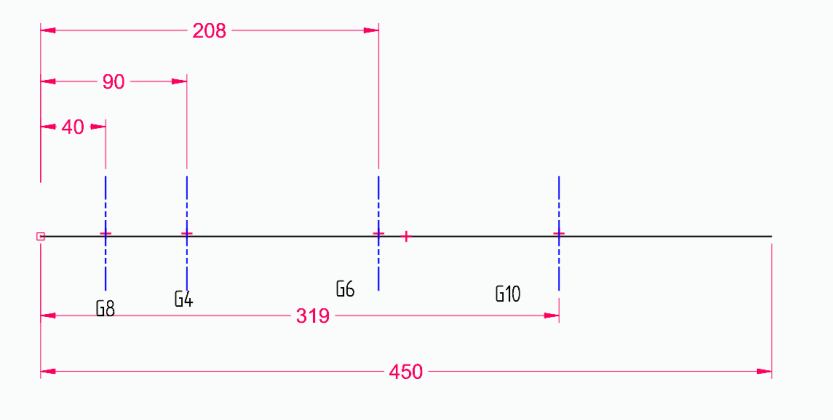


Figure 53: Gear layout for the third gear

Gear	Tangential Force(N)	Radial Force(N)	Weight(N)
Gear 6	-	-	39.20
Gear 4	-	-	23.36
Gear 8	837.15	304.70	11.60
Gear 10	-	-	2.71

Figure 54: Forces acting on gears under third gear

Vertical bending moment;

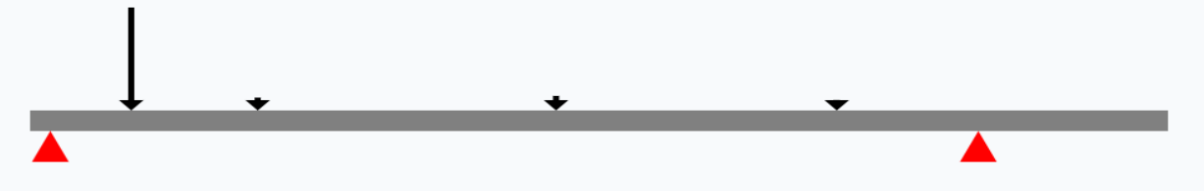


Figure 55: Free body diagram (Vertical Forces) of the Output shaft (3rd Gear Engaged)

Table 42: Beam Input Parameters of the Output Shaft

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 450	
Support-pinned	8	
Support-pinned	375	
Point Load	40	848.75
Point Load	90	23.36
Point Load	208	39.2
Point Load	319	2.71

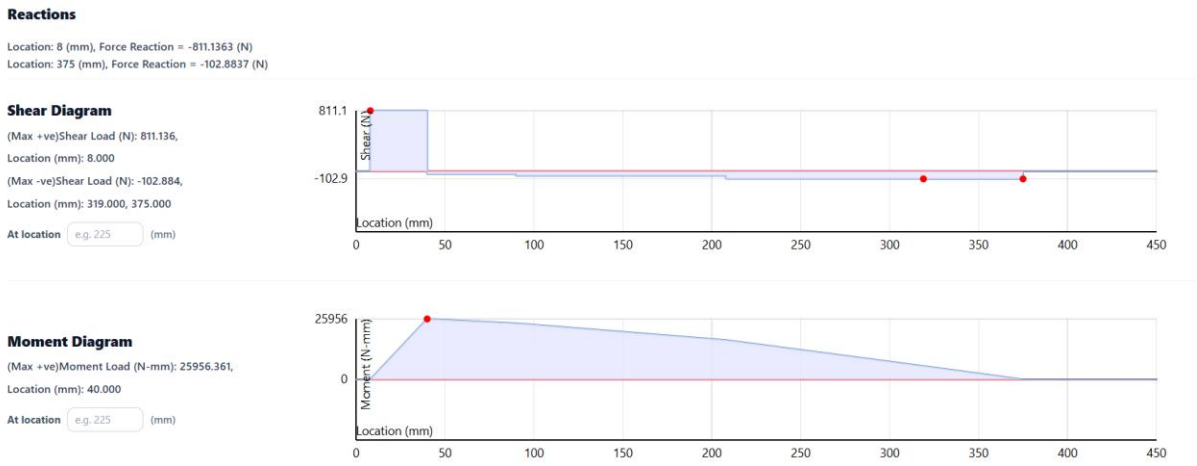


Figure 56: Shear force and bending moment diagrams (3rd Gear)

Maximum vertical bending moment= 25.96 Nm

Horizontal bending moment;



Figure 57: Free body diagram (Horizontal Forces) of the Output shaft (3rd Gear Engaged)

Table 43: Beam Input Parameters of the Output Shaft

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 450	
Support-pinned	8	
Support-pinned	375	
Point Load	40	304.7

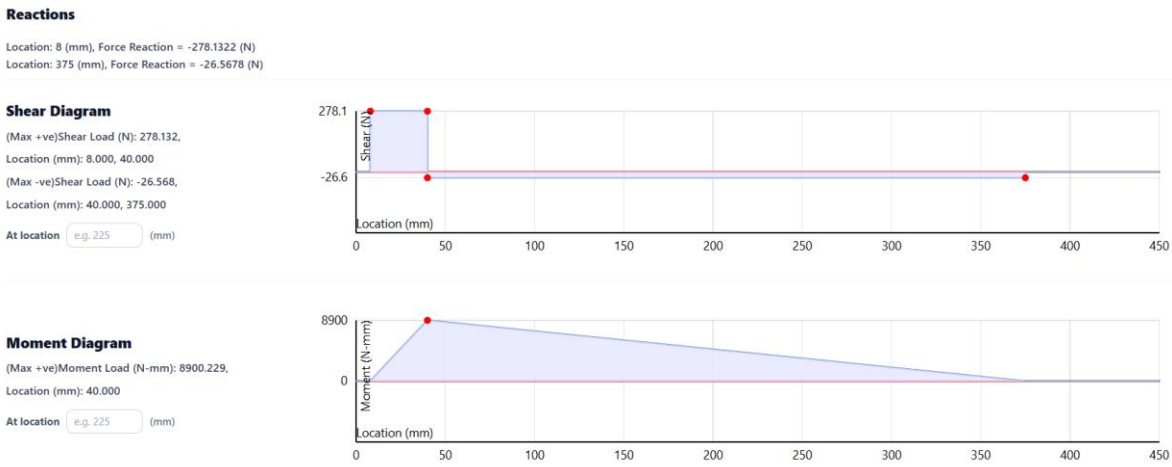


Figure 58: Shear force and bending moment diagrams (3rd Gear)

Maximum horizontal bending moment = 8.9 Nm

Maximum resultant bending moment = **<u>27.44 Nm</u>**

2.3.3.7 Bending moment analysis under the reverse gear

The small angle between idle shaft and the output shaft is neglected under this calculation.

Table 44: Forces acting on output shaft when reverse gear is engaged

Gear	Tangential Force(N)	Radial Force(N)	Weight(N)
Gear 6	-	-	39.20
Gear 4	-	-	23.36
Gear 8	-	-	11.60
Gear 10	2736.79	996.11	2.71

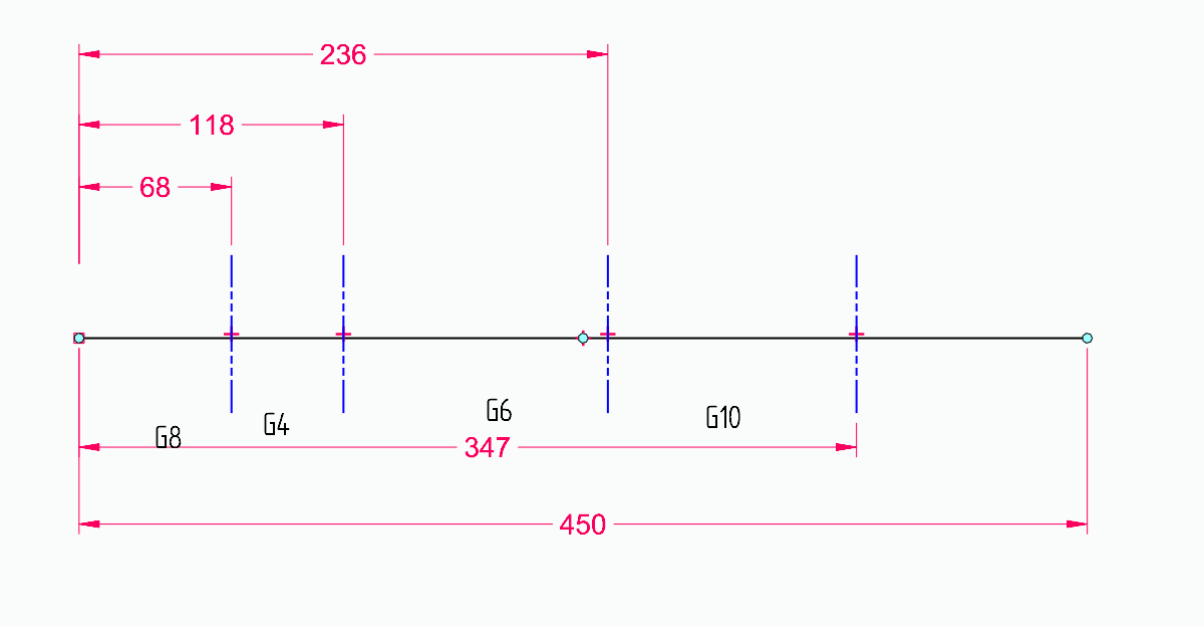


Figure 59: Output shaft gear position under reverse gear:

Vertical bending moment;

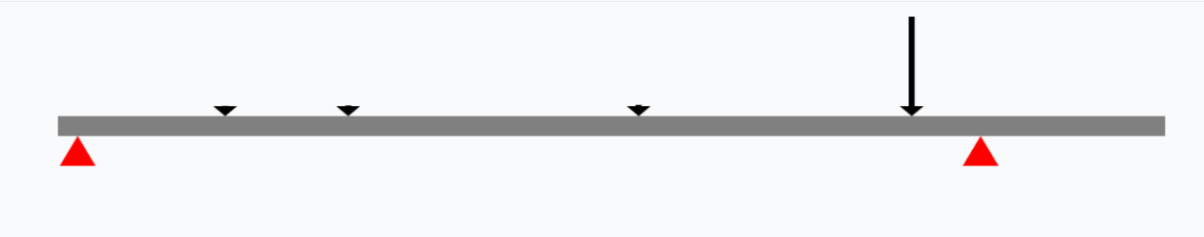


Figure 60: Free body diagram (Vertical Forces) of the Output shaft (Reverse Gear Engaged)

Table 45: Beam Input Parameters of the Output Shaft

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 450	
Support-pinned	8	
Support-pinned	375	
Point Load	68	11.6
Point Load	118	23.36
Point Load	236	39.2
Point Load	347	2737.41

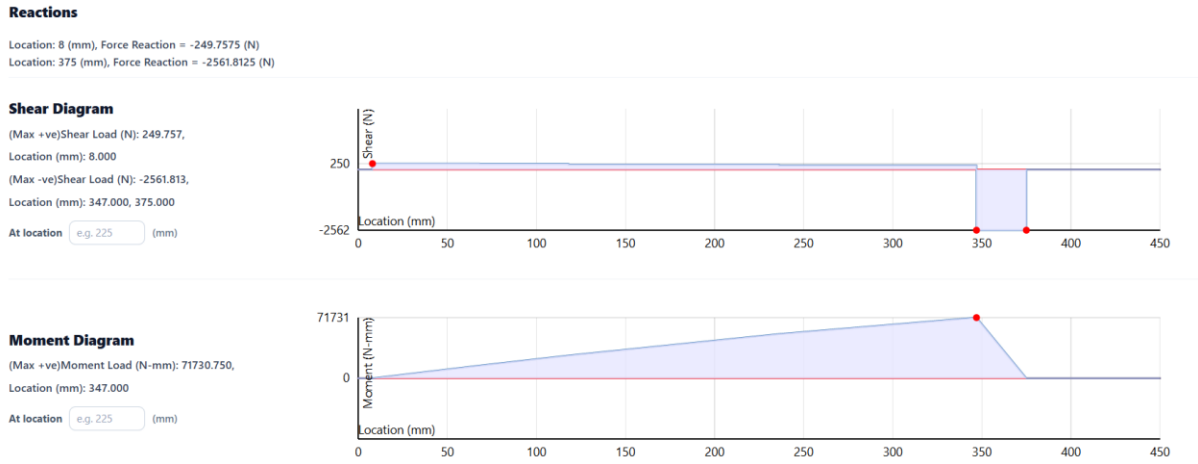


Figure 61: Shear force and bending moment diagrams (Reverse Gear)

Maximum vertical bending moment = 71.73 Nm

Horizontal bending moment;



Figure 62: Free body diagram (Horizontal Forces) of the Output shaft (3rd Gear Engaged)

Table 46: Beam Input Parameters of the Output Shaft

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 450	
Support-pinned	8	
Support-pinned	375	
Point Load	347	995.35

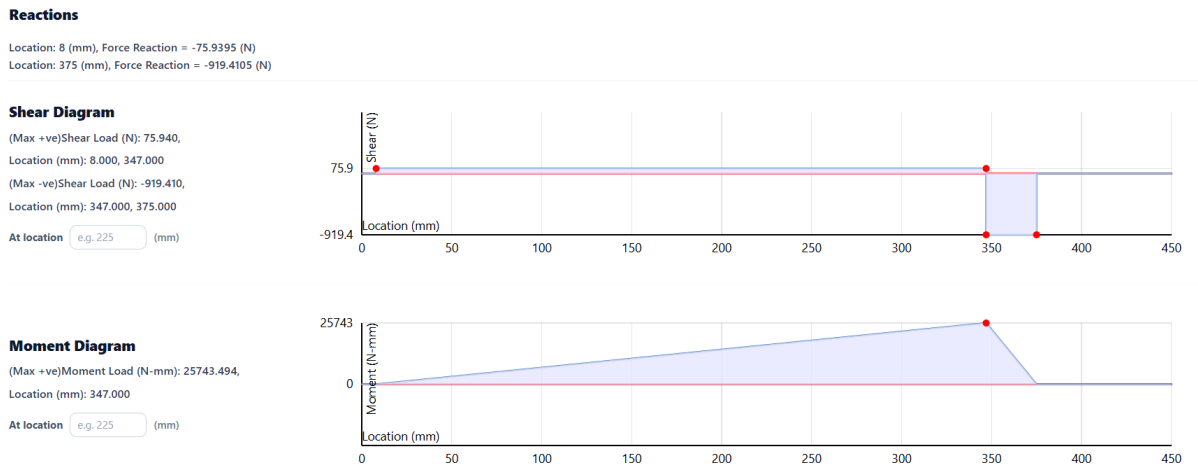


Figure 63: Shear force and bending moment diagrams (Reverse Gear)

Maximum horizontal bending moment =25.74 Nm

Maximum resultant bending moment = **<u>76.2 Nm</u>**

2.3.3.8 Equivalent Bending moment and twisting calculation

Table 47: Summary of the resulting Bending Moments under each gear setting

Gear	Maximum resulting BM(Nm)
First	178.5
Second	107.6
Third	27.44
Reverse	76.2

178.5 Nm was chosen as maximum bending moment.

Using that, equivalent bending and equivalent twisting was calculated, using the equations defined in section 1.3.2.8.

For Kt and Km values 1.5 was selected.

Shaft torque value under maximum bending moment = 152.49Nm

	Nm
Equivalent Twisting	332.2
Equivalent Bending	300.0

2.3.3.9 Shaft minimum diameter calculation

For these calculations equations defined in section 1.3.2.9 was used.

Since the shaft material is same, the maximum tensile stress allowed = 895.16 MPa

Maximum shear stress allowed = 347.58 MPa

Following minimum diameter values were obtained;

	mm
d1	16.9
d2	15.05

Output shaft diameter must be larger than 16.9 mm.

2.3.3.10 Standards and Calculations for the Design of the Splines.

At the beginning it is identified that there are two main industrial methods to manufacture a spline shaft.

- Straight-Sided (Parallel) Splines
- Involute Splines

From these two options, parallel splines were decided as ideal for this output shaft due to following reasons.

- Optimized for Axial Sliding
- Manufacturing Feasibility
- Suitability for Low-Speed Operating Conditions
- Clearance Fit Control

When designing the splined shaft, it is a technically sound decision to take the minor diameter of the splined shaft as the minimum diameter. In that case it is decided to use 17 mm diameter cross section for the portions were bearing fit and, 20 mm minimum base diameter for the area where gears slide along the shaft.

Initially during the detailed design calculations, SAE standards and ratios were used to design splines of the output shaft and gear hubs.[7]

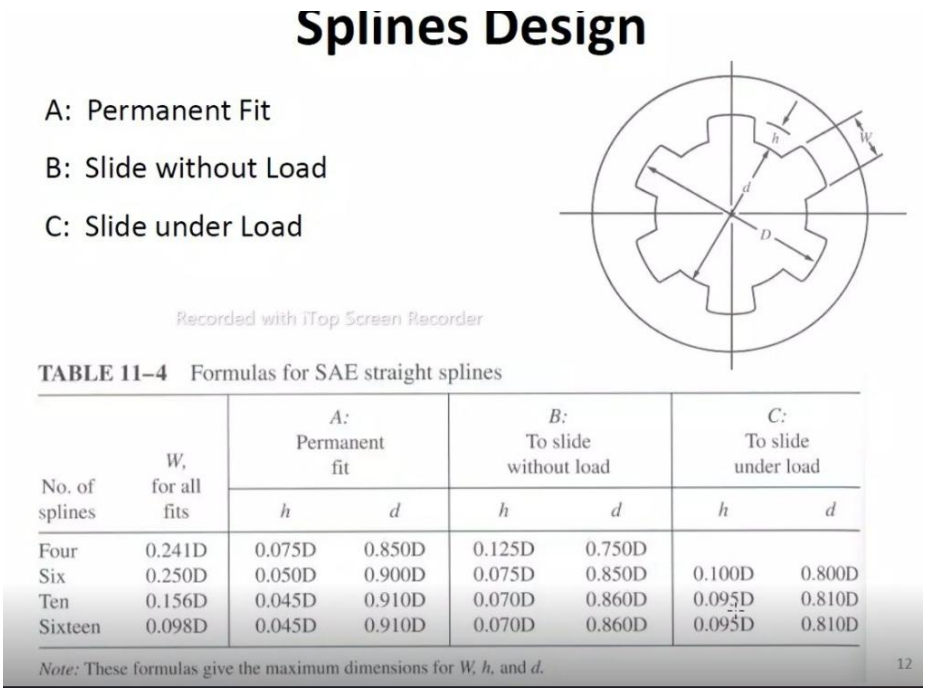


Figure 64: SAE Standards for Splines

But when designing the final splined shaft, converting the used standard more technically sound, it is decided to use the international standard ISO-14 for the design of the Straight sided splines for cylindrical shafts with internal centring - Dimensions, tolerances and verification document as a reference.[8]

corresponding gauges.

2 Dimensions

The nominal dimensions common to shaft and hub, d, D and B are given in table 1. The tolerances are indicated in tables 2 and 3.

3 Designation

The profile of a splined shaft or hub shall be designated by stating, in the following order : the number of splines N, the

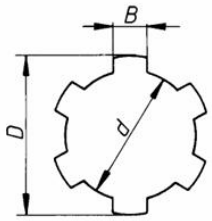


Table 1 — Nominal dimensions

d mm	Light series				Medium series			
	Designation	N	D mm	B mm	Designation	N	D mm	B mm
11					6 × 11 × 14	6	14	3
13					6 × 13 × 16	6	16	3,5
16					6 × 16 × 20	6	20	4
18					6 × 18 × 22	6	22	5
21					6 × 21 × 25	6	25	5
23	6 × 23 × 26	6	26	6	6 × 23 × 28	6	28	6
26	6 × 26 × 30	6	30	6	6 × 26 × 32	6	32	6
28	6 × 28 × 32	6	32	7	6 × 28 × 34	6	34	7
32	8 × 32 × 36	8	36	6	8 × 32 × 38	8	38	6
36	8 × 36 × 40	8	40	7	8 × 36 × 42	8	42	7
42	8 × 42 × 46	8	46	8	8 × 42 × 48	8	48	8
46	8 × 46 × 50	8	50	9	8 × 46 × 54	8	54	9
52	8 × 52 × 58	8	58	10	8 × 52 × 60	8	60	10
56	8 × 56 × 62	8	62	10	8 × 56 × 65	8	65	10
62	8 × 62 × 68	8	68	12	8 × 62 × 72	8	72	12
72	10 × 72 × 78	10	78	12	10 × 72 × 82	10	82	12
82	10 × 82 × 88	10	88	12	10 × 82 × 92	10	92	12
92	10 × 92 × 98	10	98	14	10 × 92 × 102	10	102	14
102	10 × 102 × 108	10	108	16	10 × 102 × 112	10	112	16
112	10 × 112 × 120	10	120	18	10 × 112 × 125	10	125	18

Figure 65: ISO-14 Standards for Straight Sided Splines

Since it is decided to use 20mm as the base diameter for the splined shaft, next best matching value ;21mm was chosen as the ideal for output splined shaft.

With reference to the 21 mm minor diameter value; 6×21× 25× 5, medium series spline structure was chosen.

Number of splines = 6

Minor diameter = 21mm

Major diameter = 25mm

Spline width = 5 mm

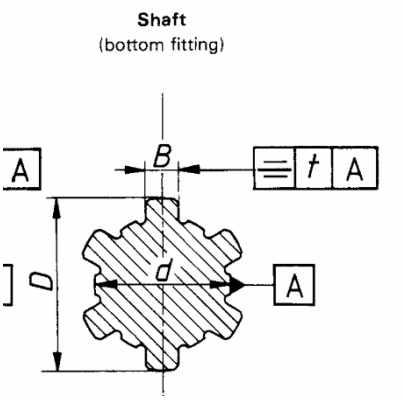


Figure 66: ISO Spline Profile

To design the spline profile following, sketch profile was used.

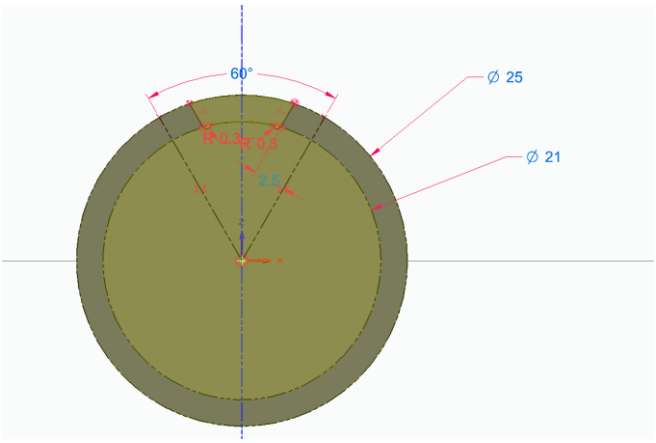


Figure 67: Sketch Profile for Splined Shaft

Since under this standard, it is unable to change the spline profile and shaft diameters according to the ratios (as in SAE standard), it is decided to connect the variable table of the CAD with an excel sheet containing the above-mentioned standard table.

When any required spline profile row is copied and pasted in the active row and update the variable table the shaft will get automatically updated without crashing the relations.

	Number of splines	Minor_D	Major_D	Spline width
Active Row	6.00	21.00	25.00	5.00

ISO Standard Table

Series	Designation PDF	N (Number of Splines) PDF	d (Minor Diameter, mm) PDF	D (Major Diameter, mm) PDF	B (Spline Width, mm) PDF
Light	6x11x14	6	11	14	3
Light	6x13x16	6	13	16	3.5
Light	6x16x20	6	16	20	4
Light	6x18x22	6	18	22	5
Light	6x21x25	6	21	25	5
Light	6x23x26	6	23	26	6
Light	6x26x30	6	26	30	6
Light	6x28x32	6	28	32	7
Light	8x32x36	8	32	36	6
Light	8x36x40	8	36	40	7
Light	8x42x46	8	42	46	8
Light	8x46x50	8	46	50	9
Light	8x52x58	8	52	58	10
Light	8x56x62	8	56	62	10
Light	8x62x68	8	62	68	12
Light	10x72x78	10	72	78	12
Light	10x82x88	10	82	88	12
Light	10x92x98	10	92	98	14
Light	10x102x108	10	102	108	16
Light	10x112x120	10	112	120	18
Medium	6x23x28	6	23	28	6
Medium	6x26x32	6	26	32	6
Medium	6x28x34	6	28	34	7
Medium	8x32x38	8	32	38	6
Medium	8x36x42	8	36	42	7
Medium	8x42x48	8	42	48	8
Medium	8x46x54	8	46	54	9
Medium	8x52x60	8	52	60	10
Medium	8x56x65	8	56	65	10
Medium	8x62x72	8	62	72	12
Medium	10x72x82	10	72	82	12
Medium	10x82x92	10	82	92	12
Medium	10x92x102	10	92	102	14
Medium	10x102x112	10	102	112	16
Medium	10x112x125	10	112	125	18

Figure 68: Excel Sheet Containing Standard ISO Spline Profiles

Filters

Type: Dimensions;User Variables;... | Named by: Both | Graphics in: File | Apply

Structure	Type	Name	Value	Units	Rule	Formula	Ran
V4304	Dim	V4304	10.00	mm			
V4309	Dim	V4309	8.50	mm			
V4425	Dim	V4425	13.00	mm			
V4519	Dim	V4519	5.00	mm			
V25545	Dim	V25545	0.00	mm			
V25547	Dim	V25547	0.00	mm			
V25896	Dim	V25896	12.50	mm	Formula	Major_D /2	
V25898	Dim	V25898	10.00	mm	Formula	(Major_D -5)/2	
V25900	Dim	V25900	8.50	mm			
V25904	Dim	V25904	81.00	mm			
V26014	Dim	V26014	80.00	mm			
Spline							
V25596	Dim	V25596	2.50	mm	Formula	Spline_W /2	
V26285	Dim	V26285	60.00	°	Formula	360/ Spline_N	
V27198	Dim	V27198	21.00	mm	Formula	Minor_D	
V27199	Dim	V27199	25.00	mm	Formula	Major_D	
Spline_FiniteDepth	Dim	Spline_Finite...	418.00	mm			
V25377	Dim	V25377	0.30	mm			
V25378	Dim	V25378	0.30	mm			
Pattern							
V31383	Var	V31383	6.000		Formula ...	Spline_N	[1.0
Round							
Chamfer 1							
Chamfer 2							
User Variables							
Minor_D	Var	Minor_D	21.00	mm	Paste Link	@'C:\Users\Dakshina\Desktop\Sem 5\Machine Design Project ME3201\5.Final 3D Model\G21\Gear Box\SplinedShaft.xlsx'!'Sheet1!R6C5'	
Major_D	Var	Major_D	25.00	mm	Paste Link	@'C:\Users\Dakshina\Desktop\Sem 5\Machine Design Project ME3201\5.Final 3D Model\G21\Gear Box\SplinedShaft.xlsx'!'Sheet1!R6C6'	
Spline_W	Var	Spline_W	5.00	mm	Paste Link	@'C:\Users\Dakshina\Desktop\Sem 5\Machine Design Project ME3201\5.Final 3D Model\G21\Gear Box\SplinedShaft.xlsx'!'Sheet1!R6C7'	
Spline_N	Var	Spline_N	6.00	mm	Paste Link	@'C:\Users\Dakshina\Desktop\Sem 5\Machine Design Project ME3201\5.Final 3D Model\G21\Gear Box\SplinedShaft.xlsx'!'Sheet1!R6C4'	
PhysicalProperties_Accuracy	Var	PhysicalPrope...	0.990		Limit		(0.0
PhysicalProperties_Density	Var	PhysicalPrope...	7833.000	kg/m^3	Limit		[0.0

Figure 69: Updated Solid Edge Variable Table Corresponding with Excel Sheet

2.3.3.11 Designing the lengths of the splines along the shafts

Length of the splines were defined by the minimum potion along the shaft where gears need to travel for effective engage and disengage with the lay shaft gear wheels. Considering this after designing the final CAD assembly and animations, splines were only applied to the limited section of the shaft, which prevents unnecessary weakening of the shafts by continuous splines along the shaft.

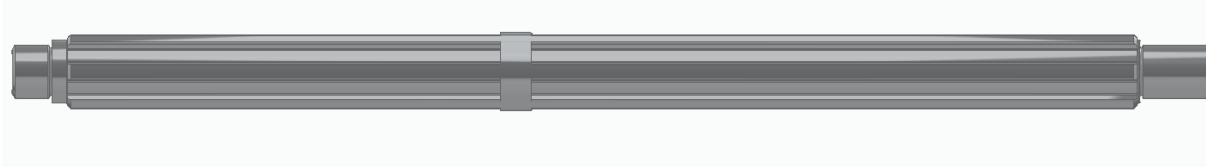


Figure 70: Splines applied into the Output Shaft

2.3.3.12 Final 3-D CAD design for the splined output shaft



Figure 71: Final Design of the Splined Output Shaft

2.3.4 Idle Shaft

Idle shaft design calculations were heavily altered with respect to the detailed design calculations because its precise position was determined after creating the final assembly and making the idle gear in tangent with the gear10 of the output shaft and gear 9 of the lay shaft.

2.3.4.1 Weight of the idle gear

	Diameter(m)	Weight(N)
Gear 11	0.045	2.71

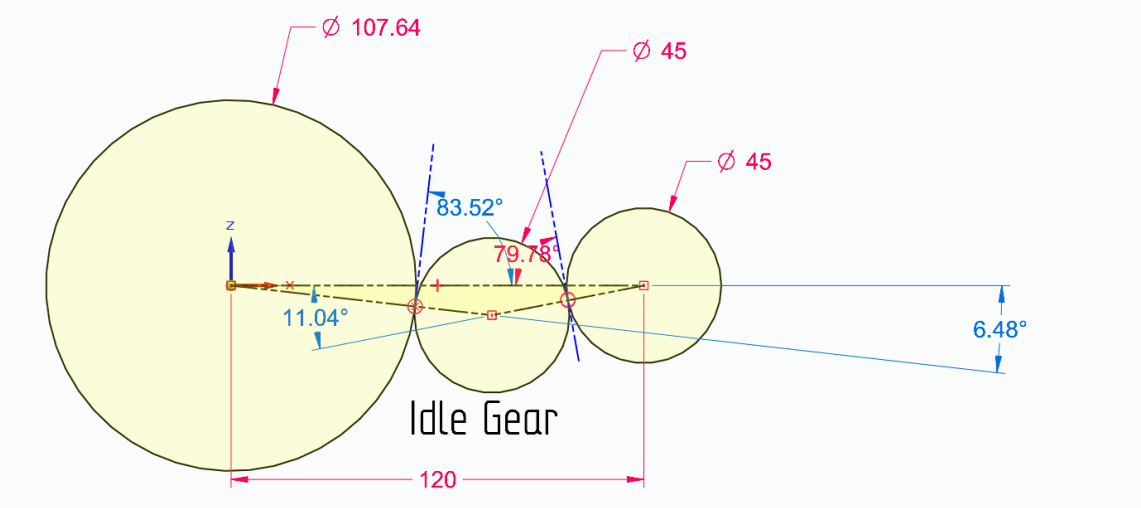


Figure 72: Idler Gear Position relative to the Layshaft and Output Shaft

**These values were obtained using calculations defined in section 1.3.2.1.

2.3.4.2 Forces acting on the shaft by idle gear

Table 48: Forces on the Idler Shaft

	Diameter(m)	Tangential Force(N)	Radial Force(N)
Gear 11 - Gear 9	0.045	2734.76	995.37
Gear 11 - Gear 10	0.045	2736.79	996.11

** These values were obtained using calculations defined in section 1.3.2.2.

2.3.4.3 Force analysis of the idle shaft

Idle gear is only engaged in reverse gear setting, and it is important to analyse idle gear wheel's position with respect to two tangent gear wheels it is in mesh with.

Horizontal forces on the idle shaft;

(right side is positive)

$2734.76 \times \cos(79.78°) - 995.37 \times \cos(11.04) + 2736.76 \times \cos(83.52) + 996.11 \times \cos(6.48)$

$= 806.88$ N

Vertical forces on the idle shaft;

(downward is positive)

$2734.76 \times \sin(79.78) + 995.37 \times \sin(11.04) + 2.71 - 2736.76 \times \sin(83.52) + 996.11 \times \sin(6.48)$

$= 277.83$N

2.3.4.4 Bending moment analysis of the idle shaft

Considering the smooth gear meshing, the length of the idle shaft was chosen as 121mm.

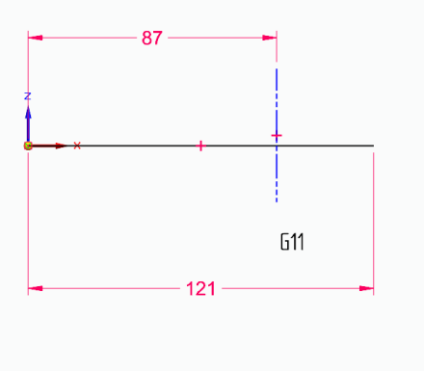


Figure 73: Idle Shaft Gear layout

Horizontal bending moments:

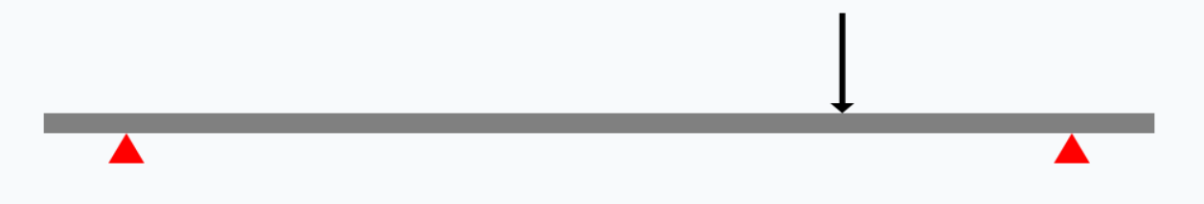


Figure 74: Free body diagram (Horizontal Forces) of the Idle shaft

Table 49: Beam Input Parameters of the Idler Shaft

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 121	
Support-pinned	9	
Support-pinned	112	
Point Load	87	806.88

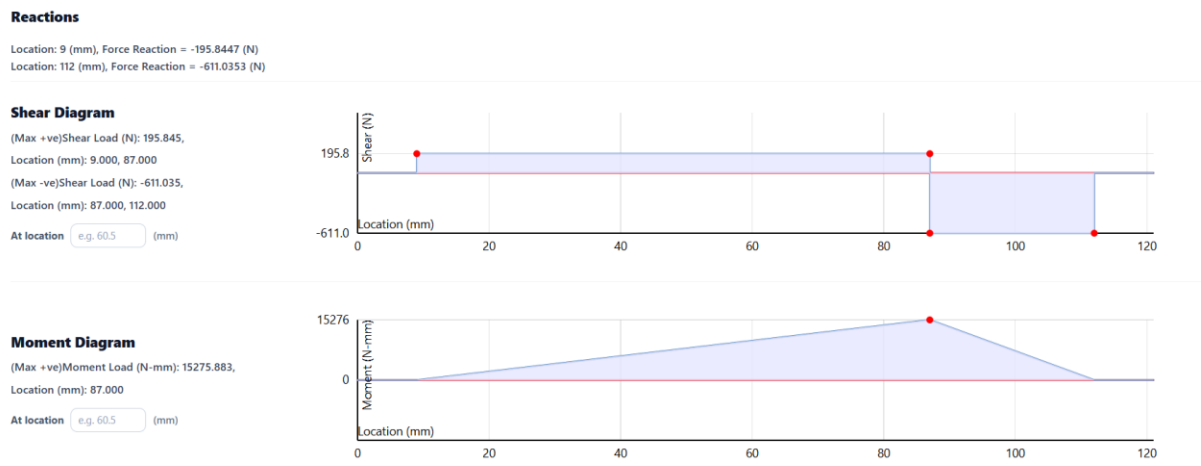


Figure 75: Bending Moment and Shear Force Diagrams

Maximum horizontal bending moment = 15.28 Nm

Vertical bending moment;

Free body diagram:



Figure 76: Free body diagram (Vertical Forces) of the Idler shaft

Table 50: Beam Input Parameters of the Idler Shaft

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 121	
Support-pinned	9	
Support-pinned	112	
Point Load	87	277.83

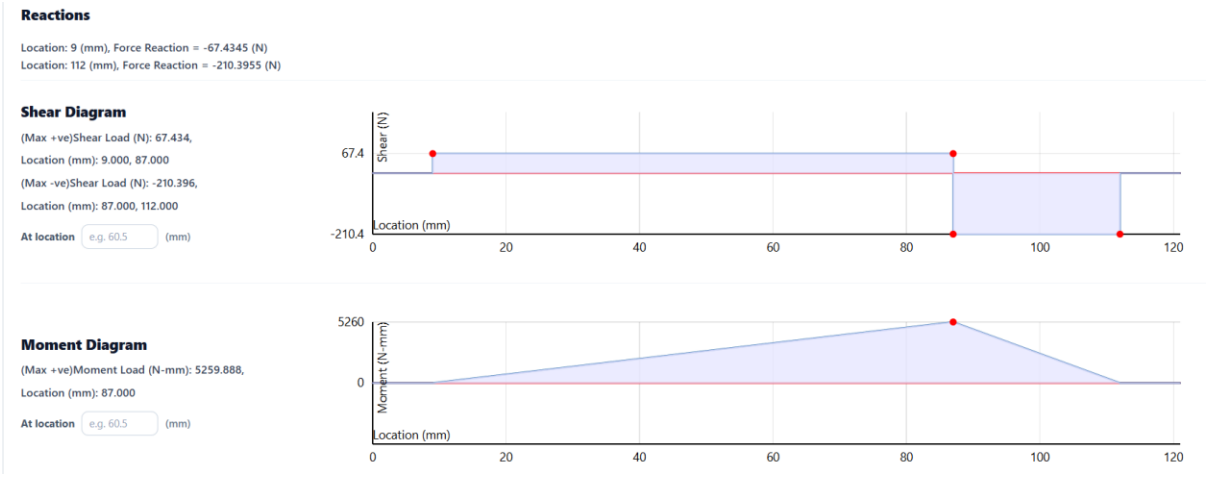


Figure 77: Bending Moment and Shear Force Diagrams

Maximum vertical bending moment = 5.26 Nm

Maximum resulting bending moment = **16.16 Nm**

2.3.4.5 Minimum diameter calculation

	(Nm)
Equivalent Twisting	95.43
Equivalent Bending	59.83

**These values were obtained using, equations defined in section 1.3.2.8.

For Kt and Km values 1.5 was selected.

Shaft torque value under maximum bending moment = 61.53 Nm

For these calculations of the diameter, equations defined in section 1.3.2.9 was used.

Since the shaft material is same, the maximum tensile stress allowed = 895.16 MPa

Maximum shear stress allowed = 347.58 MPa

Following minimum diameter values were obtained;

d1(mm)	8.797223342
d2(mm)	11.18216624

Shaft diameter must be greater than 11.2 mm. And minimum diameter of 20mm shaft was selected.

2.3.4.6 Designing the shaft steps

Steps to hold the gear wheel and the bearing was also designed following the standard methods described in section 1.3.2.

2.3.4.7 Final 3-D CAD model of the idle shaft

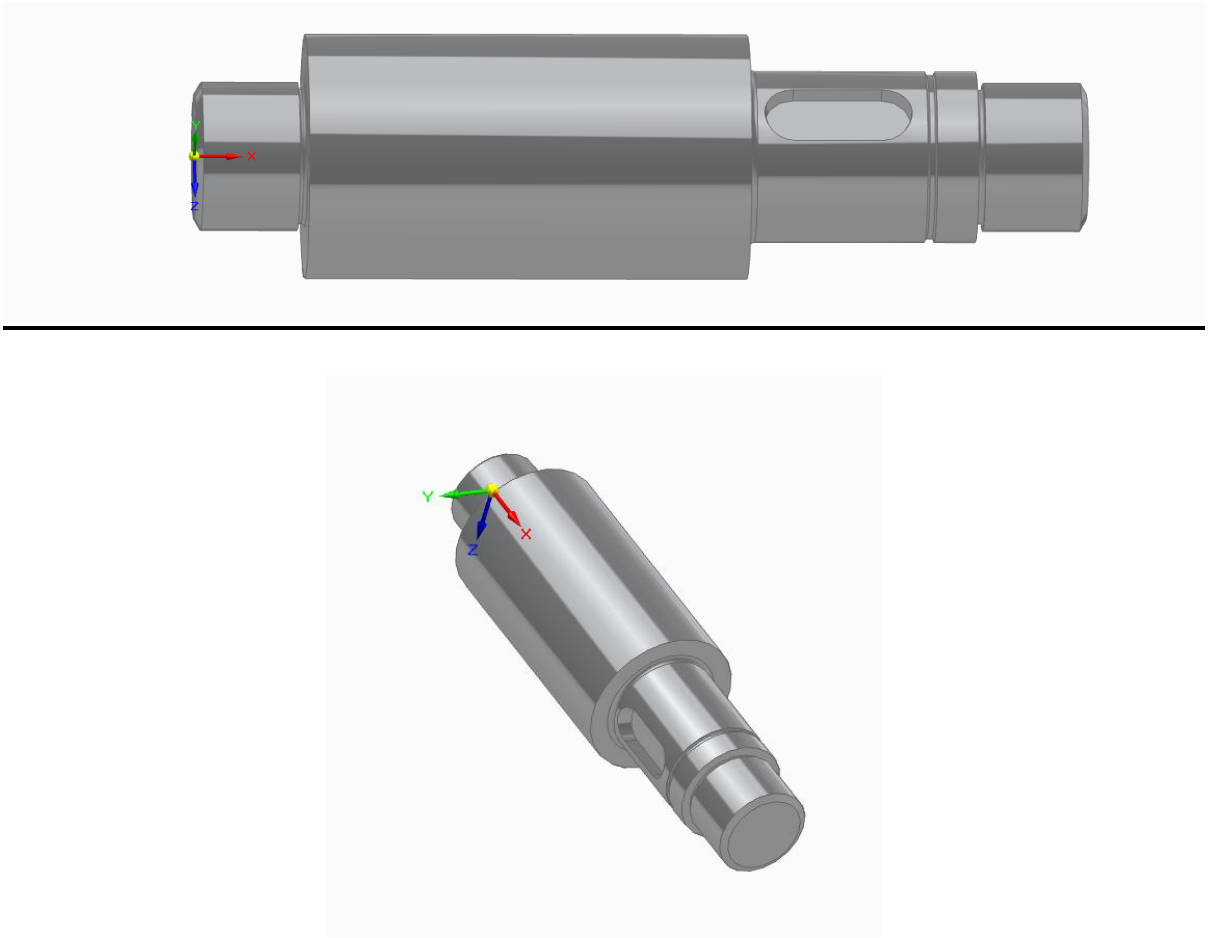


Figure 78: Final 3-D CAD Model of the Idle Shaft

2.3.5 Input Shaft

Length of the input shaft was taken as 172 mm, depending on the sufficient outward length from the housing to allow additional components to be mounted to the shaft form the outside and prioritising effective gear meshing with the gear wheel 2 of the lay shaft.

2.3.5.1 Weight of the gears and sprockets of the input shaft

Table 51: Input Shaft Assembly Overview

	Diameter(m)	Weight(N)	Face width(m)
Sprocket	0.18	5.43	0.01
Gear 1	0.069	6.38	0.024

**These values obtained using equations in section 1.3.2.1.

2.3.5.2 Forces acting on the shaft by components

Table 52: Forces on the Input Shaft

	Tangential load(N)	Radial load(N)	Weight(N)
Gear 1	719.66	261.93	6.38
Sprocket	0.00	275.87	5.43

**These values obtained using equations in section 1.3.2.2.

2.3.5.3 Bending moment analysis of the input shaft

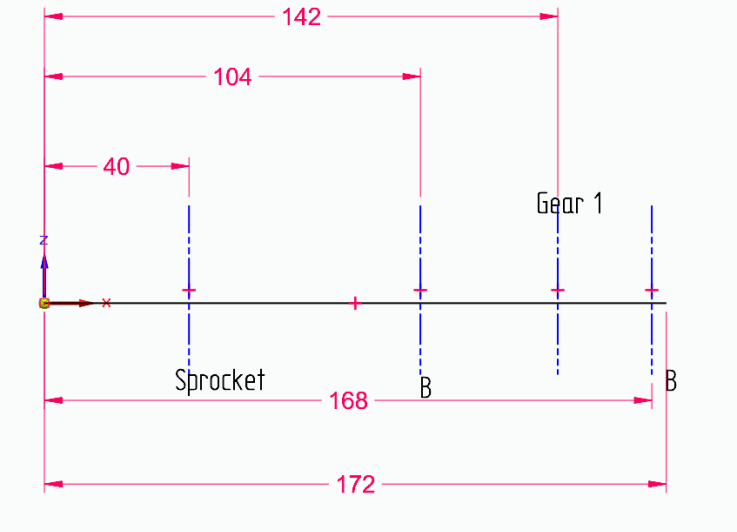


Figure 79: The gear and sprocket lay out on the input shaft

Horizontal bending moments;



Figure 80: Free body diagram (Horizontal Forces) of the Input shaft

Table 53: Beam input parameters of the input shaft

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 172	
Support-pinned	104	
Support-pinned	168	
Point Load	40	275.87
Point Load	142	-261.93

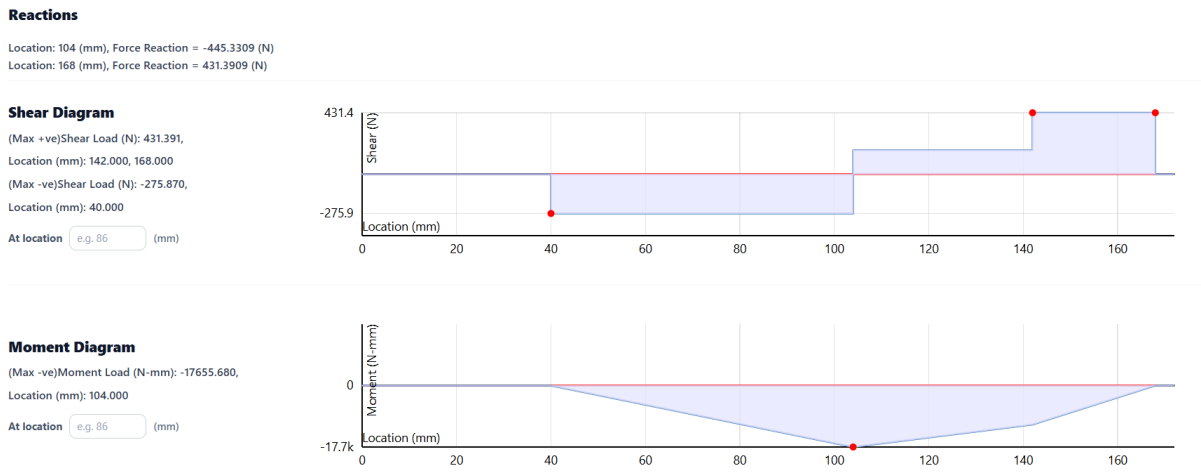


Figure 81: Bending Moment and Shear Force Diagrams

Maximum horizontal bending moment = 17.66 Nm

Vertical bending moment;

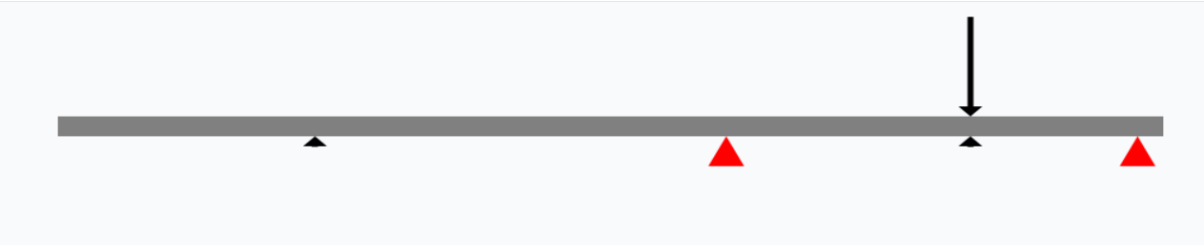


Figure 82: Free body diagram (Vertical Forces) of the Input shaft

Table 54: Beam input parameters of the input shaft

Type	Location (mm)	Load (N)|(N-mm)
Length	0 | 172	
Support-pinned	104	
Support-pinned	168	
Point Load	40	-5.43
Point Load	142	719.66
Point Load	142	-6.38

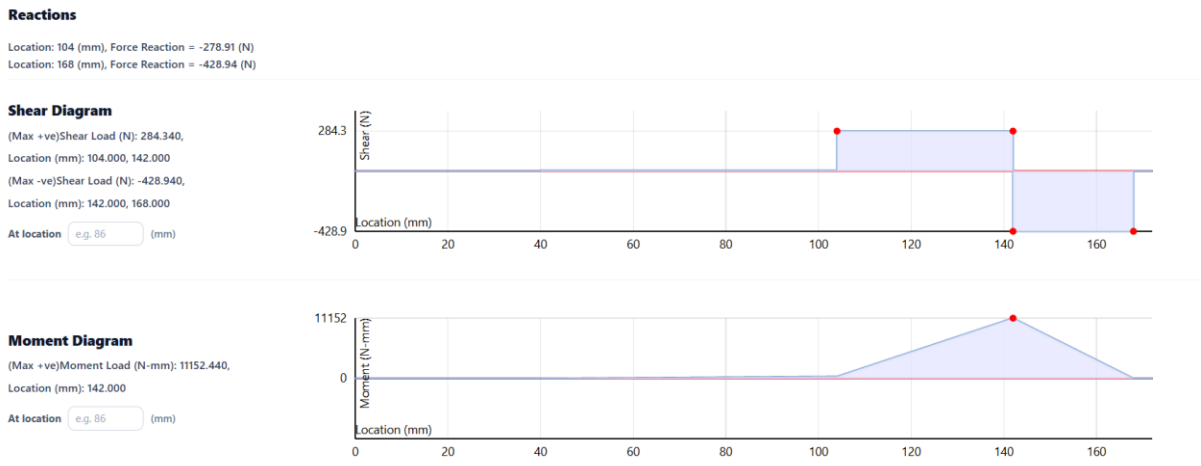


Figure 83: Bending Moment and Shear Force Diagrams

Maximum vertical bending moment = 11.15 Nm

Resultant maximum bending moment = **20.89 Nm**

2.3.5.4 Minimum diameter calculation for the input shaft

	(Nm)
Equivalent Twisting	48.67
Equivalent Bending	40.00

**These values were obtained using, equations defined in section 1.3.2.8.

For Kt and Km values 1.5 was selected.

Shaft torque value under maximum bending moment = 24.83 Nm

For these calculations of the diameter, equations defined in section 1.3.2.9 was used.

Since the shaft material is same, the maximum tensile stress allowed = 895.16 MPa

Maximum shear stress allowed = 347.58 MPa

Following minimum diameter values were obtained;

	mm
d1	8.9
d2	7.6

For a better safety factor and to match with manufacturing feasibility with other shafts, minimum diameter of 20mm shaft was selected for the input shaft.

2.3.5.5 Design of the steps of the input shaft to prevent gear wheel axial movement

This design part also done using the same standards and calculation steps done in the section 1.3.2.10.

2.3.5.6 Final 3-D CAD model of the input shaft

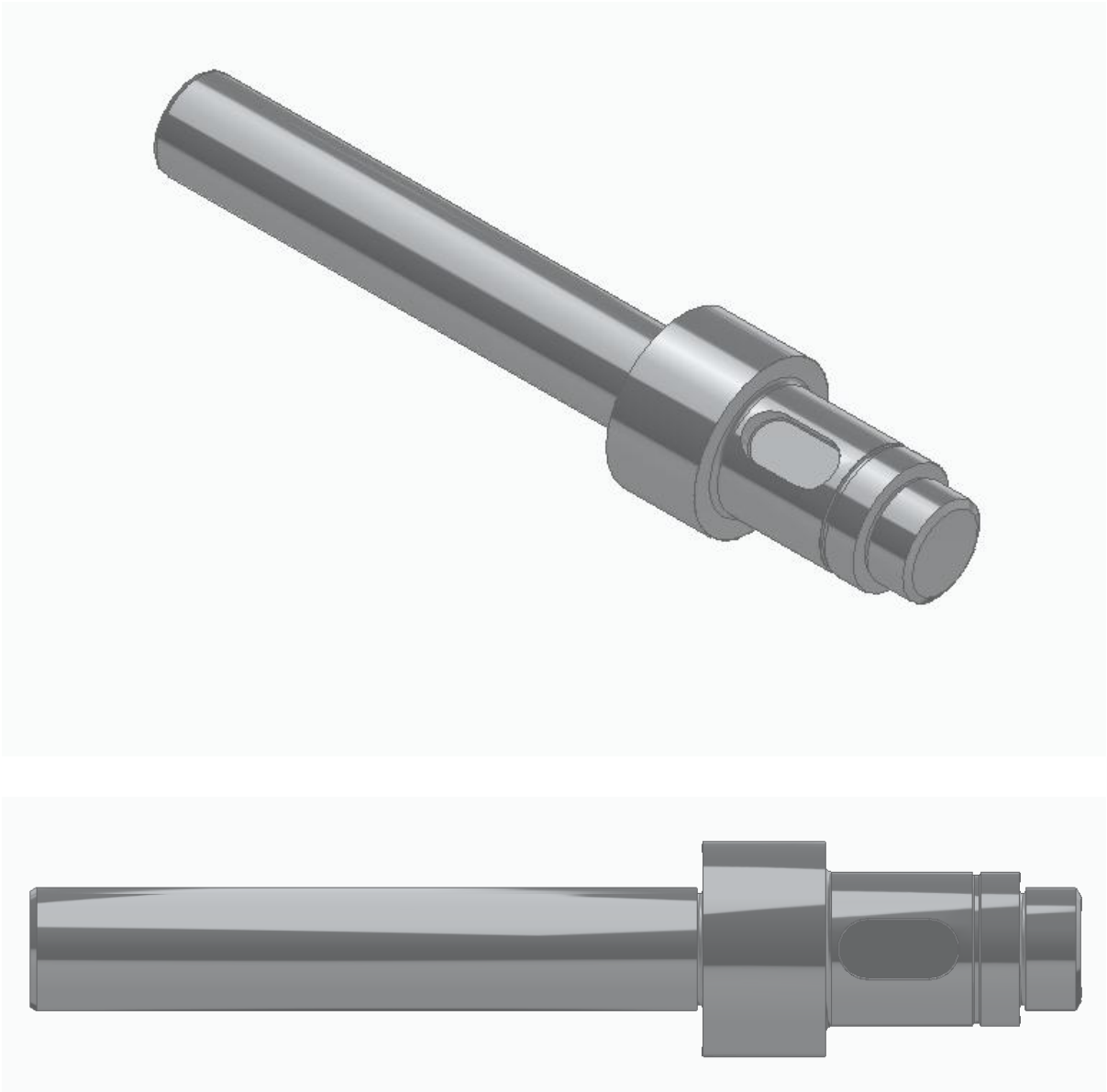


Figure 84: Final 3-D CAD model of the Input Shaft

2.4 Final Design

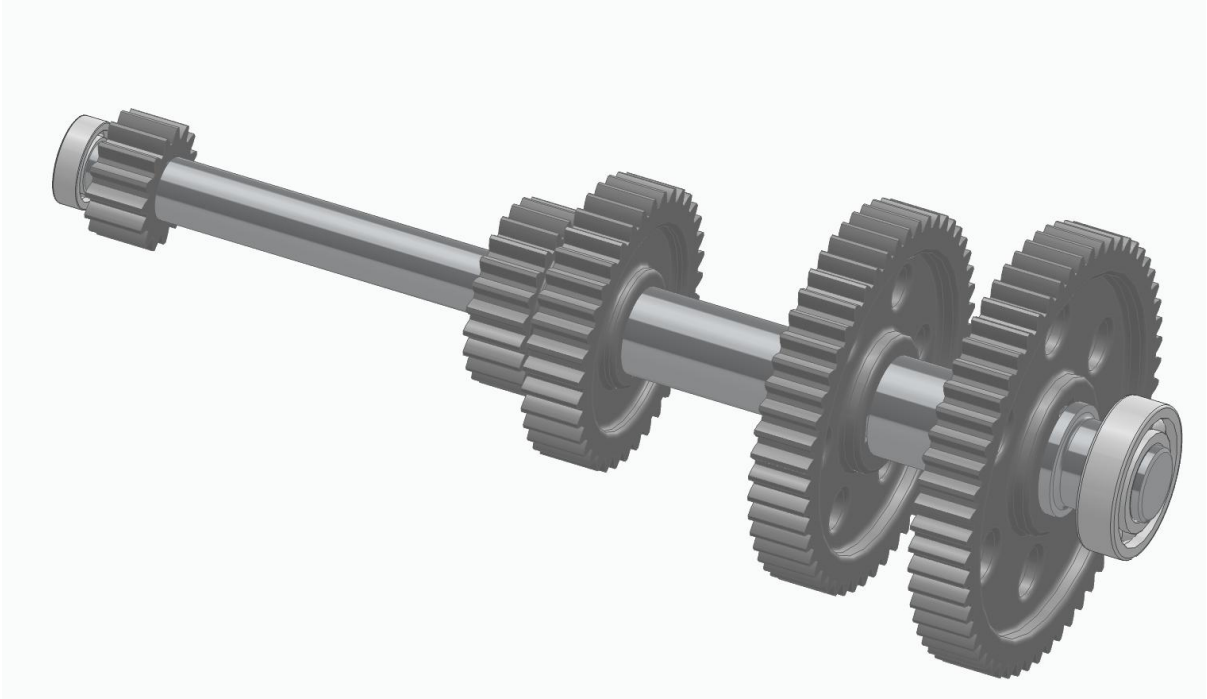


Figure 85: Final Layshaft Assembly

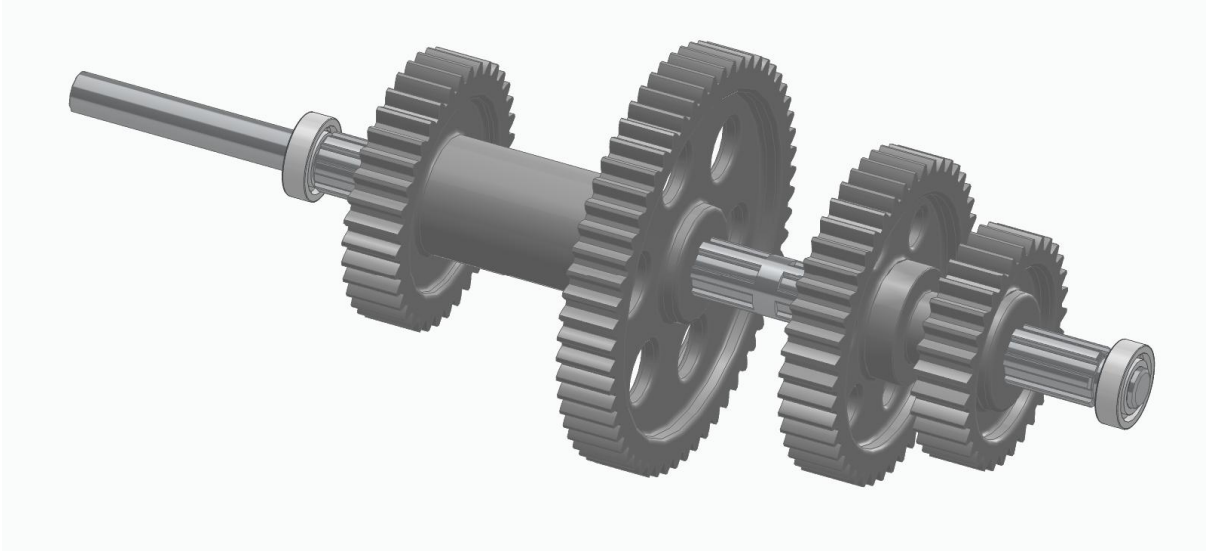


Figure 86: Final Output Shaft Assembly

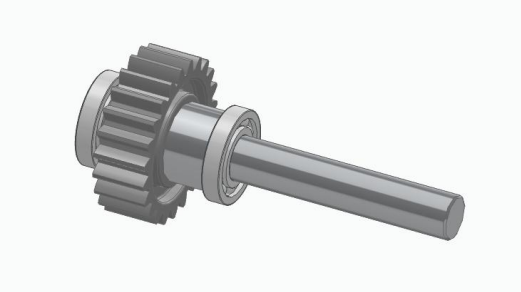


Figure 87: Input Shaft Assembly

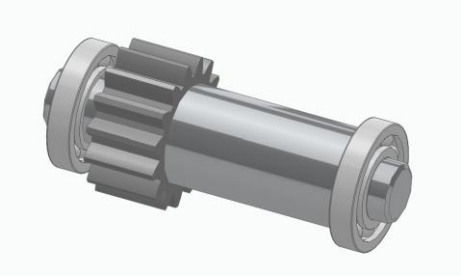


Figure 88: Idle Shaft Assembly

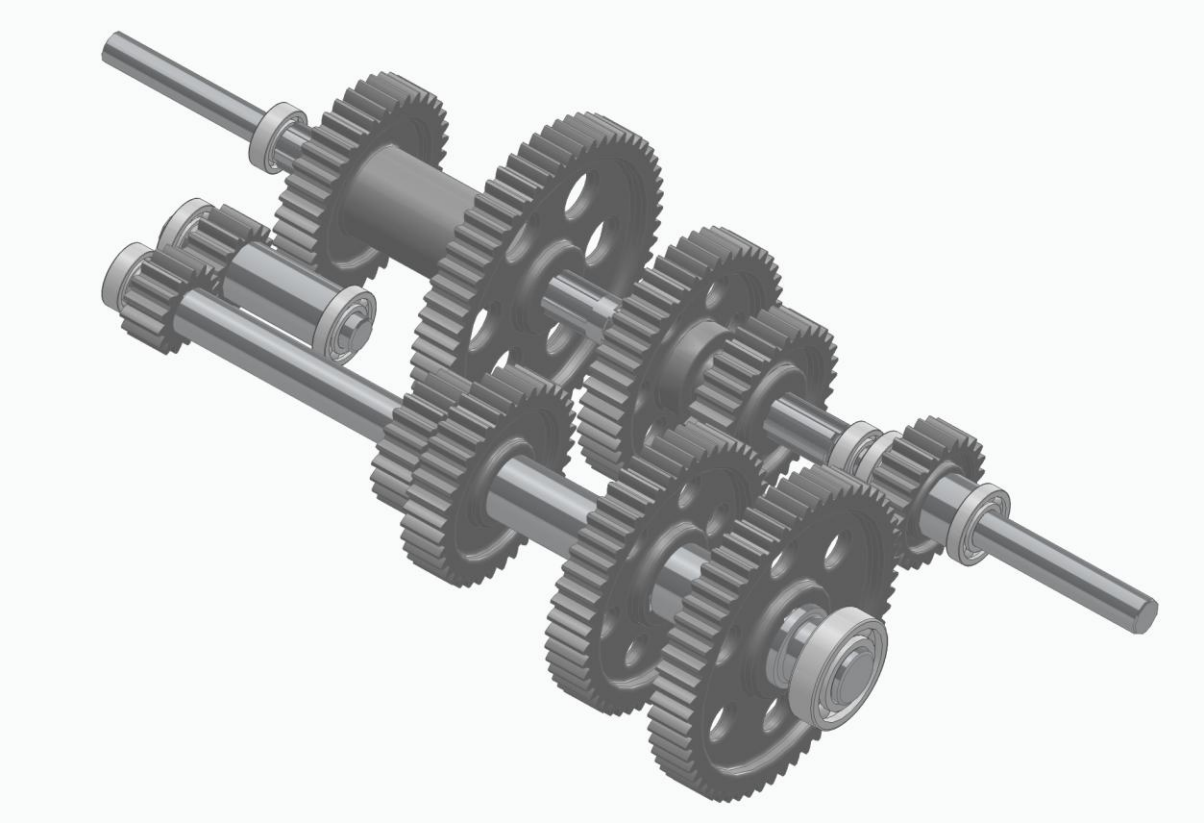


Figure 89: Complete Shaft Assembly

2.5 References

[1] Tech Steel & Materials, "Steel 300M," techsteel.net. https://www.techsteel.net/alloy/steel/300m/ (accessed Aug.30, 2026)

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[3] R. S. Khurmi and J. K. Gupta, *A Textbook of Machine Design*, 14th ed. New Delhi, India: S. Chand Publishing, 2005, p. 514.

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[7] *Straight Cylindrical Involute Splines Metric Module, Structural Dimensions, and Tolerances*, SAE Standard J499a, SAE International, Warrendale, PA

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3 Section 3 – Topic: Gearbox for Traditional Drum Lacing and Tensioning (230120A)

3.1 Introduction

Traditional Sri Lankan drums, including the Geta Bera, Yak Bera, and Davula, consist of hollowed wooden shells with rawhide drumheads secured by a zigzag-patterned cord called the ‘warapati’ [1]. Artisans manually tighten the cord while striking the drumhead to achieve the desired tone, known as the ‘kathawa’ [1]. This process is time-consuming and physically demanding, with no dedicated mechanised solution. Unlike Western drums that use motorised tension-rod systems, Sri Lankan drums rely on traditional lacing, highlighting the need for a tensioning jig that reduces physical effort while preserving artisan expertise.

Uneven manual tensioning can deform the wooden shell and affect sound quality. Additionally, rawhide's mechanical properties vary with moisture content and natural material differences, requiring different force levels during stretching and tuning. High initial force is needed to stretch the hide, whereas final tuning requires smaller, precise adjustments. Excessive force may tear the hide around the lacing holes, while insufficient control can hinder accurate tuning.

The proposed benchtop jig incorporates a clamping mechanism to secure the drum and a gearbox mounted directly onto the structure for controlled tensioning. A multi-speed gearbox is necessary to provide high torque during initial stretching and suitable speeds for fine-tuning, overcoming the limitations of a single-speed motor. The system aims to reduce physical strain, improve consistency, and preserve Sri Lankan drumming craftsmanship while retaining the artisan's role in achieving the desired tone.



Figure 90: Traditional Sri Lankan drums with lacing based tensioning

3.2 Acceptance or rejection statement

Although the project was accepted by the advisors due to its novelty and potential, the limited availability of information and the specialised nature of the concept led us to consider an alternative. We ultimately selected a carbon fibre impregnation press, as more technical information was available online. This project also offers an innovative solution within a developing field, with potential practical benefits and opportunities for further development.

3.3 Design Calculations and Materials Selection

The following calculations are based on the selected project of developing a gearbox for a mechanical carbon fibre impregnation press.

3.3.1 Lubrication

Lubrication is important because it reduces friction and wear between rotating surfaces, prevents metal to metal contact under load, and helps dissipate heat generated during operation. Without proper lubrication, bearings and gear meshes would wear out quickly and could seize, especially given the continuous operation of this gearbox.

There are 3 types of lubrications methods that can be used for this application. They are selected based on the range of tangential speeds of the spur/ bevel gears [2]. The following figure include the range of tangential speed for each of the 3 lubricating methods [2].

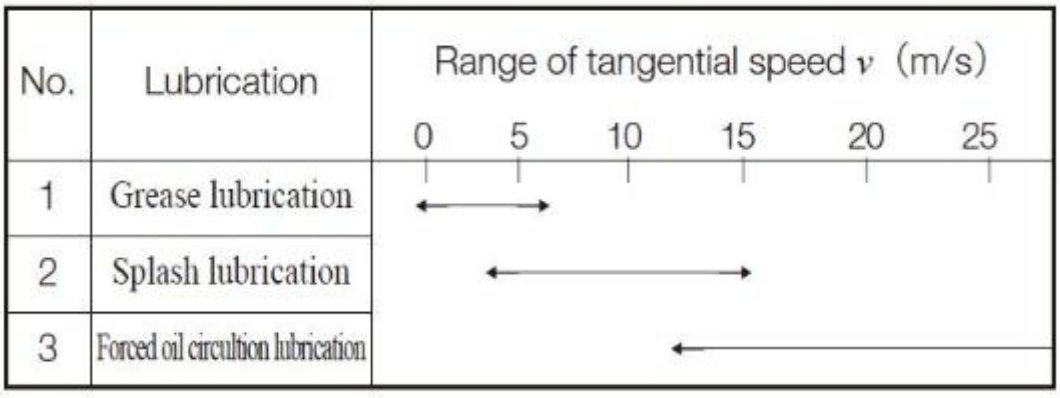


Figure 91: Methods of Lubrication and Criteria

The tangential speeds were calculated for all the gear wheels:

Table 55: Tangential Speeds of Gear Wheels

Gear	Diameter (mm)	RPM	Angular Speed (rad/s)	Tangential Speed (m/s)
G1	69	10.000	1.047	0.036
G2	171	4.035	0.422	0.036
G3	107	4.035	0.422	0.023
G4	133	3.301	0.346	0.023
G5	69	4.035	0.422	0.015
G6	171	1.628	0.170	0.015
G7	147	4.035	0.422	0.031
G8	93	6.378	0.668	0.031
G9	44	4.035	0.422	0.009
G10	109	1.614	0.169	0.009
G11	44	4.035	0.422	0.009

Since all the gear wheel tangential speeds are much less than 6m/s, **Grease Lubrication** is more than sufficient for this application [20]. Furthermore, it eliminates the leakage risk associated with an oil bath near the resin-processing area while requiring less maintenance.

3.3.2 Bearings

For bearings a 8 total bearings are needed for the gearbox to hold the shafts in place, while 2 more additional bearings are used in the Input side of the Input Shaft and the Output Side of the Output side for further support.

Bearings were selected to achieve a 5-year service life with 7 hour of operations per day for 5 days a week each month. Deep groove ball bearings were chosen as suitable for the design.

Advantages of Deep Groove Ball Bearings:

- Low noise
- Low friction
- Less vibration
- Easy to install
- Good radial load capacity

3.3.2.1 Fatigue Life Theory and L10 Calculation

The rolling contact bearings shall be selected based on their basic dynamic load rating, the rotational speed generated by material surface fatigue, commonly known as fatigue or pitting. The quality bearing frequently used for cycle rating, the engineering standard relies on the L10 nominal fatigue life concept [3].

The L_{10} life is defined as the total number of operating hours or revolutions (90 million) that 90% of a large group of identical bearings will achieve or exceed before the initial signs of fatigue appear. Consequently, it represents a 90% probability with respect to a specified load and speed conditions [3].

To establish the minimum required service life in hours (L_{10h}), the study cycle is evaluated for an operational period of 5 years, assuming 5 days per week and 7 hours per day (assuming a standard 52-week year):

$$L_{10h} = 5 \text{ years} \times 52 \frac{\text{weeks}}{\text{year}} \times 5 \frac{\text{days}}{\text{week}} \times 7 \frac{\text{hours}}{\text{day}} = 9{,}100 \text{ hours}$$

Converting this target life from hours to total revolutions (L_{10}) incorporates the rotational speed of the shaft (N in RPM):

$$L_{10} = \frac{60 \times N \times L_{10h}}{10^6}$$

The mathematical relationship linking bearing life, dynamic load capacity, and operational forces is expressed as:

$$L_{10} = \left(\frac{C}{P}\right)^p$$

Where:

- C: Basic dynamic load rating (found in manufacturer specification sheets)
- P: Equivalent dynamic bearing load
- p: Life exponent specific to the rolling-element geometry ($p=3$ for ball bearings, whereas $p = 10/3$ for cylindrical rollers) [3].

Because bearings in transmission systems frequently experience combined loading from radial and axial forces Fr and axial force Fa, the load must be converted into a single equivalent dynamic load P, expressed as:

$$P = XF_r + YF_a$$

3.3.2.2 Force Calculation and Bearing Selection

Table 56: Detailed Component Breakdown on each Shaft

Assembly Name	Component	Weight (N)
Input Shaft Assembly	Shaft	4.12
	Gear 1	6.38
	Sprocket	5.43
Lay Shaft Assembly	Shaft	34.34
	Gear 2	39.20
	Gear 3	15.64
	Gear 5	6.38
	Gear 7	28.97
	Gear 9	2.71
Splined Output Shaft Assembly	Shaft	11.97
	Gear 4	23.36
	Gear 6	39.20
	Gear 8	11.60
	Gear 10	15.64
Idle Shaft Assembly	Shaft	2.94
	Gear 11	2.71
Total Component Weight		**250.59**

Table 57: Summary of Shaft Assemblies

Assembly Name	Shaft Weight (N)	Mounted Components Weight (N)	Total Assembly Weight (N)
Input Shaft Assembly	4.12	11.81	15.90
Lay Shaft Assembly	34.34	92.90	127.24
Splined Output Shaft Assembly	11.97	89.80	101.77
Idle Shaft Assembly	2.94	2.71	5.69
Total System Weight	**53.37**	**197.22**	**250.60**

3.3.2.3 Bearing Selection

Each shaft is supported by bearings at both ends. Since the selected bearings' static and dynamic load ratings exceed the calculated shaft loads, four bearing types were selected primarily based on bore diameter and width requirements.

Table 58: Chosen Bearings and their Specification [4]

Bearing Model	Bore	Outer Diameter	Width	Basic Dynamic Load	Basic Static Load	Limiting Speed (Grease)
NSK 6003	17 mm	35 mm	10 mm	6.60 kN	3.25 kN	24,000 RPM
NSK 16004	20 mm	42 mm	8 mm	7.90 kN	4.50 kN	16,000 RPM
NSK 6004	20 mm	42 mm	12 mm	9.40 kN	5.00 kN	17,000 RPM
NSK 6206	30 mm	62 mm	16 mm	19.5kN	11.3 kN	12,000 RPM



Figure 92: Figure 21: NSK 6206 Deep Groove Ball Bearings

3.3.3 Gearbox Housing

3.3.3.1 Material Selection

Grey cast iron, grade FG 200 (ASTM A48 Class 30), was selected for both housing halves due to its castability, stiffness, vibration-damping properties, machinability, and low cost [5]. Sand casting allows the blended walls, internal ribs, bearing bosses, and integral mounting feet to be produced with minimal additional machining [5]. With a minimum tensile strength of approximately 200 MPa, the material provides sufficient strength for the intended application, making housing stiffness and bearing alignment the primary design considerations.

However, grey cast iron is brittle and has low elongation at fracture. Therefore, generous fillets and reinforcing ribs are incorporated around the mounting feet and bolt bosses to reduce stress concentrations [5]. Machining is limited to the split faces, bearing bores, and mounting surfaces, with the bores finished after both halves are bolted and dowelled together. The exterior is painted to provide corrosion protection.

3.3.3.2 Split Construction

The upper and lower housing halves are sand-cast and machined separately to allow easy assembly, inspection, and regreasing. Four dowel pins ensure accurate alignment of the mating faces, which are sealed with a gasket or liquid sealant and secured using M6 bolts. This construction simplifies maintenance in the dusty environment near the resin press.

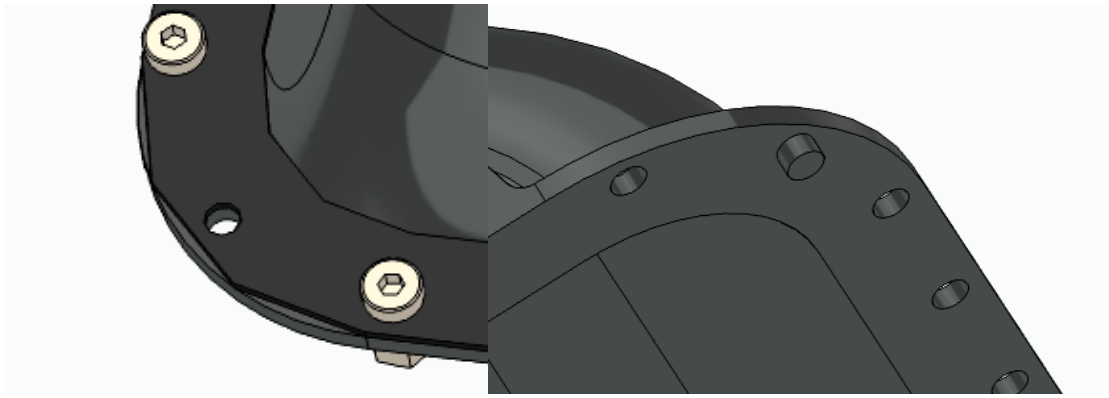


Figure 93: Dowel Pin Alignment with M6 bolt fasteners on the bottom (left) and upper (right) halves

3.3.3.3 Shaft, Bearing and Shifting Mechanism Supports

Each shaft is supported by bearings at both ends. The bearing seats are split between the upper and lower housing halves, forming complete circular bores when assembled. Shoulders locate the bearing outer rings axially, while bolted end caps secure them in place. Rotary seals fitted at the input and output shafts retain grease and prevent resin and carbon dust ingress. The bearing bores are machined with both halves bolted and dowelled together to ensure concentricity.

Internal supports accommodate the free ends of the input, output, and idler shafts, with one support also accommodating the shifting mechanism. Bearing bosses are reinforced with radial ribs that transfer gear forces into the housing walls while limiting deflection and maintaining shaft alignment. Additional ribs strengthen the side walls and split flange.

The selector rods are supported by an integrated holder in the lower housing half, allowing the sliding gears, selector forks, and rods to be installed before fitting the upper half. The shift lever passes through an opening in the upper housing and engages the selector rods, while a gate plate defines the permitted gear positions. Since gear changes occur only when the machine is stopped, the forks are not required to withstand shifting under load.

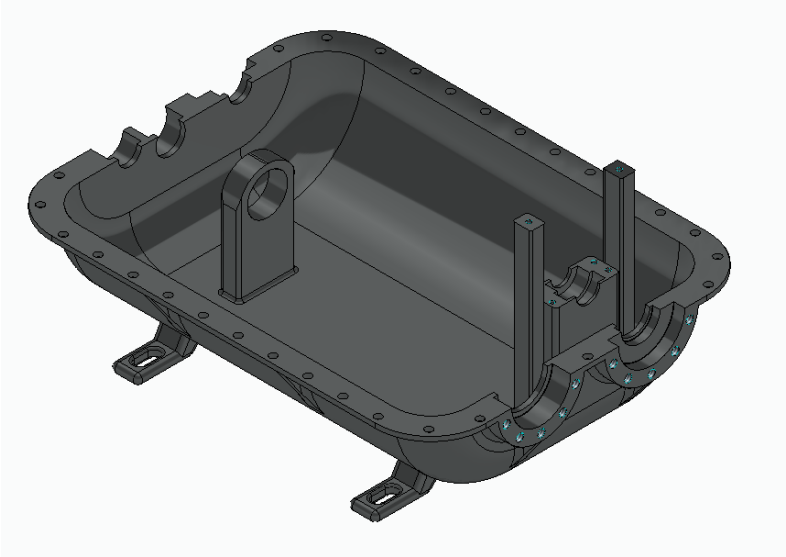


Figure 94: Bottom Half of the Gearbox Housing

3.3.4 Keys and Keyways.

3.3.4.1 Design of the keys and keyways for the gears on lay shaft

For the design of the keys, material which has less tensile and shear stress values was chosen such that the keys are acting as fail-safe mechanism.

Key Material: SAE 1045 Steel [6]

Yield Strength (σy) = 500MPa

Shear Strength (τy) = 380MPa

The selected material clearly has lesser strength than shaft material AISI 4340 Carbon Steel, hence in case of an accidental shock, the keys will fracture without transferring force to the shafts and will prevent the damages to the shafts.

For all the shaft and gear assemblies, rectangular sunk keys are decided to use;

because;

- High Torque Transmission (Positive Drive)
- Axial and Rotational Fixation
- Preservation of Shaft Cross-Sectional Strength

First to design the keyway dimensions and keys, it is identified that at least the gear root to keyway length must be greater than or equal to the 1.5 times of the module which is an empirical use of technical data for safe design.

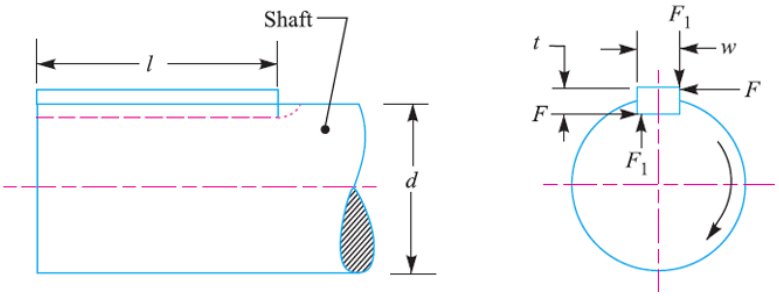


Figure 95: Keyway Design in Shafts

3.3.4.2 Sample calculation for a keyway

Shaft diameter = 40mm

Then,

$$Key\ thickness = \frac{Shaft\ diameter}{6} = 6.67mm$$

$$Key\ width = \frac{Shaft\ diameter}{4} = 10\ mm$$

Then suing following standard table;

Table 59: Standard Values for Key Design [7]

The standard values can be taken from the following table:

Shaft diameter (mm) upto and including	*Key cross-section*		*Shaft diameter (mm) upto and including*	*Key cross-section*	
	Width (mm)	*Thickness (mm)*		*Width (mm)*	*Thickness (mm)*
6	2	2	85	25	14
8	3	3	95	28	16
10	4	4	110	32	18
12	5	5	130	36	20
17	6	6	150	40	22
22	8	7	170	45	25
30	10	8	200	50	28
38	12	8	230	56	32
44	14	9	260	63	32
50	16	10	290	70	36
58	18	11	330	80	40
65	20	12	380	90	45
75	22	14	440	100	50

The proper standard key dimensions were decided.

Since the shaft diameter is 40mm, we chose up to 44mm column;

Standard key width = 14mm

Standard key thickness = 9 mm; Both values are higher than numerically obtained values, which is technically safe and sound.

Following values for each key of the lay shaft were obtained using above method and, each key dimension is checked with the root to keyway length; whether to check the key cross section is weakening the gear wheel or not.

Table 60: Key Widths and Thicknesses

Gear	Shaft Diameter at each	Key thickness	Key width	Standard key thickness using table	Standard key width using table	Gear PCD	Dedendum	Maximum Safe Shaft Diameter
G2	40	6.67	10	9.00	14.00	171	3	147
G7	35	5.83	8.75	8.00	12.00	147	3	124
G3	30	5.00	7.5	8.00	10.00	108	3	85
G5	25	4.17	6.25	8.00	10.00	69	3	46
G9	20	3.33	5	7.00	8.00	45	3	23

According to this shaft all maximum safe shaft diameter values are higher than the selected shaft diameters, because of that the design is safe.

3.3.4.3 Keyway Minimum length calculation

$$Minimum\ length\ to\ stand\ shearing = \frac{2 \times torque}{shear\ stress \times key\ width\ \times shaft\ diameter}$$

$$Minimum\ length\ to\ stand\ crushing = \frac{4 \times torque}{compressive\ stress\ \times key\ thickness \times shaft\ diameter}$$

Using above equations following values were obtained;

Table 61: Minimum Key Lengths

Gear	Shaft Diameter at each	Standard key thickness using table	Standard key width using table	Minimum length to stand Shearing	Minimum length to stand crushing
G2	40	9	14	0.578	1.367
G7	35	8	12	0.771	1.758
G3	30	8	10	1.079	2.051
G5	25	8	10	1.295	2.461
G9	20	7	8	2.024	3.516

Length of the keyway is chosen as 20mm, for it to be firm and hold the gear wheel.

3.3.4.4 Keyway profile design

For design all keyways following type of sketch profile and extrude cut option was used.

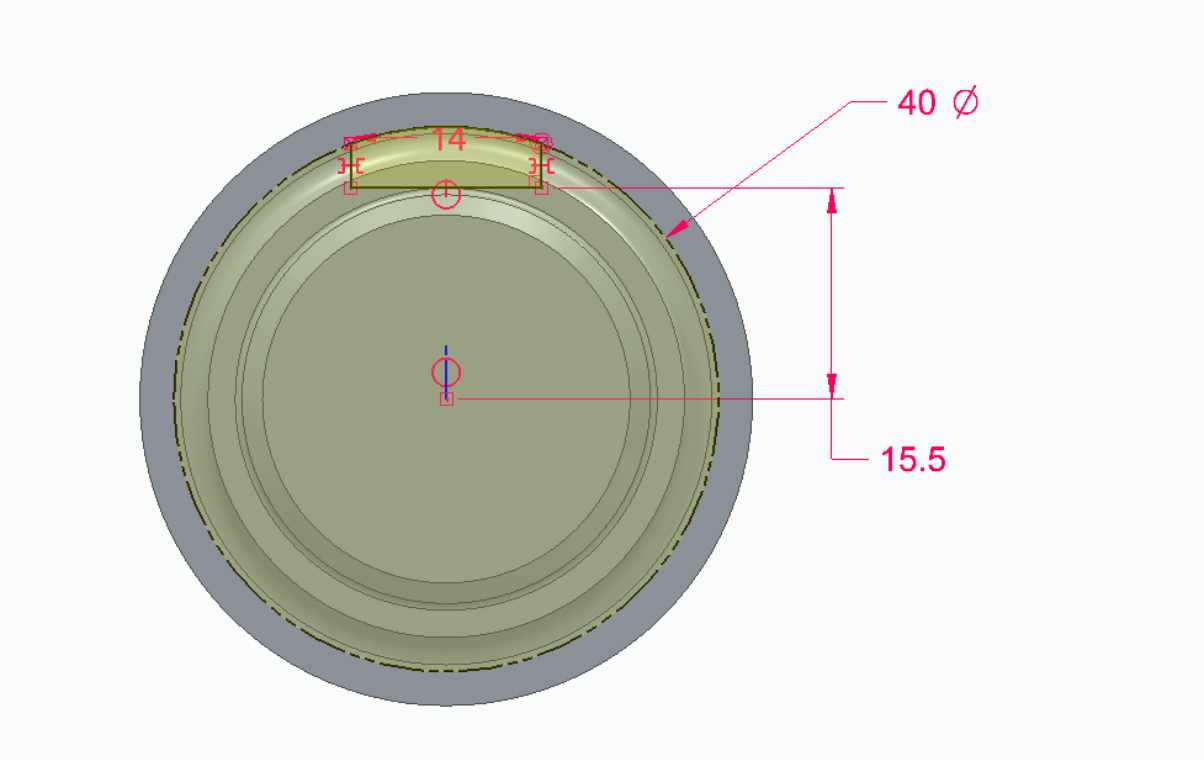


Figure 96: Keyway Sketch Profile

As did in splined output shaft, the dimensional values of the keyway profiles in combine with the shaft diameter values were connected to variable table of the lay shaft with a excel sheet and, when we change the value in the excel sheet and update the variable table, the shaft parameters will get update.

In case of a accidental relationship brake in the design tree, to prevent the harm to keyways, each of these keyways are drawn on a separate parallel plane to base planes, which can be restructured easily by changing the step size of the parallel plane.

3.3.5 Couplings

A coupling is a mechanical device used to connect two shafts at their ends to transmit power while allowing for independent assembly, maintenance, and minor alignment adjustments. In this gearbox, couplings are required at two specific points [7].

Firstly, at the Input shaft which connects the prime move

r to the gearbox which transmits a maximum torque of 24.8Nm at 10rpm.

Then the Output shaft which connects the gearbox to the driven load of 152.5Nm peak torque.

Couplings are used only at these external boundaries. Internal shafts (lay shaft and idle shaft) are connected directly through gears and bearings within the housing.

3.3.5.1 Selection of Coupling Type

A Sleeve (or Muff) Coupling fitted with a parallel key made of SAE 1045 steel was selected for this application. A sleeve coupling consists of a single hollow cylindrical sleeve that fits over the abutting ends of the two shafts, with a key at each end to prevent relative rotation between the shaft and the sleeve.

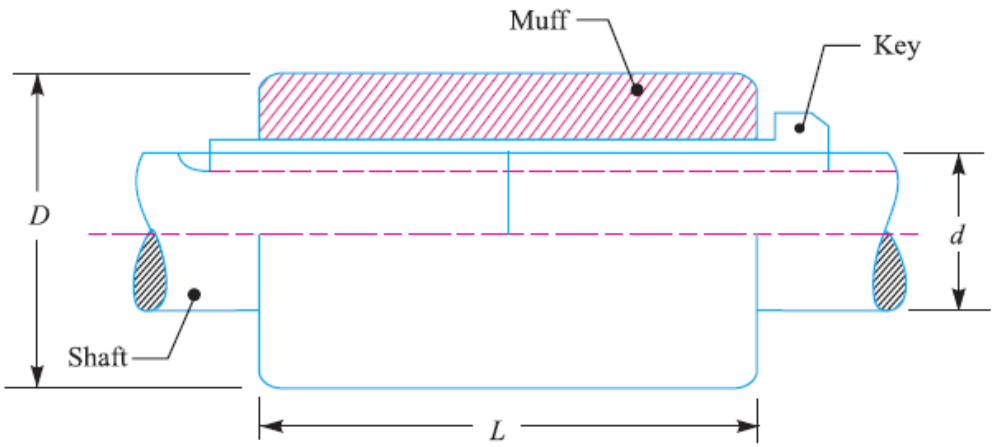


Figure 97: Sleeve or muff coupling [7]

The selection was based on the followings

- The simple design makes the coupling easy to manufacture.
- The sleeve and key can be easily checked using standard stress calculations.
- The coupling is suitable for the 20 mm shafts and their controlled alignment.
- The design follows standard machine design practices for this type of application.

Since both coupling systems connect 20 mm shafts with identical geometry, the structural sleeve and key sized for the 152.5 Nm load will easily accommodate the lower 24.8 Nm load without dimensional changes.

The values obtained for the keys are similar to the previously derived calculations under keys.

$$Width\, w \;=\; 8\, mm,$$

$$Thickness\, t \;=\; 7mm$$

$$T_{max} = 152.5 \text{ Nm} = 152.5 \times 10^3 \text{ N} \cdot \text{mm}$$

3.3.5.2 Design of the Sleeve

$$\text{Key Material: SAE 1045 Steel}$$

$$\text{Yield Strength } (\sigma_y) = 500\text{MPa}$$

$$\text{Shear Strength } (\tau_y) = 380\text{MPa}$$

$$\text{Sleeve Material: Cast Iron}$$

$$\text{Permissible Shear Stress } (\tau_c) = 14\text{MP}$$

Outer Diameter of Sleeve (D)

$$D = 2d + 13 \text{ mm} = 2(20) + 13 = 53\, mm$$

Length of Sleeve (L)

$$L = 3.5d = 3.5(20) = 70\, mm$$

The sleeve is evaluated as a hollow shaft transmitting torque T

$$T = \frac{\pi}{16} \times \tau_{ind} \times \left(\frac{D^4 - d^4}{D}\right)$$

Substitute the known parameters into the equation:

$$152.5 \times 10^3 = \frac{\pi}{16} \times \tau_{ind} \times \left(\frac{53^4 - 20^4}{53}\right)$$

$$152.5 \times 10^3 = \frac{\pi}{16} \times \tau_{ind} \times \left(\frac{7{,}890{,}481 - 160{,}000}{53}\right)$$

$$152.5 \times 10^3 = \frac{\pi}{16} \times \tau_{ind} \times 145{,}858.13$$

$$152.5 \times 10^3 = 28{,}639.26 \times \tau_{ind}$$

Therefore,

$$\tau_{ind} = \frac{152.5 \times 10^3}{28{,}639.26} = 5.32 \text{ MPa}$$

The induced shear stress in the sleeve of 5.32 MPa is less than the allowable shear stress of 14 MPa of cast iron [15]. The sleeve design is safe.

3.3.5.3 Design of the Key

Induced Shear Stress in Key

Using the key shearing equation:

$$T = l \times w \times \tau \times \frac{d}{2}$$

$$152.5 \times 10^3 = 35 \times 8 \times \tau \times \frac{20}{2}$$

$$152.5 \times 10^3 = 2800 \times \tau$$

$$\tau = \frac{152.5 \times 10^3}{2800} = 54.46 \text{ MPa}$$

The induced shear stress of 54.46 MPa is significantly lower than the shear strength of SAE 1045 steel of 380 MPa [17]. The key is safe in shear.

Induced Crushing Stress in Key (σ_c)

Using the key crushing equation:

$$T = l \times \frac{t}{2} \times \sigma_c \times \frac{d}{2}$$

$$152.5 \times 10^3 = 35 \times \frac{7}{2} \times \sigma_c \times \frac{20}{2}$$

$$152.5 \times 10^3 = 1225 \times \sigma_c$$

$$\sigma_c = \frac{152.5 \times 10^3}{1225} = 124.49 \text{ MPa}$$

The induced crushing stress of 124.49 MPa is well below the yield strength of SAE 1045 steel of 500 MPa [67]. The key is safe in crushing.

3.4 Final Design

3.4.1 Keys and Keyways



Figure 98: Keyways in the Layshaft

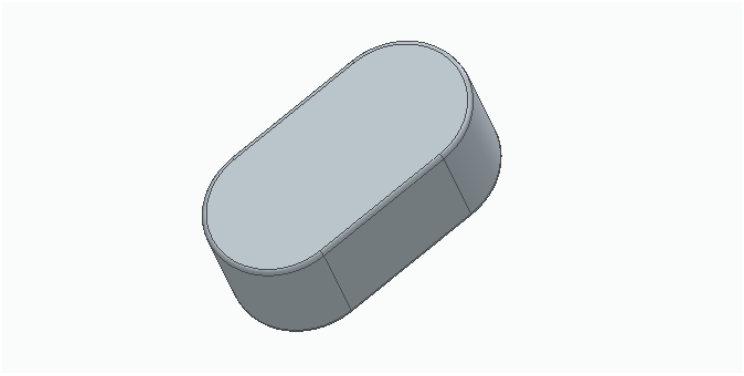


Figure 99: Keys

3.4.2 Gearbox Housing

The housing locates the four shafts in their correct parallel positions, supports the bearings, retains the grease, keeps resin and carbon dust out, and passes the reaction torque to the machine frame. Because the gearbox is grease lubricated, no oil sump, oil level or splash clearance was needed, so the casing follows the gear layout closely and stays compact. The highest pitch line speed is only 0.036 m/s, so heat generation is very low and no breather or cooling features were provided. The design is symmetrical except at the shifter opening, which keeps fabrication simple and loads evenly shared.

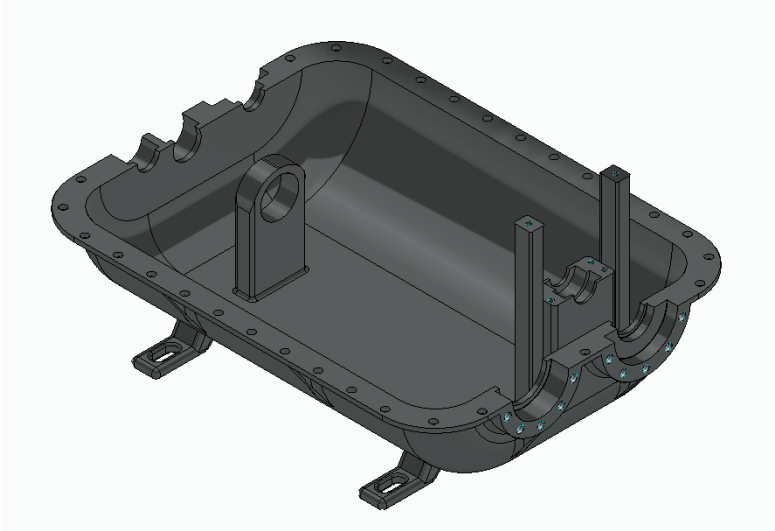


Figure 100: Lower Half of the Casing

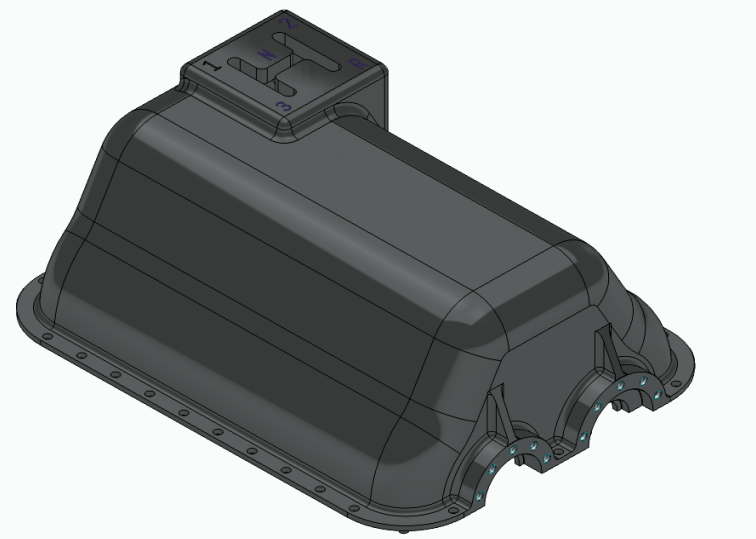


Figure 101: Upper Half of the Casing

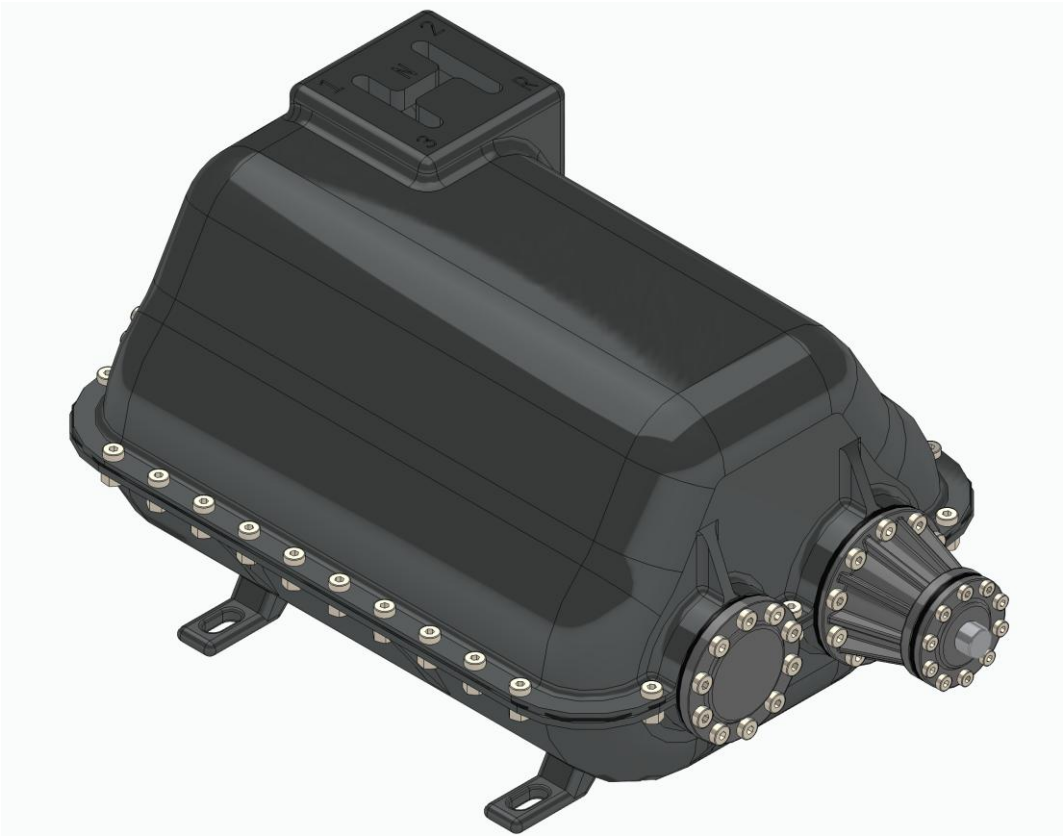


Figure 102: Complete Gearbox Housing

3.4.3 Gear Shifter Mechanism

The four gears on the output shaft are connected in pairs and can slide along the splined output shaft. Each gear pair is connected to a selector fork made out of cast iron, which moves the pair back and forth along the shaft to engage and disengage the gears. Selector fork sits in a groove of 12mm in width and 10mm depth the hub that connects each gear pairings.

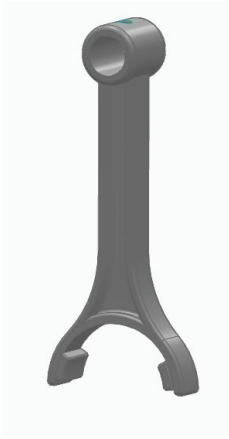


Figure 103: Selector Fork

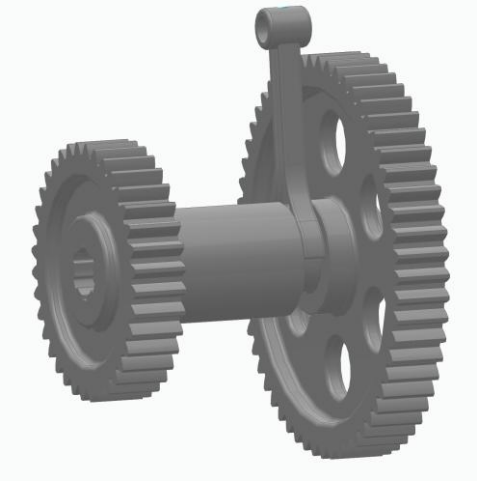


Figure 104: Selector Fork in the Grooved Gear Hub

Each gear wheel has a face width of 24 mm, and at the neutral position, each meshing gear pair is separated by a 4 mm gap. Therefore, the H-pattern gearbox housing is designed to accommodate 28 mm of travel in each direction to ensure proper gear engagement. The two selector forks are tilted about the vertical axis to provide a separation of 50 mm between them at their uppermost points. Both forks are then screwed at the top to a 12 mm diameter and 335mm long selector rods made out of AISI 1020 that run longitudinally through the upper portion of the gearbox. The rods are supported and secured by connecting them to the lower portion of the gearbox housing, allowing easy assembly and maintenance.



Figure 105: H Pattern Casted in the Gearbox Housing



Figure 106: Gear Selector Mechanism Assembly

A Gear Selector is used to shift the gears, and it is held in place by the groove of the selector, which fits on either side of the 15 mm outer casing H pattern. The diameter of the selector fork is 12 mm, and it travels exactly within the H pattern. The Gear Selector is covered by a flexible rubber sleeve, which moves with the lever and prevents dust and carbon fibre particles from entering the gearbox through the H pattern opening.



Figure 107: Gear Selector Fork

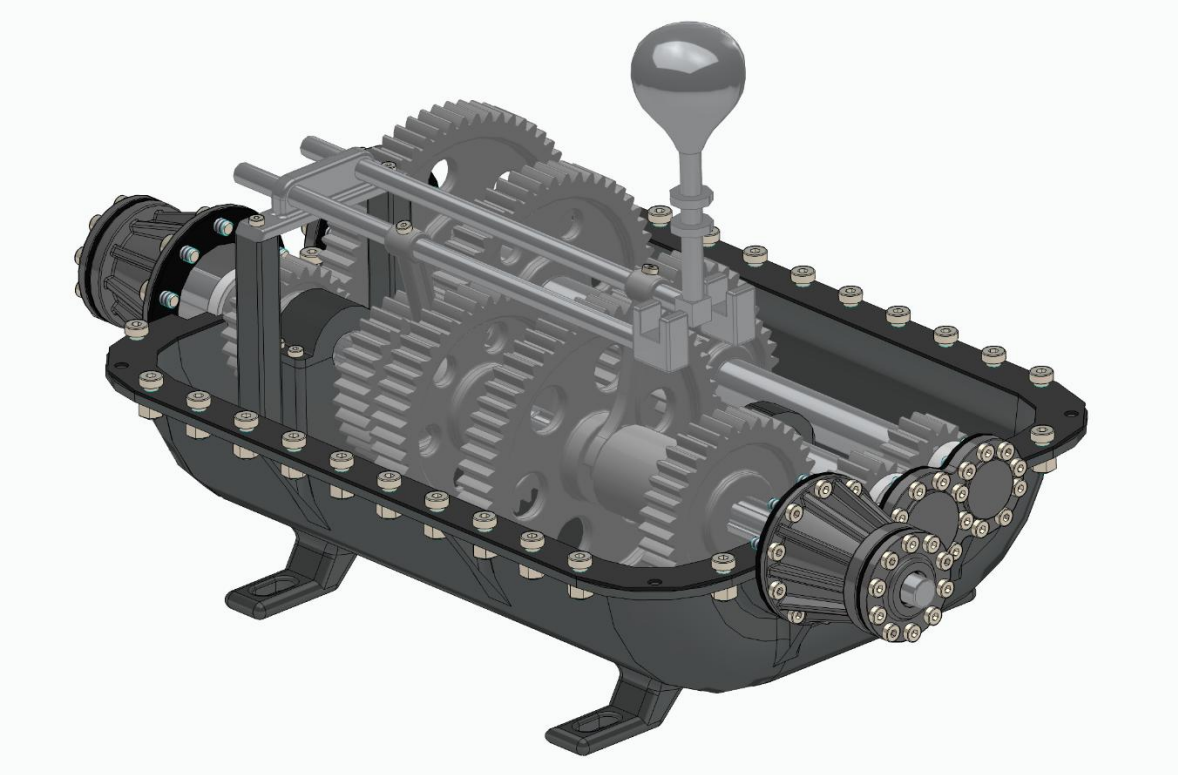


Figure 108: Full Gear Shifting Mechanism Integrated with the Output Shaft and Assembled in the Housing

3.4.3.1 Gear Engagement

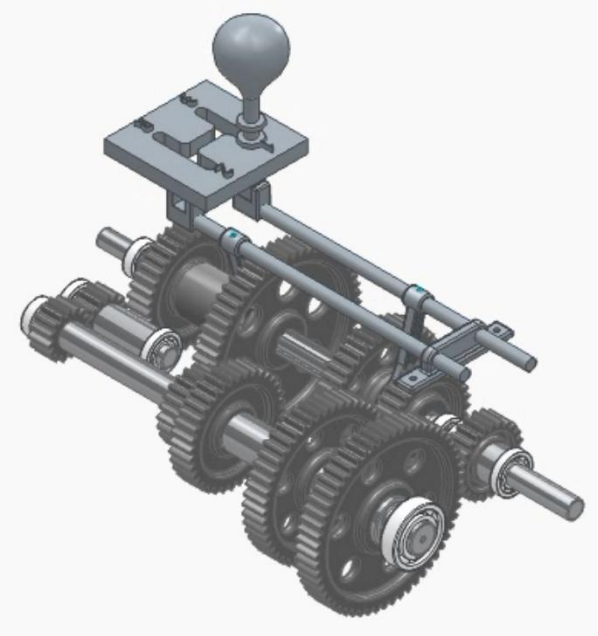


Figure 109: Gear 1 in Mesh

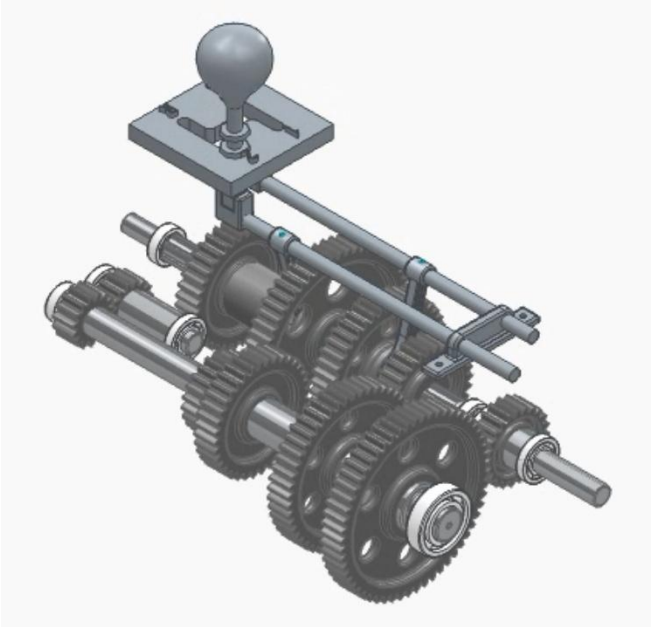


Figure 110: Gear 2 in Mesh

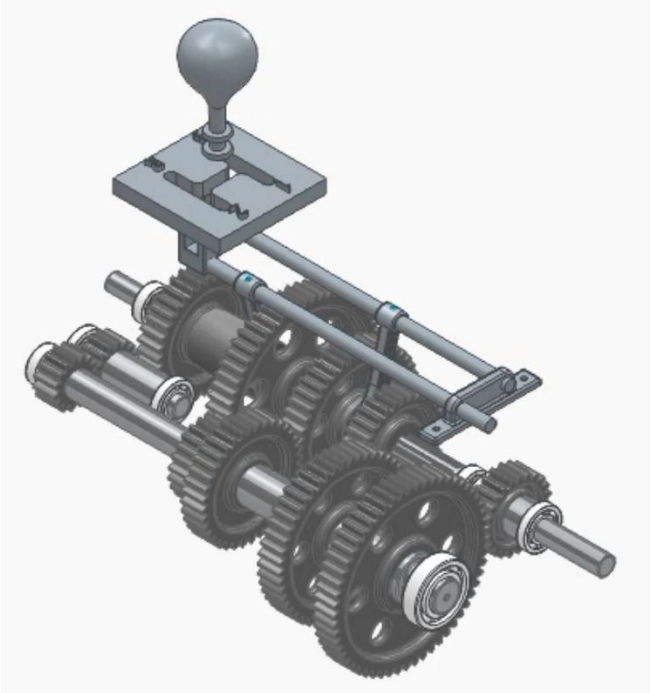


Figure 111: Gear 3 in Mesh

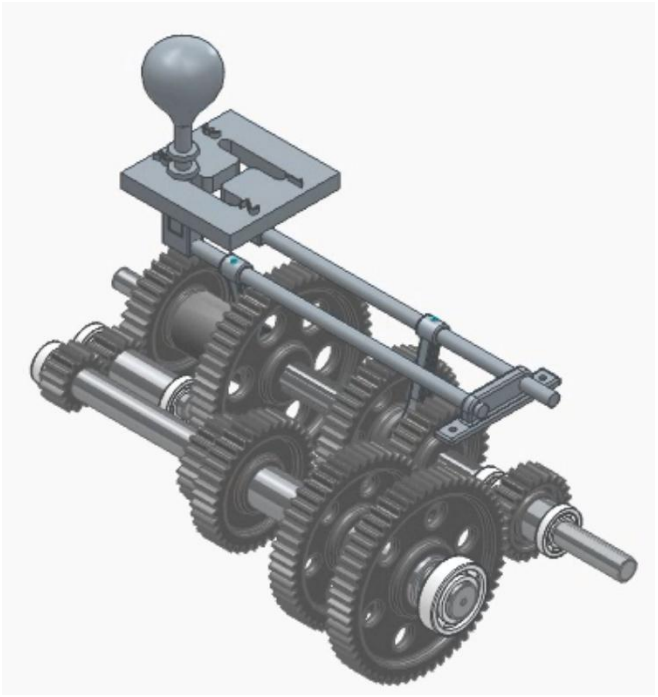


Figure 112: Reverse Gear in Mesh

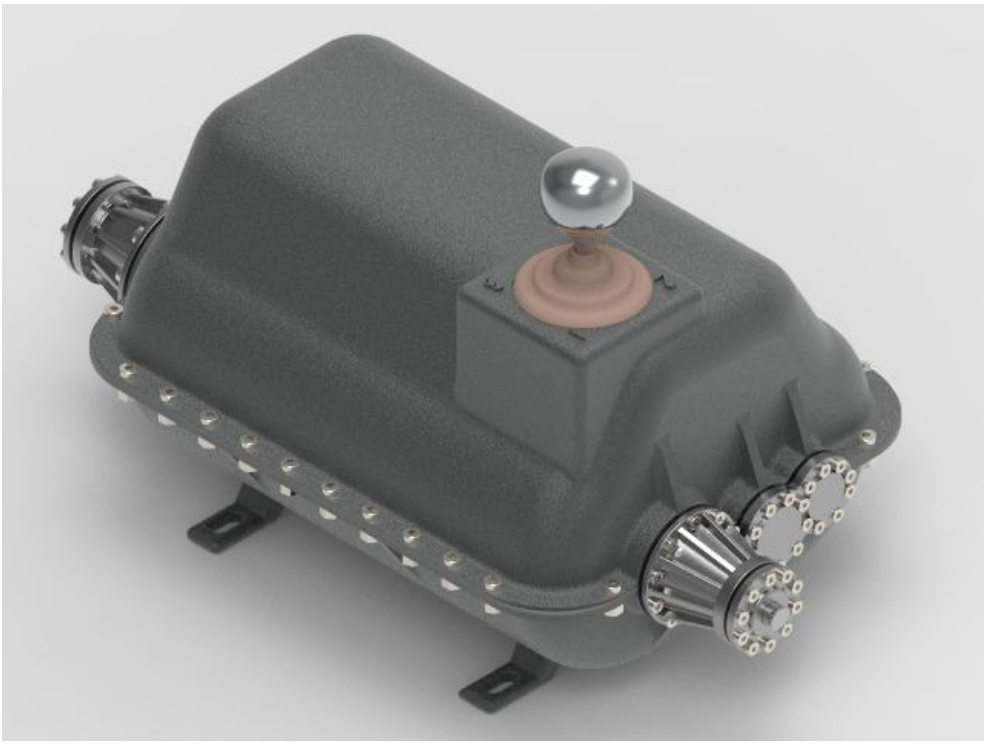


Figure 113: Full Gearbox Assembly

3.5 Future work

The shifter opening in the upper casing is currently covered by a flexible rubber gaiter, similar to the boot on a car gear lever. It allows the lever to move while keeping resin and carbon dust out of the housing. A pivoting lever mounting would be a natural next step, since it gives the lever a defined pivot point and keeps the opening sealed. The lever could then be sized by analysing the forces and moments acting on it and choosing a leverage ratio that keeps the effort needed for a gear change low.

A finite element analysis of the housing could be carried out to verify stress and deflection under all gear loading cases, particularly at the bearing seats, the ribs and the split flange. The same analysis could be applied to the mounting feet and bolts, which carry the reaction torque of the gearbox. The results could also be used to reduce wall thickness and weight.

3.6 References

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